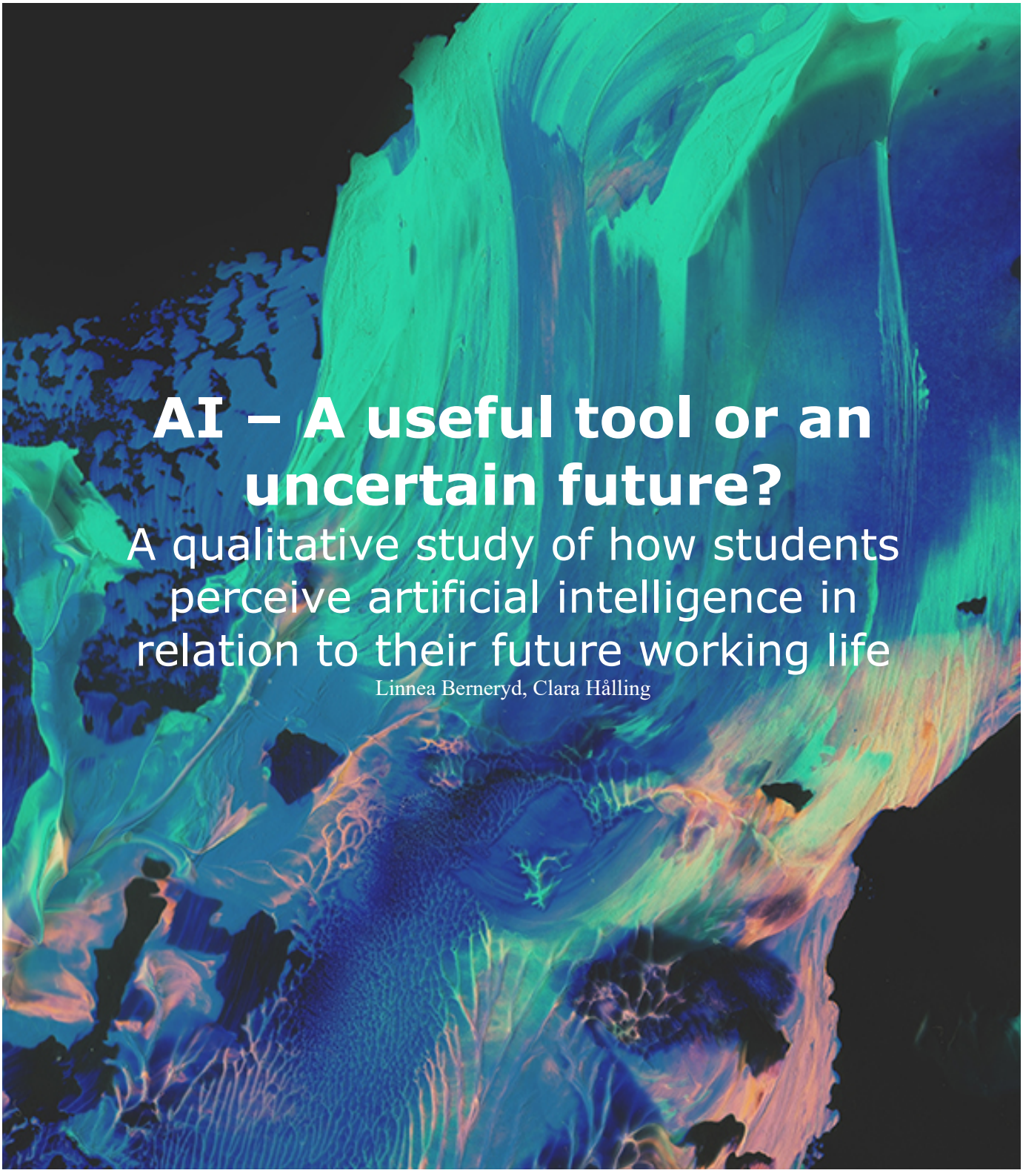
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AI – A useful tool or an uncertain future?

A qualitative study of how students perceive artificial intelligence in relation to their future working life

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Clara Hålling



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Abstract

Artificial intelligence is reshaping how tasks are performed across industries, which raises questions about which competencies will remain valuable in the future labour market. Students are a relevant group in this context, as they are currently developing knowledge and skills in preparations for their future working life that AI is expected to influence. Despite the growing research on AI and its implications on the labour market, little attention has been paid to how the future labour market entrants form their perceptions of AI before entering organizations, and how those perceptions shape the human capital their develop.

This study addresses the gap by researching how students' perceptions of AI in relation to their future working life is formed. A qualitative research design has been used, where the findings are based on 15 semi-structured interviews with business students and civil engineering students at Umeå University. A thematic analysis has been conducted to identify patterns across the two groups.

The findings show that the formation of students' perception of AI in relation to the future working life is shaped by the interaction of three conditions: educational background, social environment and prior experience with AI tools. These conditions determine what aspects of AI students attend to and what they view in terms of potential gain and loss and perceived control. Both student groups express a concern about skill devaluation, but differ to some extent in their defensive strategy: civil engineering students argue that staying competitive means deepening their technical knowledge and developing AI literacy, and business students argue that what will protect them is being good at things AI cannot replicate such as communication, relationships and judgement. Both groups manage uncertainty through distancing by locating the threat another occupational fields or further into the future. The dominant perception pattern across both groups is a split view by viewing it as an opportunity on an individual level, and a threat at the societal of macro level. Perceived control has a consistent split across both groups where the respondents perceive high control over their own usage but near zero control over how AI develops at the macro level. Finally, formal AI education produces different perception formations. Civil engineering students, with greater formal exposure to AI, develop more analytically grounded views while business students rely more heavily on media and peer networks.

This study contributes to business administration research by extending perception theory, specifically König et al.'s (2021) two-factor appraisal framework and Dearborn and Simon's (1958) selective attention concept beyond their original organizational contexts. It demonstrates that the educational field can function as a shaping condition for how individuals form and structure perception. Further, the study contributes to human capital theory (Becker, 1962; Schultz, 1961) by introducing perception formation as a variable in investment decisions for future labour market entrants. It shows that human capital investment decisions is not a rational calculation, but rather shaped by how they perceive and appraise AI.

Key words: Artificial intelligence, perception, creative destruction, business students, civil engineering students, human capital, selective attention, perceived employability, opportunity-threat appraisal

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1. Introduction

1.1 Background

Artificial intelligence (AI) has developed significantly since its conceptual origins in the 1950s, when the idea that machines could be capable of intelligent behaviour first emerged and the term “artificial intelligence” was introduced in 1956 (OECD, 2019, p.15). Since then, the development of artificial intelligence has accelerated, particularly through advances in machine learning that enable systems to use historical data to generate predictions, recommendations and decisions (OECD, 2019, p.15-17; Mucci, n.d.). This allows for more accurate predictions at a cheaper cost, promising productivity gains and help in addressing more complex problems (OECD, 2019, p.15).

In recent years, artificial intelligence has gained increasing economic importance. Investments in AI technologies have grown rapidly, and the number of AI start-ups increased substantially after 2016 as investments in the sector accelerated (OECD, 2019, p.38). The global AI market also show a strong expansion and was valued at approximately €130 billion in 2023 and is projected to grow to nearly €1.9 trillion by 2030 (Madiega & Ilnicki, 2024, p.1). Similar trends are reported in the Stanford AI Index (Maslej et al., 2025) which highlights the substantial growth in private investment, research activity and industrial adoption of AI technologies during the past decade. Together, this indicates that artificial intelligence is likely to become increasingly important in organizations, industries and the future labour market.

As AI technologies become more integrated into organizations it is expected to influence the labour market more broadly by changing how tasks are performed and how work is allocated between humans and technology. Technological change has historically reshaped labour markets by replacing certain tasks but also creating new ones (Schumpeter, 2003, p.83). AI can either automate tasks previously performed by labour or create new tasks and activities in which humans can be employed (Acemoglu & Restrepo, 2020, p.25-26). This means that the consequences of AI for work are not predetermined but instead depend on how the technology will develop and applied in practice (Acemoglu & Restrepo, 2020, p.26). In this way, AI may both automate certain human activities and augment others, thereby reshaping skill demands and changing the conditions for future employment (Raisch & Krakowski, 2021, p.192-193).

This broader expected transformation can already be seen through developments in specific professions where AI is increasingly used to support, automate or restructure work tasks. For instance, in financial services AI is used in areas such as credit underwriting, credit decisioning and automated fraud detection (OECD, 2024, p.13-15). Another profession is auditing, where AI supports automated analysis of large datasets and anomaly detection, which allows auditors to identify potential risks more efficiently (Celestin & Vanitha, 2019, p.46-48). Taken together, this suggests that AI is likely to alter not only how work is performed, but also which skills and capabilities become valuable in the labour market. What skills will be valuable in future is therefore uncertain, raising the question of individuals interpretation of what AI implies for their own competencies and competitiveness in their future working life. This is particularly relevant for students, since they are in the process of developing competencies and preparing for future careers in a labour market increasingly shaped by AI.

AIs implementation in an organization does, as mentioned, affect different occupations differently, depending on task context and nature. Research often uses augmentation and automation as its paradigms to describe and categorize AI usage (Agrawal et al., 2019; Raisch & Krakowski, 2021). An augmented AI usage supports and complements the human labour to further improve decision-making, productivity and effectiveness (Agrawal et al., 2019, p.39). This complement can therefore process and analyse data faster and more accurately than a human, surpassing the human cognitive limitations (Enholm et al., 2022. p.1720). Automation on the other hand refers to when AI machines control the input as well as the output of a task with little human involvement (Agrawal et al., 2019, p.37). However, automation of a task can lead to adjacent tasks to an augmented usage style as a result of the need of human judgement (Raich & Krakowski, 2021). Thereby, AI changes and restructured how tasks and workflows are handled and shifts where the human control is needed.

This perspective highlights that students may not necessarily perceive AI as only a threat or opportunity (Raisch & Krakowski, 2021). Instead of the students having a simple perception of how AI can be implemented, this suggests that a student can view AI from multiple angles, depending on how AI may affect their future work. AI can automate tasks as well as augment tasks, affecting occupations differently. Therefore, students' perceptions are affected by their interpretation of the relationship between AI and their future competencies. This perspective shows the importance of gaining the students perception of AI and a reflection of their understanding how AI can both automate and augment capabilities. How AI is integrated into organizations influences what skills students think will continue to be valuable and therefore influence how students will prepare for their future employment.

1.1.1 Students and how they use AI

Students can be considered an especially relevant group in the context on artificial intelligence, since they are in the process of developing knowledge, skills and expectations about their future working life. At the same time, they are preparing to enter the labour market in which AI is expected to influence how work is performed, how tasks are allocated and which competencies become valuable (OECD, 2024, p.12). According to OECD, the integration of AI affects high-skill and high-wage occupations substantially (OECD, 2024, p.17). These occupations include financial examiners and budget analysts which AI can excel and therefore outperform a human (OECD, 2024, p.17). Simultaneously, AI has and is predicted to take over more routine tasks, making them less valuable to develop (OECD, 2024, p.14-15). This makes students an important group for understanding how AI is perceived in relation to future careers as they navigate an AI-influenced labour market.

Furthermore, this also indicates the importance of AI competence beyond technical skills, but also an understanding how to handle an AI-influenced society. According to UNESCO, AI competency among students is important because it enables them to understand human agency in relation to AI such as human judgement and decision making which is crucial (Miao et al., 2024, p.14). Students need to develop competencies regarding human-centred mindset and how to ethically handle AI to effectively use AI responsibly (Miao et al., 2024, p.14-15). This entails that students' perceptions may not only be shaped by the labour markets expectations but also by the nature of their education, if it provides understanding how to use AI critically and responsibly.

A study conducted on higher-education students in Sweden explored the adoption and attitudes of chatbots. Result showed a broad awareness and use of AI tools such as OpenAI's

ChatGPT (Stöhr et al., 2024, p.9). As for engineering students, their AI usage was reported to be among the highest which also was consistent with a greater optimism towards AI tools (Stöhr et al., 2024, p.8). Moreover, social science students, including business administration students, displayed a slightly lower level of usage, in combination with a lower level of optimism towards AI (Stöhr et al., 2024, p.8). The study also displayed a difference among students regarding their attitudes and how they work with AI tools, varying between gender and field of study (Stöhr et al., 2024, p.1). Students' usage can be connected to their perception, higher usage is according to Stöhr et al. accompanied with a positive attitude towards it (Stöhr et al., 2024, p. 8). Students may therefore be impacted by the rapid developments of AI and likewise influenced in how they choose to educate themselves. They can reflect future skill development and technology adoption, which is a valuable insight to understand labour market changes. Students entering the workforce are joining a changing labour market, affecting the value of their capabilities and skills. Recent studies emphasise that individuals actively take actions of their value to sustain the competitiveness against AI (Kong et al., 2024, p.229). This is a result of the individuals' perception of AI, to which shapes and creates their actions against a change. An opportunistic perception of AI is associated with career resilience and an improved informal learning, enabling adaptive behaviours (Kong et al., 2024, p.228). From an organizational perspective, having knowledge about future employees' perception can improve the understanding of their potential individual development and adaptability towards the workplace. These indications can predict not only individual performances and results but also on an organizational level.

1.2 Problem discussion

The labour market is subject to change due to new conditions such as demographical, economic cycles and technological changes. The technological advancements of AI have in the last decade gained significant importance across multiple sectors of the labour market, providing productive and effective solutions (OECD, 2019, p.15). This rapid and fundamental change has accelerated the interest in AI rapidly with investments going from 200 to 1400 deals over the years of 2011 to 2017 (OECD, 2019, p.41). As AI is becoming more widespread, occupations are experiencing changes in task processes as well as task characteristics. Additionally, the uncertainty of how AI will ultimately affect occupations remains unclear with researchers reaching different conclusions. Even though AI is often associated with efficiency and an increased productivity, occupations can suffer the consequences when competencies are being altered.

Furthermore, the developing labour market is creating uncertainty among multiple occupations, as AI is affecting task processes and ownership (OECD, 2024, p.12). AI has exhibited to transform how tasks are done and who bear the responsibility of task performance, either by fully taking control or supporting human labour. Automating tasks implies automating tasks to not require human input and the performance relies on the AI technology. This suggests that AI replaces individuals in their work, making human labour unnecessary in certain areas (Enholm et al., 2022, p.19). Augmentation on the other hand act as a complement to the human labour, supporting and improving the performance of the task. Previous research has shown a divide on how the integration of AI will affect occupations either by abolishing or shifting the labour demand (Orrell & Veldran, 2024, p.1). The ambiguity of how the future labour market will look like affects skills and knowledge value, as some tasks can be more effectively made by AI tools.

In organizations, companies seek knowledgeable and skilled workers to gain organizational success. By leveraging on individual traits and attributes of workers, organizations can on a unit-level become factors that give competitive advantages (Ployhart & Moliterno, 2011, p.142-143). With this view, companies should be interested in individuals' development in terms of creating value and to remain competitive on the market. The individuals own competency development is not only relevant for their own employability but is also crucial for organizations. Researchers have discussed the value of individual capabilities and how it translates to unit-level benefits, emphasising the accumulated human capital enables organizational learning (Youndt & Snell, 2004, p.339). However, AI's integration in work processes creates problems for this exchange. Individuals are currently investing in education that is expected to generate value in the future, however, AI complicates what competencies will still be valuable as it augments and automates tasks. Skills in areas that AI can perform faster, more precise and cheaper will become less valuable to develop. On the other hand, some competencies that AI cannot match may be the human performance and will therefore become more valuable to develop, such as judgement and ability to collaborate with AI. AI is therefore a central reason to why students experience uncertainty regarding what to put their efforts and investments in. This will, in turn, affect organizations that all are dependent on human competencies.

1.3 Research gap

Despite AI and its effects on occupations being a largely studied area, there are still more perspectives to discover. Human capital studies have mainly focused on determining investments in education, skills and knowledge to be grounded in an understanding that they will bring value in the future (Becker, 1962, p.48). The logic is as followed: investments in education and training now will in the future generate a higher salary and competitive advantages (Becker, 1962, p.48). The assumption is that the individual can make rational judgements regarding which skills and knowledge will be of value in the future. This in relation to AI becomes much more difficult and uncertain since AI have the ability to change the demand of competencies. Previous human capital research has explored how human capital is accumulated through training and education, while not exploring the perception of individuals, how they appraise the value of their investments with a changing competence demand. This is of high relevance and a central focus of this study because students are individuals that are currently developing their competencies they expect to utilize in their future working life.

Previous research of AI disruption has focused on how artificial intelligence can and will shape the labour market and alter task compositions (Acemoglu & Restrepo, 2019; Autor et al., 2003; Agrawal et al., 2019). Research has also discussed the difference of augmented and automated AI usage, describing the consequences on occupations, labour demand and skills. Although these studies are important to be able to understand how AI can change work, it does not explore the interpretation of these changes in relation to human capital accumulation by those who currently are developing their skills and knowledge.

Simultaneously, how individuals respond and act towards new technology have through research shown that it is affected by their own interpretation of whether it is a threat or an opportunity (De Freitas et al., 2023, p. 1848-1849). The focus has been on already active workers or people in general, emphasizing the impact their perceptions have (Jackson & Allen, 2024; De Freitas et al., 2023). The findings display the impact individual perceptions may have on organizational success, inducing a relevance to gain understanding of future labour market entrants' perspective of AI's effect on their working life in the future. This

creates value in exploring future labour market entrants who are in the process of developing competencies and attitudes, having great impact on organizations. Similar research discusses how perceptions are shaped by perceived control over a technological change (König et al., 2021, p.791). This particular study determined that a change which is framed as a gain, the outcomes are mainly considered positive, and the attention prefers the benefits rather than the negative effects (König et al., 2021, p.790). These perspectives are however rarely connected to future labour market entrants, whom are in the process of choosing investments in skills and knowledge. These perceptions affect how they understand the labour market, as well as how they evaluate their competence development.

Taken together, these three strands of research reveal a gap. Human capital theory assumes that individuals are able to make informed decisions about their skills and knowledge, AI disruption research show that those conditions are changing and perception research establishes that individual interpretations of change can shape behaviour. The latter perspective is however something that has not been applied to future labour market entrants navigating an AI-influenced labour market. Our thesis aims to address this gap by examining how students form their perceptions of AI in relation to their future working life, and how those perceptions shape the human capital they develop. Students are of particular interest, since they are currently accumulating human capital, while also forming their expectations of how their education will translate into value in the future. Jung et al. (2024, p.542) emphasise the need to explore how students are affected by technological disruption, specifically in relation to their skills development, which supports the relevance of this gap. This contributes to a deeper understanding of how future labour market entrants navigate the changing competence demands while simultaneously developing their human capital, and how it shifts the focus on different competencies which shapes their expectations and capabilities they will bring to their future workplace.

1.4 Research question

The research question is derived from the gap identified above. While previous research has examined AI's impact on labour market conditions and how active workers respond to technological change, less attention has been paid to how future labour market entrants form their perceptions of AI before entering the organizations, and further how those perceptions shape the human capital they develop. This study therefore addresses both the formation of perception and its consequences for skills and knowledge development among near-graduate students at a critical point before labour market entry. The research question is formulated as the following:

"How are students perceptions of AI in relation to their future working life formed, and how do these perceptions shape their development of skills and knowledge as future labour market entrants?"

1.5 Contribution

This study aims to contribute towards research within business administration by increasing understanding of how future labour market entrants perceive artificial intelligence in relation to their future working life, and how those perceptions shape the human capital they develop. The study explores how perceptions of AI are formed, and why they differ between students with different educational backgrounds. In this way, the thesis contributes to organizational research on technological change by focusing on how future labour market

entrants make sense on AI before entering working life. This contrasts with previous research, which have focused on AI's impact on already active workers, labour market demand as well as how tasks are structured. Understanding the pre-entry stage is therefore relevant for organizations, as the competencies that future labour market entrants develop before entering the labour market will ultimately shape the human capital available to the organization.

From a theoretical perspective, the study contributes to a shared insight between technological disruption, perception and human capital research. Human capital theory assumes that individuals make relatively informed decisions about their skills and knowledge (Becker, 1962). However, this assumption has not been examined in the context of future labout market entrants and technological change such as AI. By combining human capital with perception research, specifically selective attention (Dearborn & Simon, 1958), sensemaking (Weick, 1995) and König et al.'s (2021) two-factor appraisal framework, the study offers a theoretical contribution by including perception formation as a factor in human capital investment for students prior to entering the labour market.

From a practical view, the study contributes to businesses and organizations, since students represent the future employees and workforce. The findings may help organizations understand how future employees approach AI-related change, competence demands, and uncertainty which can be valuable for organizational processes such as recruiting, onboarding or upskilling, or in change-management processes. This therefore becomes especially important for businesses that seek to recruit employees in the near future while they are also in a work environment where AI is becoming increasingly important.

1.6 Delimitations

This study is delimited to students at Umeå University and focuses on two different student groups: business students and civil engineering students. The civil engineering student group will be used as a comparative element against the business students, this allows for a deeper analysis of two specific fields which may work in the same organizations in the future. Business students are likely to enter occupations that AI is expected to influence work processes and decision-making such as management, administration or finance. Civil engineering students on the other hand are likely to encounter Ai from a more technical perspective, leading to a difference in AI exposure among the fields. The study is further limited to students in the later stages in their education, as these students are considered more likely to reflect on future career opportunities, threats and competence demands related to artificial intelligence. This provides a relevant perspective to explore, as the population seeks to formalize their skills and knowledge for their future workplace.

The study is also delimited to students' perceptions of artificial intelligence in relation to their future careers. We thereby exclude the actual effects of AI on labour market outcomes or career development in practice. Previous research, that has briefly been presented in the introduction and will be presented in the theoretical framework, has shown that different perceptions of change can be associated with individual and/or organizational outcomes, which emphasises the importance to study perceptions. However, such outcomes are not examined in this thesis. The focus of this thesis is limited to how students perceive AI in relation to their future career.

Importantly, this study is delimited by theoretical scope. The thesis covers theories within technological change, human capital and perception, three theoretical fields that are broad

and complex. To fully understand each of these areas would require an extensive amount of research. This study is therefore limited to aspects within these fields that are relevant to the thesis, which is how AI changes task and competence demand, human capital and how perceptions are formed under AI. The aim is not to cover these fields entirely, but to the selected aspects that are of relevance to the research question and purpose.

1.7 Definitions

Table 1 presents relevant definitions that will be used in the thesis.

Table 1: Definitions.

Term	Definition
Artificial Intelligence (AI)	Digital systems that perform and process tasks, such as learning from data, generating content and making predictions, that normally are done by human intelligence.
AI tools	The system that with the help of artificial intelligence can perform tasks, managed by human interaction.
Generative artificial intelligence (Gen AI)	Generative artificial intelligence refers to the type of AI that creates new content such as images, text, summaries and ideas which are based on a user generated prompt.
Perception	Refers to the interpretation and meaning assigned to a certain phenomenon.
Augmented AI usage	Augmented AI usage refers to when AI supports the human user in completing a task. The user remains in control regarding judgement, decision-making and interpretation.
Automated AI usage	Automated AI usage refers to situations where AI performs the larger part of a task with little human involvement.
Formal AI education	Structured teaching of AI concepts, methods or applications within an academic programme.
AI literacy	Refers to the individual's broad ability to understand, use and evaluate AI and AI tools in a relevant context. The ability can come from formal education, work experience, self-learning or private use.

Sensemaking	The process where individuals interpret and make sense of ambiguous or uncertain situations by drawing on prior experience, social interaction and available information.
Future labour market entrant	Refers to individuals that are close to completing their education and are getting ready to enter the labour market in their intended occupational field.

Source: the authors.

Note: In this thesis, AI will be used as the broader concept, while AI tools represent the specific applications student may encounter and use any form of AI, where Gen AI is an example of.

2. Theoretical background

The theoretical background presents the reader with a cohesive examination of relevant research and theories in order to understand the ambiguity of the perceptions of AI by students. Firstly, an assessment of technological discontinuity (AI) mechanisms and the prediction on jobs in the future describes the complexity of technological development in the labour market. Secondly, Human capital theory frame skills and knowledge as investments which emphasises the threat and/or opportunity perception on AI on future work opportunities. Lastly, perceptions are treated in internal processes, how individuals respond, react and behave towards uncertainty to assemble a comprehensive framing for the research. The chapter ends with a conceptual framework, section 2.5, that integrates the theoretical perspectives presented in the chapter into a unified model, which guides the analysis of the study.

2.1 Creative destruction

Through innovation and technological development, the external and internal processes of an industry abruptly change, according to the economist Schumpeter (Schumpeter, 2003, p.114). In the early 20th century, Joseph Schumpeter developed an economic concept which emphasised the disruptions by innovative solutions and entrepreneurship that changes industry patterns and promote economic growth (Schumpeter, 2003, p.11). His earliest work is the Theory of Economic Development (TED), which introduced entrepreneurship as an "engine of growth" in the economy (1942, cited in Mehmood et al., 2019, p.5). Innovation is essential for an entrepreneur to contribute to the development of a company or a whole economy, driving the economic growth and creating new business opportunities (Mehmood et al., 2019, p.5). This leads to the economy to constantly revolutionize and change from old patterns. Thus, Schumpeter viewed the economy as a cycle that have reoccurring phases than span over time (Kuznets, 1940, p.260). Prosperity, recession, depression and revival are the identified phases that the economy passes through at the pace of innovation intensity (Kuznets, 1940, p .259). The process of each phase depends to which extent innovations emerges and wipes out old workways or entire industries (Kuznets, 1940, p.263).

Schumpeter conceptualizes the process of reinventions as a *creative destruction* where new innovations replace current economic structures continuously which transforms entire industries as well as relocate resources (Schumpeter, 2003, p.81). Creative destruction is according to Schumpeter a process that changes the economy from within, erasing old structures while also creating new more productive workways (Schumpeter, 2003, p.83). This implies that the outcome of technological innovations can generate opportunities by promoting productivity and threatening conditions when labour demands shift or disappears. The technological changes further both gains and losses to an economy depending on the point of view. From a labour market perspective, innovation comes with a shift in the demand of certain skills and capabilities which forces a revaluation of skill acquisitions of future workers. It also simultaneously makes certain capabilities or tasks depreciate in the labour market. In the setting of AI disruption and perception, this contributes to describe how humans can view new technologies as valuable career tool as well as a threat to acquired skills and knowledge. Innovations create an obsolescence of skills, while simultaneously facilitating new and improved processes that need another skillset.

2.1.1 Exploration vs exploitation

To explain how individuals as well as organizations react to technological discontinuity, it is important to understand the different approaches they may use to encounter it. As mentioned, technological discontinuity according to Schumpeter enforces an adaptation of new procedures while it alters the old procedures (Schumpeter, 2003, p.83). New technology such as AI can enable an explorative and curious approach as well as a more conservative, where known routines are utilized. March explains these approaches as either *explorative* or *exploitive* in terms of an organizations resource allocation (1991, p.71). Exploration is the experimentation and learning of new potential without the certainty of benefits in return (March 1991, p.71). Exploitation on the other hand explain the refinement and usage of existing procedures (March 1991, p.71). A condition for both exploration and exploitation are their struggle for the same resources within an organization. According to March, organizations may struggle with balancing the two factors because they distribute costs and return on different time-horizons (1991, p.85). Exploitation is seen as predictable in outcome since it based on previous knowledge, which reduces outcome volatility and generate results faster (March 1991, p.84-85). Exploration in contrast enables innovations that can improve outcome, nevertheless, do take longer time to generate results (March 1991, p.84).

Gupta et al further explores the relation between exploration and exploitation and how it is not necessarily linear, and how organizations and individuals can balance them (Gupta et al., 2006, p.693). The number of systems that are subject to the analysis decides whether or not exploration and exploitation will act as competing or complementary (Gupta et al., 2006, p.698). This ambiguity is described as either a balancing or *ambidexterity,* which explains the simultaneous usage of both exploration and exploitation (Gupta et al., 2006, p.693). In the context of AI at an individual level, this portrays that students can both use their current education and learning related to AI without having to choose and therefore combine these two in regards to their own development. Exploitation can generate faster results and therefore be seen as the more reliable. However, exploration can sometimes be viewed as more trustworthy if it demonstrates obvious advantages that rule out not adapting to it (March 1991, p.83). Exploration can however restrain productivity due to its limited background and therefore create uncertainty in whether it will be profitable. This also implies that exploration can create uncertainty regarding competence investments, impacting the students perception if their investment are going to remain valuable.

2.2 Task based view of AI

As previously mentioned, technological advancement does not distort in the same way across all occupations. Research about the impact of AI on work and occupations commonly use a task-based view of how AI changes specific activities or allocates labour into new areas (Acemoglu & Restrepo, 2019, p.4; Orrell & Veldran, 2024, p.1; Autor et al., 2003, p.1279). According to Acemoglu and Restrepo's framework on technology's impact on labour demand and productivity, AI contributes to two key forces: *displacement effect* and *reinstatement effect* (Acemoglu & Restrepo, 2019, p.3). The displacement effect refers to tasks being seized by AI that was previously done by labour while the reinstatement effect counteracts the decreasing labour by the creation of new tasks, allowing an allocation for tasks called the *productivity effect* (Acemoglu & Restrepo, 2019, p.4). Similarly, Autor et al. emphasises that routine tasks are subject to a decrease with advancements in computer technology, all while an increase of demand on labour for non-routine tasks (Autor et al.,

2003, p.1322). They highlight that the alteration of labour demand occurs inside occupations, reducing the human input in routine tasks and raises input in tasks that are nonlinear (Autor et al., 2003, p.1322). Taken together, AI can reshape how occupations are constructed and not necessarily abolish whole professions. Routine heavy occupations such as accounting or data programmer may therefore be interpreted as disappearing occupations, which both authors argue does not have to be realistic because of AI altering effect on labour demand (Acemoglu & Restrepo, 2019, p.27 ; Autor et al., 2003,p.1322).

2.2.1 Augmented and automated AI usage

AI can be used in multiple ways to reach the same outcome and can therefore not be discussed as a uniformed usage type. Hence it is relevant to discuss the different types of AI usage practices to understand the behaviours and perceptions of AI from the students depending on which their usage style. To create a clear structural discussion of different usage types, the thesis will use "augmented" and "automated" as its definitions to distinguish the different perceptions that may come from each usage style, translating into individual human capital investment strategies. According to multiple researchers, the distinction between augmented and automated use is the human participation and the level of reliance on the AI tool (Nguyen & Elbanna, 2025, p.6; Enholm et al., 2022, p.1720). The reliance of an AI tool is dependent on factors such as perceived control, understanding and personal compatibility (De Freitas et al., 2023 p.1848). Attitudes are shaped by the sense of trust individuals have towards AI, which affects whether they choose to use a performative automated AI tool or assisting augmented usage type (De Freitas et al., 2023 p.1845). This reflects the research purpose of gaining understanding of student's perception of AI tools, providing a distinguishment of how the perceptions diverge.

Augmented AI usage

A common assumption of artificial intelligence is that it only fully automates and fails to discuss an augmented usage type (Agrawal et al., 2019, p.39). However, an augmented usage is more recently discussed as a prominent AI usage type among multiple fields in the labour market (Agrawal et al., 2019, p.30). Augmented AI usage is where the usage complements the process of finishing a task, rather than fully completing it without any human interaction (Agrawal et al., 2019, p.39). AI act as a support and operate part of the task and therefore gain an assist role to the human (Enholm et al., 2022, p.1720). According to Enholm et al, augmentation has the possibility to process larger amounts of data and information at a higher speed that surpasses the human cognitive limitations (Enholm et al., 2022, p.1721). It is therefore common for AI augmentation to be implemented to process and read data in organizations, since it aids the human decision making effectively (Enholm et al., 2022, p.1721). The human judgement and context considerations is still present with the support of technology, that enhances and clarifies conditions. It is therefore critical for the AI user possess skills such as judgement and interpretation reflecting from a human capital perspective, where these skills are considered valuable investments (Becker, 1962, p.11). In Agrawal et al.'s framework about prediction tasks, augmented AI's positive impact on prediction tasks is its ability to decrease costs and improve quality (Agrawal et al., 2019, p.32). Augmented support in decision tasks such as supplier choices or raising interest rates, can help improve the human decision-making by facilitating competitor outlooks, stakeholders and economic conditions that affect the decision outcome (Agrawal et al., 2019, p.39-40). With the suitable data and prediction, the decision-making process is more time-effective while also being more accurately estimated. This type of augmented use is common among policework, where they can reduce crime rates by accurately predict risks (Agrawal et al., 2019, p.39).

Moreover, AI augmentation use is as mentioned highly dependent on context, as the nature of certain work fields or existing structures come with specific norms and requirements (Lei & Kim, 2024, p.261). Applying the same function of an AI tools across different disciplines does not consider the contextual information of the field which affect how AI can be embedded and used (Lei & Kim, 2024, p.261). It is still needed for expertise, human judgement and contextual interpretation to have a central role and to have the responsibility of the outcome. The value of the output is therefore not only due to the artificial intelligence abilities, but rather on the capacity of human expertise and judgement to utilize the tool. Augmented AI usage main contribution is therefore its ability to enhance the productivity by minimizing costs and improving the information interpretation whilst still requiring human proficiency.

Automated AI usage

Researcher either define the exchange of human contribution as automation or substitution AI usage. The thesis will use automation as its terminology since it reflects a realistic usage of AI tools. Automation does not exclude human participation completely in all cases and does therefore not substitute human interaction, instead the human role is managing and observing the output (Agrawal et al., 2019, p.34-35). AI therefore replaces the human input, and performance relies on the AI to perform and produce. Automated AI usage is common among routine and prediction tasks with precedents or a clear framework (Agrawal et al., 2019, p.35) This type of usage can according to Agrawal et al streamline tasks and make them more productive and cost effective (Agrawal et al., 2019, p.33). By automating processes and tasks, organizations can also reduce workload on employees and shorten the process of completing tasks (Enholm et al., 2022, p.19). Thus far in the development, automated AI is mainly used for routine and repetitive tasks that needs little to no human interaction or judgement. The research made by Agrawal et al show that automation with AI exceed in prediction tasks where the expected outcome can be accurately predicted by data (Agrawal et al., 2019, p.33). However, with the rapid development of AI technologies, this practice is becoming more common across more business tasks. The author further argues that this is due to tasks being modified to prediction-oriented, making it possible for AI to take over (Agrawal et al., 2019, p.32).

Because of the automation of tasks, there are major effects on multiple occupations and work task skills that can be replaced by AI tools. Technological development has historically led to labour demand changes, where the outcome has resulted in a reorganization and shift in the labour demand. Human labour demand can therefore be relocated rather than erased (Acemoglu & Restrepo, 2019, p.28). In recent years, the advancements in technology have made the labour productiveness optimized to the point where the labour demand has not shifted but decreased (Acemoglu & Restrepo, 2019, p.28). As a result, certain skills and knowledge are becoming less valuable and necessary in the future. According to De Freitas et al, people try to form a causal chain to predict outcomes and conditions to create comfort and sense for themselves (De Freitas et al., 2023 p.1846). Not knowing how AI works or will affect them, can lead to discomfort (De Freitas et al., 2023 p.1846). The discomfort can also cause a perceived replaceability, where AI can take over tasks from humans, leading to a shift in what knowledge and skills are valuable and therefore contribute to how one decided to invest in education and experiences (Becker, 1962, p.41). The incentives to invest in specific skills will change in relation to how AI automation progresses, leading to this type of AI usage affecting the perception of one's long-term abilities.

Augmentation and automation paradox

Augmentation and automation can be seen as categorically opposite to each other, while recent studies argue for them to not be fully separated. Raisch and Krakowski define automation and augmentation as paradoxical, where they cannot be seen as separate phenomena but instead as interdependent (2021, p.203). They argue for augmentation and automation to be interchangeable for tasks. As mentioned previously, augmentation refers to a collaboration between humans and machines versus automation refers to the machine fully taking over decisions. Certain tasks may originally require augmentation in the form of human judgement and contextual understanding where the human expertise is the starting point, to subsequently become more routine and codifiable which enables automation (Raisch & Krakowski, 2021, p.195). This may also result in other tasks adjacent to the task to require an augmented usage to supervise and control outputs (Raisch & Krakowski, 2021, p. 197). This paradoxical phenomenon explains the ambiguity of AI integration in an organization where context and task characteristics impact how tasks can be managed. Therefore, AI usage can induce positive and negative perceptions largely depending on the understanding of how capabilities can change in value.

2.2.2 The future of AI in the labour market

The forecast of the future of AI in the labour market is heavily discussed among researchers across multiple research fields. In a study made by Grace et al., surveyed 2,778 AI researchers in 2023 about their forecast about the pace, nature and impacts of AI (2025, p.1). The survey found that the researchers were divided on AI impact in the future, with the 68% believing positive outcomes from high-level machine intelligence AI was more likely (Grace et al., 2025, p.9). However, almost half (48%) of the respondents that where dominantly positive towards AI scored a 5% likelihood of extremely bad outcomes which were exemplified as human extinction etc. (Grace et al., 2025, p.9). The majority showed either a “substantial” or “extreme” concern when asked about AI’s impact on fundamental scenarios such as AI spreading false information and misuse of powerful tools (Grace et al., 2025, p.8). The uncertainty and division emphasise the ambiguity of AI research and how it continuously changes and provide different perceptions of how AI will eventually impact the future.

A report conducted by Brend Orrell and David Veldran in 2024 discusses how AI will affect society, skills and jobs (2024). The authors created a table regarding the impact on employment and skills by AI and present the opposing views among authors (Orrell & Veldran, 2024, p. 9). The table is highly relevant to discuss in the thesis since it provides a structured overview of the different predictions, which is of high relevance when discussing perceptions of future workers. The debate of conflicting expectations can explore students’ perception of AI as an opportunity or a threat. Table 2 contextualizes the conflictions between AI and work to further the understanding of the students. However, because AI is rapidly changing and developing, it is relevant to adapt the table with current studies to adhere to the fast developments.

Table 2 is adapted on table 1 in Orrell and Veldran (2024, p,10-11) and reworked by the authors. The reworked table will preserve the original concepts used and supplement the table with peer-reviewed, current concepts and articles to ensure contemporary relevance. The original table is considered a secondary source, however the authors have reviewed the sources used in the table to ensure quality and authenticity.

Table 2. Summary of AI's projected impacts on employment and skills (Adapted)

Researchers' names	Year	Prediction
Acemoglu and Autor	2011	Many jobs are at risk of automation, especially medium-skilled and highly routinized ones.
Frey and Osborne	2013	Forty-seven percent of US jobs, especially lower skilled, are at high risk from automation by machine learning and robotics over the next decade.
Autor	2015	AI will substitute for human labor in some domains but complement it in others, such as in nonroutine tasks that require advanced cognitive skills and creativity.
Martin Ford	2015	As AI advances, it raises the possibility of a "jobless future"
Arntz, Gregory, and Zierahn	2016	Around 9 percent of jobs across developed countries are automatable with current AI technologies
Manyika et al.	2017	Between 75 million and 375 million people, up to 14 percent of the global workforce, might need to switch occupations by 2030; AI may create up to 890 million new jobs by 2030.
Till Alexander Leopold, Vesselina Ratcheva, and Saadia Zahidi	2018	Technology will displace 75 million jobs, but 133 million new jobs will emerge by 2022.
Bessen	2018	AI will likely substitute for human labor in some domains, but it will create more jobs than it destroys by increasing productivity in key markets.
Jacques Bughin et al.	2018	AI automation will put a premium on basic digital and advanced programming skills.
Brynjolfsson, Mitchell, and Rock; Webb; Felten, Raj, and Seamans; and Acemoglu et al.	2018-22	Many occupations most exposed to AI are white-collar and require high education levels and complex cognitive ability.
Acemoglu and Restrepo	2019	AI may boost productivity and demand for labor, which could result in job gains.
Susskind	2020	In the long term, AI could destroy more jobs than it creates.
Zahidi et al.	2020	By 2025, AI may displace 85 million workers but create 97 million jobs.
Lane and Saint-Martin	2021	AI can perform some nonroutine cognitive tasks, but many jobs, especially those requiring social interaction and physical labor, cannot yet be automated.
Michael Chui, Eric Hazan, et al and Chui, Lareina Yee, et al.	2023	Generative AI and other technologies could automate work activities that absorb 60 to 70 percent of employees' time, AI technology could automate half of today's work between 2030 and 2060, and AI can increasingly perform high-skilled tasks involving management, application of expertise, and decision-making.

Table 2. Summary of AI's projected impacts on employment and skills (Adapted) (*Continued)*		
Eloundou et al.	2023	Generative AI could affect some tasks in 80 percent of jobs and half or more of component tasks in 20 percent of jobs.
Valoir	2023	AI could automate 40 percent of activities in the average workday.
Goldman Sachs	2023	AI will expose two-thirds of US occupations to automation, but will more likely complement than substitute for most jobs.
Brynjolfsson, Li, and Raymond; Shakked Noy and Whitney Zhang; and Ajay K. Agrawal, Joshua S, Gans, and Avi Goldfarb	2023	AI may help level the playing field for lower-skilled workers by boosting their skills' value.
Addition by authors		
Bayan and Nawaf Ali	2024	AI adoption significantly affects job displacements while at the same time creates new job opportunities, creating a shift in employment as well as in skills transformation.
Hellsten et al.	2025	Hiring in white-collar AI-related roles will increase as a result of AI adoption, while at the same time does not increase in overall employment.
Jeong and Jeong	2025	Organizations investments in AI raises employees' perceptions of insecurity regarding their job.
Brynjolfsson, Li, and Raymond	2025	AI assistance tool could improve productivity of workers, specifically for lower-skilled workers that benefit their output.

Source: Orrell and Veldran (2024), adapted by the authors

2.3 Human capital theory: Foundations

The Human capital theory emerged in the 1960s and was conceptualized by G.S Becker and T. Schultz. Schultz firstly constructed the view of education as an investment," By investing in themselves, people can enlarge the range of choice available to them. It is one way free men can enhance their welfare" - (Schultz, 1961, p.2). Viewing training and education as investments and not consumption promotes knowledge and skill development, meaning that the productive capacity of the individual is an asset (Schultz, 1961, p.2). Becker on the other hand considered human capital as an economic investment, evaluating costs and returns (Becker., 1962, p.1). These investments, which are ranging from education to training, are expected accumulate to earnings meaning that the people that invest more in themselves tend to earn more in the future (Becker, 1962, p.48). The investments can be in the form of time, effort or money and is a possible way to improve the future earnings (Becker, 1962, p.11). The actual value of the competence relies heavily on labour markets and its demand in terms of what is considered valuable and desirable which is reflected in wages (Becker, 1962, p.11). The investments generating human capital are knowledge, skills and competence gained through training, education and experience on the job as well as outside the job. By doing so, the individual increases their productivity and earnings for their future (Becker, 1962, p.11). Training and education can come with a tuition cost and other expenses but also requires actual experience (Becker, 1962, p.25). Becker also notes the

forgone earnings accumulated during the actual training and education, which affects the result of whether the educations can be considered valuable for future earnings (Becker, 1962, p.26).

2.3.1 Human capital as a unit-level resource

Human capital theory has historically focused on the impact of education and training to increase future earnings and productivity. Ployhart and Moliterno furthers the human capital theory to a multilevel model by addressing its relevance for organizational progress (2011). The theory connects the micro-level and macro-level research of human capital to form a cohesive model of how these two factors come together and form a resource at an organizational level (Ployhart & Moliterno, 2011, p.145). The authors present a "unit-level resource" that emerges from the individual's traits known as the micro-level of research, which are defined as knowledge, skills, abilities and other characteristics (KSAOs) (2011). They combine micro, intermediate and macro levels of processes that emphasise the importance of individual attributes, providing valuable human capital to which have strategic relevance at a macro level (Ployhart & Moliterno, 2011, p.127). The individual traits can on a unit-level perspective gain strategic and competitive advantages by interaction and collaboration between individuals that transforms into a collective performance (Ployhart & Moliterno, 2011, p.142-143). This view of the value of accumulating human capital being a core stone to organizational success is shared among researchers, highlighting that the created knowledge can enhance organizational learning as well as knowledge acquirements (Youndt & Snell, 2004, p.339). This perspective displays the importance of embedded individual competencies in organizations and how they need to be integrated properly to gain full potential in an organization.

2.3.2 The KSAO framework

The different forms of capability are valued differently among different disciplines, where some occupations require more specialized training and experience on-the-job than other (Becker, 1962, p.25). Therefore, it is important to specify the capabilities to further understand how the labour market values different individual traits in different fields. The thesis will use Ployhart and Moliterno's conceptualization of how human capital emerges from individuals' knowledge, skills, abilities and other characteristics (KSAOs). As mentioned, the Ployhart and Moliterno's KSAO's separates characteristics and capabilities to which can bring understanding to how AI-usage influences the KSAO's in separate manners.

Knowledge

In the KSAO framework, knowledge is described as the comprehension of processes and concepts by an individual. The knowledge is what the individual understands of task procedures, specific software's or principles (Ployhart & Moliterno, 2011, p.134). Knowledge is a cognitive "can do" which means that it can be acquired or learned through education and/or experience (Ployhart & Moliterno 2011 p.134). According to Ployhart and Moliterno, having a high general cognitive ability support learning, as it advances knowledge and skills acquisition to be quicker and thoroughly (Ployhart & Moliterno, 2011, p.133).

However, knowledge possession does not create value automatically. For the knowledge to be of value, it is reliant on how it is embedded and applied in organizational and task-

specific contexts. In a study exploring knowledge-intensive companies, human capital was determined to be influencing competitiveness if it was combined with knowledge capabilities, which suggests that knowledge is strategically important when it becomes an embedded capability within an organization and its context (Mohammad Shafiee et al., 2024, p.138). Knowledge is therefore impacted by AI, as it changes what individuals need to know and more importantly how knowledge is used. An AI automation of a task shifts the human knowledge from performing routine tasks and reproducing information to contextual and evaluation knowledge. This perspective is relevant, since it emphasises the students' individual ability to use the knowledge they possess in interaction with AI, as an important factor of their accumulating human capital. How the student uses their knowledge is a determinant of how well their human capital will contribute to a changing labour market.

Knowledge can be segmented into two dimensions, tacit and explicit knowledge (Nonaka, 1994, p.16). *Tacit knowledge* comes largely from experience and is therefore difficult to perfectly formalize. It involves both cognitive and technical elements, that can be in the form of observation, education or a shared experience (Nonaka, 1994, p.19). Individual contribution is according to Smith, an important factor in one's knowledge and thus making the transferability limited (2001, p.314). However, tacit knowledge can be transferable under certain circumstances, majorly depending on the context. The author argue that all knowledge has a tacit component, exemplifying that all knowledge has an element of personal interpretation (2001, p.314). *Explicit knowledge* on the other hand is more easily transferable knowledge since it can be codified and systemised (Nonaka, 1994, p.19). Explicit knowledge can be formalized into manuals, models and other documents that are understandable by others. This type of knowledge creation is according to Smith a resource that can be utilized and reused for solving multiple problems or even contribute to knowledge sharing (2001, p.315). Tasks that are grounded in explicit knowledge and are predictable or routine can therefore be more suitable with an automated AI usage.

Skills

An individual's skills are explained by Ployhart and Moliterno as the capacity to acquire information and the practical capabilities (2011, p.134). They suggest that skills can be bound to larger context-generic domains such as social or leadership skills as well as context specific and narrow domains such as firm specific cultural functions (2011, p.134). Skills can further be divided into two subsections, soft and hard skills. Levasseur classifies *soft skills* as a combination of self-awareness, communication, collaboration, judgement and leadership abilities (2013, p.566). The author further argues that soft skills are less widespread than hard skills since it requires interpersonal and self-awareness which can be hard to transfer due to its personal nature. In other words, soft skills are personal and might appear differently among individuals. Personal motivation is an important factor in developing soft skills, because it enables a desire to understand complex mechanisms and behaviours (Levasseur, 2013, p.569).

Furthermore, *hard skills* are skills acquired through task specific training and education such as mathematics and accounting (Levasseur, 2013, p.569). Similarly to explicit knowledge, the hard skills are more often transferable due to its tangible nature. Hard skills also therefore also more exposed to being automated (Agrawal et al., 2019, p.35). Both types of skills are needed and valuable across all work fields in varying amounts. Disciplines such as accounting and data analysists display a higher level of hard skills and a lower need of soft skills to complete work tasks. Levasseur argues that individuals choosing a career in the disciplines that require a lower level of interpersonal contact tend to carry a lower level of soft skills and often do not feel the drive to acquire them (2013, p.569). With this in mind, soft and hard skills are both subject to be affected by the AI tools integration and thereby

change the accumulation of human capital. Tasks that require hard skills are exposed to AI automation whereas soft skills namely communication and leadership are less susceptible because they are difficult to replicate and codify.

Abilities

Abilities according to Ployhart and Moliterno's model are the general cognitive ability to understand, create and retain information (2011, p.134). It refers to the innate potential of how well an individual can learn and execute tasks. According to the authors and other studies on the subject, present that ability is context-generic, referring to it affecting multiple situations in a similar way (Jensen, Kanfer in Ployhart & Moliterno 134). This contrasts with skills and knowledge which are developed through training or education that can be advanced over time, all while abilities are connected with deep-seated traits. Ability further has an impact on risk taking and decision-making under uncertainty according to a study made by Dohmen et al. (2018) In their study, they identify that a high cognitive ability is correspond to a lower level of harmful risk-taking (Dohmen et al., 2018, p.117). Therefore, ability can explain how well an individual can retain and comprehend information which affect skills and knowledge retainment that consequently affects task performance. In the context of AI, students' abilities can affect how they learn, understand and use AI knowledge. This is of relevance, because AI creates unclear competence demands making students ability to evaluate information and adapt under this uncertainty influence their perception to AI in relation to their future working life.

Other characteristics

Lastly, other characteristics refers to personality, values and interests that affect the behaviours and actions from individuals (Ployhart & Moliterno, 2011, p.133). These are individual noncognitive traits that impact behaviour and reactions to the environment. The most commonly used theory to explain personality characteristics is the Five-factor model which conceptualizes traits into five major dimensions (Digman, 1990, p.418). Woods et al. uses this concept to explain that choice of occupations dependent on personal characteristics that are developed from a young age (Woods et al., 2013, p.17). In the context of AI, other characteristics have the ability to influence how student respond and react to AI discontinuity. Personality traits and values can affect if they students choose to interact with and learn AI.

2.3.3 Human capital in strategic management

Individuals impact on an organizational level is discussed by Lepak and Snell, where they emphasize the difference of how human capital is evaluated across organizations (Lepak & Snell, 1999, p.45). They implement an architectural view of an organizations HR work that highlight that human capital varies in different organizations in respect to their uniqueness and strategic benefit (Lepak & Snell, 1999, p.36). The ability of the individual is highly valuable as well as hard to replace, since the accumulation of an organizations human capital is highly linked to employees working well together and creating a unique effective structure (Lepak & Snell, 1999, p.36). These bonds form due to long-term involvements and internal investments, welcoming commitment and firm-specific human capital (Lepak & Snell, 1999, p.38). Schultz also highlights this by stating: *"The observed growth in productivity per unit of labour is simply a consequence of holding the unit of labour constant over time although in fact this unit of labour has been increasing as a result of a steadily growing amount of human capital per worker"* (1961, p.6). The productivity is improved not only due to better technology but is also impacted by the quality of the work done.

Knowing this, there is a clear connection to classical human capital as explain by Becker and Schultz. Individuals invest in training and education to gain greater earnings for themselves in the future, which the organizations interpret and value in different way depending on their own unique structure. Capturing the right competences to fit into an organization determines the outcome. With the perspective of a student, it explains the motivating factors to invest in an education to better their employability in the future. However, with the increasing influence of AI in multiple arenas of the economy and society, this directly affects what in the future will be considered valuable skills (Becker.,1962, p.11). AI is rapidly increasing the pace of technology development, changing the demand of skills.

The human capital of a firm can be seen as a source of competitive advantage when the employees have capabilities that are valuable, rare, imperfectly imitable and non-substitutable. Nevertheless, Campbell et al. argues that human capital does not automatically create firm specific advantages (2012, p.389). The employee capabilities are individual to each employee and not always embedded in the organization which makes the firms ability to accumulate human capital important (Campbell et al., 2012, p.389). This indicates the strategic value of human capital must be taken care of and extracted from the skills and knowledge to make them valuable, rare, non-imitable and non-substitutable in the organization. Simply acquiring capable employees does not generate competitive advantages. In relation to AI's integration in the workplace, this means that different AI usage practices may generate different types of human capital. An augmented AI usage type still requires human collaboration, which allows development of capabilities such as judgement and expertise (Agrawal et al., 2019, p.39). Substitutional AI usage on the other hand has the possibility to stimulate more substitutable skills, since the human involvement does not require actual judgement or interaction.

2.4 Perception

The following section presents the theoretical framework for understanding perception as it is applied in this thesis. The section presents three interconnected themes. First the fundamental understanding of what perception is and what affects it will be presented. Secondly, perception is discussed in the context of discontinuous change and technology. Third, the section provides understanding on how perception influences individuals actions.

2.4.1 What perception is and what shapes it

Perception is a fundamental process when individuals assign meaning to information, situations and events in their environment. Rather than being a passive and objective when receiving facts, perception is an active and interpretive process shaped by prior experience, existing knowledge and contextual factors (Walsh, 1995, p.281). This means that two individuals can encounter the same situation, such as the integration of AI, and have constructed fundamentally different understandings of what it means and what it implies for them.

A foundational concept for understanding how perception operates is through sensemaking. Weick (1995, p.13) explains sensemaking as the process through which individuals form plausible interpretations of unclear or unfamiliar situations, which draws from existing knowledge and social cues to form an understanding of what is happening and what is required from them. Therefore, sensemaking is not about finding the most correct

interpretation that is objective, but about forming an interpretation that is plausible enough to guide action (Weick, 1995, p.17). Maitlis and Christianson (2014, p.67) develop this further develop this concept further by showing that sensemaking is both an individual and social process, which means that when perceptions are formed it does not happen in isolation. It is shaped by different narratives, conversations and shared interpretations within the groups and environment that the individual belongs to. Applying this reasoning to this thesis, it suggests that the students' perception of AI is not only formed through individual reasoning. It is also formed through their educational environment and social network about what AI means for their future work.

An important mechanism for perception is selective attention. As explained by Dearborn and Simon (1958, p.140), it means that individuals attend selectively to information that aligns with their existing role, background and prior experiences and knowledge, which leads to filtering out the information that falls outside the individuals frame of reference. While sensemaking describes what individuals do with the information that is encountered, selective attention operates at an earlier stage. It shapes what information reaches the individual in the first place. In this way the two concepts are connected, where selective attention filters out the information and sensemaking transforms that information into an interpretation. This concept is relevant to this thesis as it provides an understanding for how students with different educational backgrounds and different levels of AI exposure may attend to different aspects of AI.

In addition to sensemaking and selective attention, a further factor shaping perception that is particularly important to this thesis is perceived employability. Berntson and Marklund (2007, p.280) define perceived employability as an individual's subjective assessment of their possibilities to obtain employment in the labour market. This perception is not only formed by formal qualifications, but also how individuals assess the relevance and market value of their competencies in relation to the labour market conditions (Berntson et al., 2006).

Taken together, these perspectives illustrate that perception is neither the same for everyone encountering the same phenomenon, nor random. Instead, it is shaped by earlier experiences, existing knowledge and how individuals understand their own their own situation in a changing environment. This is important to keep in mind when analysing how students perceive AI in relation to their future careers, because their perceptions cannot be explained by one single factor.

2.4.2 Perception in discontinuous change and technology

Discontinuous change refers to external changes that requires an internal adoption in ways that break from the firms or individuals established trajectory along a nonlinear path, and it is described as challenging the assumptions on how value is created and captured (König et al., 2021, p.774). Such change is associated with, for instance, potential shifts in competence requirements and increased uncertainty about what strategies and skills will remain valuable (König et al., 2021, p.774-775). This aligns with technological discontinuities, where breakthroughs disrupt prior patters of development and are associated with heightened turbulence and uncertainty (Tushman & Anderson, 1986, p.439). In this thesis, AI can be approached as a discontinuous change in the sense of rapid integration into organizational processes and may alter competence demands (König et al., 2021, p.774–775; Tushman & Anderson, 1986, p.439), which may increase uncertainty about future career requirements.

A key insight from research on discontinuous change is that perception becomes particularly consequential when the situation is uncertain. Gilbert (2005) argues that when individuals face discontinuous change, existing resource and rigid routines can constrain how the change is interpreted and responded to. Prior investments in skills, knowledge and ways of working shape the perception through which new developments are assessed, which can lead to either inertia or adaption. This is relevant to the students where prior educational investments and existing knowledge structures are likely to shape how they interpret AI and what implied for their future careers. Building on this, König et al. (2021, p.789) argues that perception of discontinuous change should not be treated as one distinct appraisal, instead there are two independent factors: gain/loss framing and perceived control. Gain/loss framing refers to whether a change is interpreted generating potential benefits or potential harm. When a change is framed as a gain, attention is directed towards positive outcomes and improvements, making an opportunity perception more likely. On the other hand, when it is framed as a loss, attention is guided towards risks and negative consequences, making a threat perception more likely (König et al., 2021, p.790). The second appraisal, perceived control, explains to which extent individuals believe they can influence the outcomes of a change and respond to it effectively. Higher perceived control tends to support opportunity perceptions since the change appears more manageable, while a lower perceived control supports threat perceptions since the situation seems difficult to affect (König et al., 2021, p.791). Further, gain/loss framing and perceived control can vary independently from each other. A person may frame a change as a potential gain while perceiving low control over it at the same time leading to ambivalent or constrained opportunity perceptions. Conversely, a person can frame potential losses while still perceiving high control of the change, which can lead to a more manageable threat where coping is possible (König et al., 2021, p.789-792). In practice, students' perceptions of AI are likely to be more nuanced than a straight opportunity or threat.

Research examining how individuals perceive AI and related technologies in organizational contexts supports the picture presented above. Brougham and Haar (2018, p.241) find that awareness of smart technology, AI, robotics and algorithms is associated with concerns about job displacements and uncertainty about the future, but that these perceptions vary depending on individual characteristics and occupational context.

2.4.3 How perception influences action

Perception does not only shape how change is understood, it also influences how individuals respond to it. Findings within organizational research show that the way a situation is perceived, whether it is a threat or opportunity, has consequential effects on the actions that follow (Staw et al., 1981, p. 502; Gioia & Thomas, 1996, p.372). Understanding these consequences clarifies why the perceptions that students form of AI are likely to matter for how they prepare for and navigate an AI-influenced labour market.

When a situation is perceived as a threat, research suggests that responses tens to become more constrained and defensive. Staw et al (1981, p.50) describe this as threat rigidity, where perceived threats lead individuals to restrict their information processing, narrow their focus and fall back on well-established routines rather than exploring new approaches. This narrowing of attention and behaviour means that a threat perception does not only produce discomfort, is shapes what actions feel possible and reasonable. Connelly and Shi (2022, p.1371) develop this understanding of threat responses by identifying three forms of responses to perceived threats. The first one is fight, where the individual takes active steps to mitigate the threat. Secondly, flee, where the individual tries to avoid the exposure.

Finally, freeze, where the individual takes a defensive stance to protect from potential harm (Connelly & Shi, 2022, p.1372). The response that is most likely depends on how the individual assesses the threat in terms of certainty, vulnerability and severity (Connelly & Shi, 2022, p.1371). Fugate et al. (2012, p.6) further explains that when the threat perception is combined with low perceived control, passive and avoidant responses tend to dominate, whereas higher perceived control can enable more active coping.

In contrast, when a situation is perceived as an opportunity, behavioural consequences tend to show up differently. Gioia and Thomas (1996, p.390) show that opportunity framing leads to positive outcomes and encourages exploratory responses and investments in further skills. Kong et al (2024, p.585) similarly find that perceiving AI as an opportunity is linked to greater career resilience and a higher tendency to engage in learning, such as independently seeking out new skills.

Together these findings show that perception can drive actions and behaviour. Whether students perceive AI as a threat or opportunity is therefore likely to shape attitude, but also their actions, such as human capital investments.

2.5 Conceptual framework

The conceptual framework presented in Figure 1 compiles the theoretical perspectives discussed in this chapter into an integrated model and will guide the analysis of this study. Rather than treating the theories presented as separate lenses, the framework illustrates how they are connected and guide this study. The framework is structured around four layers that follow the research question to fulfil the purpose of this study.

Top layer
This layer presents the external context where creative destruction (Schumpeter, 2003) and task-based view (Acemoglu & Restrepo, 2019; Autor et al., 2003) together describe the labour market conditions that generate the uncertainty that one must respond to. AI both displaces existing tasks and creates new ones. This layer does not explain perception itself but establishes the conditions that make perception consequential.

Second and third layer
The first part of the research question is addressed in the second layer as it explains how perceptions are formed according to the theories presented. This layer is divided into two connected parts. The left side presents the conditions that shape what students notice and how they interpret it, where sensemaking (Weick, 1995; Maitlis & Christianson, 2014) explains how individuals create plausible interpretations drawing on previous experiences and social environment. Selective attention (Dearborn & Simon, 1958) explains how prior background determines which aspects of AI an individual attends to and which filters out. The right side applies König et al.'s (2021) two-factor appraisal framework, which assesses the perception under a discontinuous change. Gain/loss framing refers to whether the individual interpret the phenomenon as potential harm or benefit, and perceived control refer to whether the individual believes that they can influence the outcomes of the change. As a result of these two parts that together form the perception, the third layer presents the perceptual outcome.

Fourth layer
The final layer addresses the second part of the research question in how the perceptual outcome shape the skills and knowledge development. First, the perception can trigger a

revaluation of the human capital investments by assessing the future returns of competencies they develop, which draws on human capital theory (Becker, 1962; Schultz, 1961), including the KSAO framework (Ployhart & Moliterno, 2011). This revaluation drives an assessment of the perceived employability (Berntson & Marklund, 2007). This thereby creates a behavioural response, which can be active or passive depending on the perception and the previous assessments (Staw et al., 1981; Gioia & Thomas, 1996; Connelly & Shi, 2022; Fugate et al., 2012).

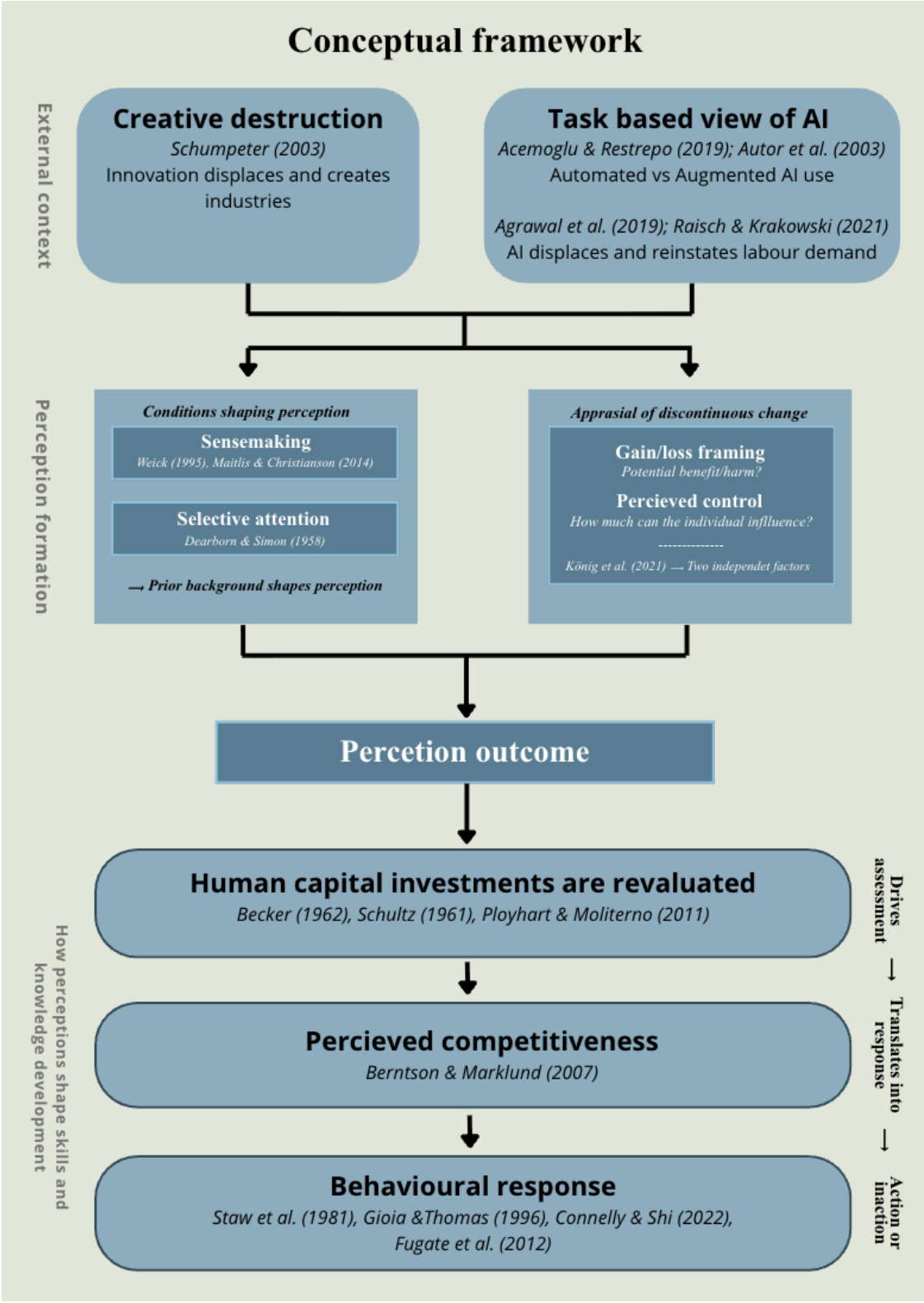


Figure 1. *Sources: The authors.*

3. Methodology

This chapter presents the methodological framework of this thesis and explains how the research was designed and conducted. It outlines the study's philosophical assumptions, research approach, and research design followed by a description of the literature search, data collection, sampling and data analysis. In this chapter, ethical considerations, trustworthiness of the study and methodological limitations will also be addressed.

3.1 Research philosophy

The research philosophy refers to the beliefs and assumptions about the development of knowledge, and shape how the thesis and study is designed and conducted. Saunders et al. (2023, p.131-134) explain that research philosophy reflects the view of the world where our research takes place, which influences the research questions asked, the methods selected and the interpretation of findings. The choice of research paradigm is therefore the foundation for the methodological decisions (Collis & Hussey, 2021, p. 38-45). This study seeks to understand how students perceive artificial intelligence, and such perceptions are shaped by meanings, experiences and interpretations rather than fixed and measurable facts. The study adopts an *interpretivist paradigm* because it seeks to understand these socially constructed meanings and perceptions. Within the interpretivist research paradigm, researchers adopt a philosophy concerned with socially constructed realities, emphasizing narratives, perceptions and interpretations (Saunders et al., 2023, p.146; 150-151). This supports our approach towards students perceptions on AI as an opportunity or threat to be subjective and allows for contextual understanding.

3.1.1 Ontology

Ontology is the assumptions about the nature of reality and shapes the way research objects are seen and studied (Saunders et al., 2023, p.134). Ontology can be defined as the assumptions researchers make about the reality they study and note that these assumptions shape both how the research object is understood and what is considered relevant to investigate (Saunders et al., 2023, p.134-135). The ontological position in this study is interpretivist, as interpretivists believe that the social reality is socially constructed and therefore it is subjective (Collis & Hussey, 2021, p.43). This means that reality is not understood as one single, fixed objective phenomenon, rather that reality is created through perceptions and actions of social actors. Applied to this thesis, this means that perceptions of artificial intelligence are not treated as objectively given. Instead, they are understood as shaped by how individuals interpret AI in relation to their work and studies, environment and prior experiences. The study therefore assumes that AI can be perceived differently by different individuals, and that these differing perceptions represent meaningful realities.

3.1.2 Epistemology

Epistemology concerns the assumptions about knowledge, including what counts as acceptable and legitimate knowledge and how such knowledge can be generated. This is

important since different epistemological assumptions lead researchers to value different forms of evidence and different methods (Saunders et al., 2023, p.134-135; Collis & Hussey, 2021, p.42-43) Interpretivists seek to minimise the distance between the researcher and the researched phenomenon which contrasts the positivists who maintains a detached and fully objective stance (Collis & Hussey, 2021, p.43). The epistemological position of this study is aligned with interpretivism. Rather than seeking objective and generalizable facts about AI perceptions, the study aims to understand how respondents themselves make sense of AI and how they attach meaning to it in different settings. The study places an emphasis on opinions, narratives, perceptions and interpretations and seeks to create a richer understanding by studying human meanings, which is aligned with the interpretivists view on knowledge (Collis & Hussey, 2021, p.42-43; Saunders et al., 2023, p.134-135).

The epistemological position also supports the use of qualitative interviews, since such a method makes it possible to access the respondents' reflections, interpretations and perceptions. The aim is therefore not to determine one objective truth about AI, but to generate a deeper understanding of how perceptions of AI are formed and expressed.

3.1.3 Axiology

Axiology concerns the role of ethics and values in the research process where a key axiological choice is to which extent the researchers' own values and beliefs will impact the study (Saunders et al., 2023, p.135). Unlike positivists who view research as value-free, interpretivists recognise that researchers have values which influence what are regarded as facts and how interpretations are drawn (Collis & Hussey, 2021, p.43). In this study, this means that the authors acknowledge that their own interests, experiences and prior familiarity with artificial intelligence may have influenced the choice of topic and the interpretation of the empirical material. The authors aim to acknowledge this through transparency rather than claiming full neutrality. It therefore becomes important to disclose and clarify those influences. The following section therefore presents the authors' prior understanding of the subject to provide transparency regarding the perspectives brought into this study.

3.2 Prior understanding

The authors are near-graduate business students and have entered this study with a relatively moderate familiarity with artificial intelligence. Although AI has not been a formal part of the university programme, it has been used independently by the authors in their private lives, academic work and other professional contexts. This prior experience has contributed to an existing understanding of AI as a tool and its growing relevance in organizational settings. The familiarity with AI has also influenced the development of the research interest. Through prior discussions and reflections, the authors observed that artificial intelligence can be perceived in different ways, mostly as either a threat or opportunity. These observations have contributed to the formulation of the research scope and the interest in why these perceptions differ.

At the same time, the authors acknowledge that their prior experiences and existing knowledge may have influenced the research process. In line with the interpretivist perspective presented above, the researchers cannot be regarded as entirely detached from the phenomenon that is studied, as their own background and experiences shape the choice of topic and the interpretation of the empirical material. The purpose of clarifying this prior

understanding is therefore to provide transparency to the reader regarding the perspectives brought into this study by the authors. However, to reduce the risk of bias, the authors were aware that their own perspectives could influence the research process and therefore grounded their interpretations in the theoretical framework presented rather than personal assumptions.

3.3 Research approach

This study adopts an abductive research approach. According to Saunders et al. (2023, p.155; p.158-159), the abductive approach involves moving between theory and empirical material and combining elements of deduction and induction. This means that existing theory can be used as a starting point to explore a phenomenon or explain patterns, while the researcher remains open to new themes and interpretations emerging from the data. An abductive approach was considered suitable for thesis study because the topic is not chosen is not entirely unexplored. Previous research has already addressed artificial intelligence, how individuals respond to change and perceptions of change as either a threat or opportunity. However, this study seeks to deepen the understanding of how such perceptions are formed under a specific context. Saunders et al. (2023, p. 160) also explains that abduction is appropriate when there is existing literature, but the theory can be adapted or refined in a specific context.

The study did not follow a strictly deductive approach, as the purpose was not to test a predetermined hypothesis derived from a theoretical structure (Collis & Hussey, 2021, p.7-8). Nor did it follow a strictly inductive approach as the study was informed by prior theory rather than exploring a phenomenon first to generate a theory (Saunders et al., 2023, p. 155). Instead, the theoretical framework guided a formulation of the research problem and the interview guide, while the empirical material was used to explore and refine the theoretical understanding. In this way, the study moved between theory and empirical findings, which is the characteristic of an abductive research approach (Saunders et al., 2023, p.156-160).

3.4 Research design and research strategy

The research design provides the plan for conducting the study to obtain the most valid findings and answer the research question in a coherent and credible way (Collis & Hussey, 2021, p.88). Further Collis & Hussey (2021, p.88) note that a good research design should provide a clear framework for the study, apply appropriate methods and enable the researcher to address the research question and meet the aims of the study. In line with this, the study was designed as a qualitative and exploratory study with a comparative element. A qualitative design was considered appropriate as the purpose of the study is to develop a deeper understanding of how individuals perceive artificial intelligence and why it might be interpreted as either a threat or opportunity. Rather than seeking measurable and generalisable findings, the study aims to explore meanings, experiences and interpretations, which is consistent with the interpretivist research philosophy presented above. A quantitative design was therefore not considered suitable, as the purpose of the study was not to measure the variables statistically, but to gain a deeper understanding of the respondents' subjective perceptions and interpretations.

The study could also be regarded as exploratory, as it seeks to deepen the understanding of a phenomenon by examining how it is perceived and interpreted in a particular context. This is in line with Collis & Hussey's (2021, p.4-5) explanation of an exploratory study, which

they mean is distinguished by its purpose of investigating a phenomenon to gain insight and understanding. Although Collis and Hussey (2021, p.5) note that exploratory research is especially relevant when there are few or no previous studies, they also point out that such research aims to look for patterns and develop ideas rather than testing propositions. For this study, previous research does exist, but the study is still exploratory in the sense that is seeks to generate a deeper understanding on how perceptions of AI are formed in the specific context rather than test predetermined propositions. The comparative element comes from our two respondent groups, Business and economics students and Civil engineering students. The aim is to explore how they differentiate and agree on their perception of AI as a threat or opportunity. A comparative design is according to Bell et al. when using the same methods for two or more cases to be able to understand social phenomenon's more clearly (Bell et al., 2019, p.68). This element allows the thesis to explore interpretations and perceptions of the students coming from different educational backgrounds and exposures to AI, that can change how they feel towards the tool.

The research was based on semi-structured qualitative interviews. Saunders et al., (2023, p.450) presents interviews among the main strategies for collecting primary qualitative data and highlights the potential of semi-structured interviews when the aim is to explore respondents' views, meanings and experiences. Similarly, Collis & Hussey (2021, p.121) explain that under the interpretivist paradigm, unstructured or semi-structured interviews are typically used, whereas structured interviews are more associated with the positivist paradigm. This strategy was considered suitable for this study as it allowed for the researchers to access the respondent's own reflection and perception of artificial intelligence, while still having enough structure in the interviews to address the relevant themes for the research purpose. The qualitative exploratory design presented provided the overall structure of the study, while the interviews provided the practical strategy in which the empirical material was collected.

3.5 Literature search and source evaluation

3.5.1 Literature search

The literature search was conducted systematically thoughtfully to have a good theoretical base for the study and identify a relevant research gap. Collis & Hussey (2021, p.67-69) points out that a literature search is a systematic process aimed at identifying the existing body of knowledge on a particular topic and argue that the search should be described transparently in the methodology chapter. They further note that the search process should include identifying relevant sources, defining the scope of search, determining keywords and to keep records of the search process. In this section, the authors will present how the literature search was organized and conducted.

The primary search platform used in this study was Umeå University Library's online databases. Google Scholar was also used to a limited extent as a complementary search engine to identify relevant academic publications. This is consistent with Collis & Hussey's (2021, p.69) recommendation that researchers use the resources that the institution provides, but also other academic search engines, where Google Scholar is specifically mentioned. The search process was focused primarily on literature related to artificial intelligence, technological discontinuity, creative destruction, human capital theory and perception. In accordance with Collis & Hussey (2021, p.69-73), keywords were developed and used to capture the core elements of the themes. The keywords gradually developed and changes as the literature review progressed, since additional relevant concepts, synonyms and related

terms and topics were identified. Table 3 gives a summary of keywords used to find literature connected to the different themes.

Table 3: Summary of keywords. *Source: the authors*

Subject area	**Keywords**
Artificial intelligence and AI use	Artificial intelligence, AI, generative AI, AI use, AI usage, AI adoption, AI implementation, AI usage, AI automation, augmentation, augmented AI, AI substitution, AI in practice, human-AI augmentation
Technological discontinuity and creative destruction	Technological discontinuity, discontinuous change, disruptive technology, technological change, disruption, creative destruction, Schumpeter, Schumpeterian theory, inertia
Human capital theory	Human capital, human capital theory, human capital investment, labour demand, skills, knowledge, competence, education, training, productivity, tacit knowledge, explicit knowledge, intellectual capital
Perception	Perception, appraisal, threat appraisal, threat perception, opportunity perception, gain/loss framing, perceived control, reactions to change, response to change, perception of AI

The search focused primarily on peer-reviewed journals published in English. Since the thesis is written within the department of business administration, the main emphasis was placed on literature from journals within business administration, management and organization. Sources outside these fields were only included to a limited extent when they provided important support for specific arguments or points. This aligns with Collis & Hussey (2021, p.67-69) who argues that only literature relevant to the topic, theory and methodology should be collected. In some cases, non-peer-reviewed sources were also included. These were used restrictively and only when they were considered highly relevant and credible. This was particularly the case for recent institutional reports, such as OECD publications, which were used to illustrate the current developments in artificial intelligence and the change in demand for skills in the labour market. Given the rapid pace of change and development in the AI field, these sources were considered valuable for providing up-to-date information and context to support where peer-reviewed research may not yet have fully captured the most recent developments.

In areas related to artificial intelligence specifically, priority was given to more recent publications because of the fast-changing nature of the field. However, older sources have been included in the thesis as they, for the most part, represent foundational theories or theoretical contributions that still is relevant for understanding recent phenomena, such as artificial intelligence. One example of this is that discontinuous change in technology has been applied to artificial intelligence. Time delimitation is, according to Collis & Hussey, especially appropriate for literature searches within newer technologies, while also allowing flexibility where older literature remains important to the study.

The selection process when searching for literature followed several stages. First, titles were screened in relation to the purpose of the study. Secondly, abstracts were reviewed to assess relevance. Third, full texts were read for the sources that were considered the most relevant. In addition to this, the reference list of the most relevant sources was examined to identify further relevant literature within the field.

3.5.2 Source evaluation

The sources used in this thesis were evaluated based on credibility, relevance, recency and academic quality. Collis & Hussey (2021, p.84) explain that reviewing the literature involves not only locating previous studies, but also to read and evaluate them, it requires a critical analysis rather than a simple description of what has been found (Collis & Hussey, 2021, p.83). First, peer-reviewed journal articles were prioritized, since these are generally subject to a greater scrutiny, and therefore provide a stronger basis for the theoretical framework. Emphasis was placed on articles published in established journals within business administration. To support this, journal rankings provided by the university and was used in assessing journal quality, and sources with a higher number of citations were generally regarded as more influential in the field. However, the number of citations was not treated as a criterion on its own, rather as one indicator among others.

Relevance was treated as a decisive evaluation criterion. Sources were selected primarily when they were clearly connected to the thesis topic and the theoretical framework. Since the thesis is written within business administration, the ambition was to ground the theoretical discussion mainly in business administration research. Sources from outside this field were therefore used only to a limited extent and primarily when they contributed support for a specific concept or argument that was directly relevant to the study. In such cases, extra consideration was taken into account when choosing these, such as that they were published by institutions.

Recency was especially important in areas related to artificial intelligence, because AI is a rapidly developing field. For this reason, more recent studies were prioritised when discussing AI use, labour market implications and changing competence demands. At the same time, older sources were retained when they represented foundational theories or other contributions that is relevant for understanding more recent phenomenon.

Non-peer-reviewed sources were used restrictively. In practice, these consisted of mainly institutional reports such as publications from OECD and the Stanford AI Index. These were considered acceptable because they were produced by recognised and credible institutions, were highly relevant to the topic and provided recent contextual material on the developments of artificial intelligence, investments and skills demand. These non-peer reviewed sources were not used as the main theoretical contribution to the thesis but rather seen as a complementary source to provide information on recent development that peer-reviewed literature may not yet have fully captured.

At the same time, the authors were aware of some limitations when evaluating sources. Institutional reports may have provided relevant insights with more recent information but may also reflect some bias of the organizations that use them. On the other hand, older foundational sources remain theoretically important but may not reflect the current development if artificial intelligence. In addition to this, sources outside business administration may offer support, but heavy reliance on them could weaken the consistency

of the thesis. For this reason, such sources were used with caution and had a supportive role. Overall, the source evaluation aimed to ensure that the theoretical framework was based on sources that were academically credible, relevant to the topic of out thesis, and appropriate in relation to the evolving technology in the AI field.

3.6 Data collection

3.6.1 Semi-structured interviews

After a qualitative approach as the research design had been decided for the study, different data collection methods were evaluated. Semi-structured interviews were chosen as the primary data collection method in this study. According to Bell et al. (2019, p.211) a semi-structured interview is where interviews involve some prepared questions related to the main topics. The questions tend to be more of a general nature, and even though there is typically an interview schedule the interviewer is able to vary the sequence of questions and is also able to ask further questions in response to the interviewee. This flexibility makes semi-structured interviews appropriate for this study since the aim is to gain a deeper understanding of students' perceptions. It allows the researcher to understand the student's world regarding their opinions and perceptions.

When different types of interview strategies were evaluated, structured interviews were one of the options. Structured interviews are characterized by questions that are planned in advance and asked in the same order to each respondent (Collis & Hussey, 2021, p.191-193). The aim of using this type of interview style is that the answers for each question can be aggregated and compared since each respondent is given the exact same context of questioning (Brylan et al., 2019, p.209). This format was deemed less suitable for the present study, as the purpose is not to obtain standardised responses, and rather to explore how participants interpret a complex phenomenon.

Comparing semi-structured interviews with unstructured interviews, the latter are considered time-consuming and potentially problematic since it might be hard to control the topics in the interview. Prior to the interview, no questions are prepared and instead the questions develop and evolve during the interview (Bell et al., 2019, p.211; Collis & Hussey, 2021, p. 121). Semi-structured interviews was considered more appropriate as they allow for the researcher to maintain a direction relating to the purpose of the study, while still having some flexibility to allow respondents to elaborate on experience and perceptions.

3.6.2 Interview guide

When conducting semi-structured interviews, the aim is to create a natural and conversational dynamic between the respondent and the interviewer. To ensure that all essential topics were covered and addressed, the questions in the interview guide were mostly structured to promote reflective and elaborative answers rather than yes or no responses. This guide is grounded in the theoretical framework and is structured into four different phases serving a specific purpose, the interview guide can be found in Appendix 1. The cursive questions are to be considered as follow-up questions, intended to be used if needed. The interviews were conducted in the respondent's native language, which in this case was Swedish. This allowed for the respondents to feel comfortable speaking and using expressions that reflected their perceptions and attitudes.

The interview begins with an introductory phase. Here the purpose of the study is presented, and the consent for the study and to record the interview is obtained. The consent for the study is grounded in the consent form which have been sent out to all respondents in advance so that the respondent can make an informed decision on whether they consent to be a part of the study. This is then followed by a background phase consisting of questions about their current programme, semester and a general familiarity with AI tools. These questions serve two purposes; first we need to ensure that they fit our selected population (near graduate student and is studying the business programme or civil engineering programme), and secondly it is easing the respondent into conversations before more complex and reflective questions are introduced. Collis & Hussey (2021, p.123) recommend placing such background questions in the beginning so that respondents gain confidence before progressing to more challenging topics.

The remaining phases are organized around the core of the study and derived from the theoretical framework. The first thematic phase, *AI usage*, is to understand the respondents' habits and patterns of AI engagement without introducing the theoretical terminology such as augmentation or automation. The questions in this phase explore which tasks respondents delegate to AI, how their usage has evolved over tile and to what extent they rely on AI-generated outputs. This gives an understanding of how AI is practically integrated into the respondents' academic and daily lives. The second phase, *Human Capital Implications*, focuses on how respondents' reason about their own skills and knowledge development related to AI. Here, the questions explores whether respondents perceive certain competencies as becoming less valuable, whether their educational environment prepares them for an AI-influenced labour market, and how AI has affected the way they prioritize their time and efforts in their studies. The third phase, *Perceptions of AI*, explores the respondent's perceptions and attitudes toward AI in toward AI in relation to their future working lives. Questions in this phase are designed to explore mechanisms underlying these perceptions, including perceived control, sensemaking and selective attention. Respondents are reflecting on both the opportunities and threats they associate with AI, including how respondents perceive their own ability to influence and adapt to the role that AI will play in their future work. Finally, the fourth phase, *Sensemaking*, examines how the respondents' social environments, such as education, peers, family and media, shape their understanding and attitudes toward AI. This phase seeks to understand how external factors in the respondents' surroundings have shaped the views and attitudes that were explored in the previous phases.

The language of the interview guide was kept accessible and free of academic or technical terminology to ensure that questions were easy to understand and did not steer respondents toward particular framings and reflections. Collis & Hussey (2021, p.123) recommends using words that are non-threatening and familiar to the respondent. No interview questions were shared with the respondent in advance, in order to ensure that the answers were spontaneous and authentic.

3.7 Data sampling

The following section explains our sampling strategy, how respondents were selected and recruited, and the limitations of the sampling.

3.7.1 Sampling strategy

This study applied a combination of purposive sampling and networking as its sampling strategy. Under an interpretivist paradigm, Collis and Hussey (2021, p.118) explain that researchers do not need to select a random sample, since the purpose is not a statistical generalization from the sample to the population. Instead, they describe several methods for selecting a non-random sample, which includes networking, snowball sampling, purposive sampling, convenience sampling and piggyback sampling. Networking is defined as making use of existing contacts who fit the sampling criteria, which purposive sampling involves selecting participants based on their experience or relevance in relation to the phenomenon under study. The combination of networking and purposive sampling was considered appropriate because the study required respondents who fulfilled specific criteria, but since the authors are near-graduate students existing networks and contacts are in place. The purposive element of the sampling strategy are predefined selection criterias which will be presented in the next subsection.

Other non-random sampling methods were considered less suitable. Snowball sampling was not the main strategy, since the purpose was not to expand the sample through referrals but rather to identify respondents who fit the predetermined criteria. The referral was also not necessary to find respondents due to the authors existing network. Convenience sampling was also not considered appropriate, since it risks producing a sample based on availability and not relevance. In contrast, the combination of purposive sampling and networking made it possible to align the sample with the purpose of the study while also providing access to relevant participants.

3.7.2 Respondents and selection criteria

The sample consisted of 15 respondents in total, seven civil engineering students and eight business students. The business students in this sample were enrolled in one of the four-year long business programmes at Umeå University: the general business programme, the specialisation in Retail and Supply Chain Management or the specialisation in Service Management. Across these programmes, students study a broad base of business administration, economics, statistics and with the possibility to specialise in areas such as accounting, finance, marketing and management (Umeå University, n.d.-a – n.d.-c). The civil engineering students in this sample were enrolled in one of the seven different engineering specialisations at Umeå University, which covers areas such as energy technology, industrial economics, interaction and design, computer science, technical physics, biotechnology and chemical engineering (Umeå University, n.d.-e, n.d.-f, n.d.-g, n.d.-h, n.d.-I, n.d.-j). Even though they have different orientation, the programmes share a common foundation in technical and scientific problem-solving and prepare students for roles where technical expertise is central. The two student groups were chosen by the researchers to add educational diversity. Since the two student groups have different starting points regarding the technical knowledge in general but also within the field of AI, the researchers expected different analytical perspectives to answer the research question. The researchers considered having other student groups as well. However, the two groups that were chosen where deemed to be the most relevant due to their distinction and in the level of AI integration in their education. The authors have acknowledged that there are other fields of education that would be interesting to study, this is discussed in the future research section in chapter 6.

The respondents that were selected were at near-graduate stage of their education, since this was considered relevant in relation to the study's focus on artificial intelligence, future working life and competence development. We have defined near-graduate students as

students who does not have more than three semesters until they graduate. For students in the Business and Economics program (in Swedish; "Civilekonomprogrammet"), this means respondents in semester 6-8 out of 8 semesters. For students in civil engineering programmes, this meant respondents in semester 8-10 out of 10 semesters. Students at earlier stages of their education, semester 1-5 in business and economics and 1-7 in civil engineering, were not included as they were considered less likely to have begun reflecting on their future professional role, future work related competencies, and what impact AI might have on their future work to the same extent as near-graduates.

The two student groups, business students and civil engineering students, were selected because they were expected to have a difference in formal education and exposure to artificial intelligence within their programme. This assumption is partially based on the program information on Umeå University's website. For all business programmes AI was not mentioned in any way on their information page and not explicitly included in the syllabus. On the other hand, when looking at the engineering programmes, all of the different orientations explicitly mentions AI as a part of the programme or mentioned in the syllabus (Umeå University, n.d.-a, n.d.-b, n.d.-c, n.d.-d, n.d.-e, n.d.-f, n.d.-g, n.d.-h, n.d.-I, n.d.-j). Since the business programmes has not integrated artificial intelligence formally, the authors have taken part of the program evaluation for the programs. The summarized evaluation states a concern about the shortage and lack of AI in the curriculum. The concern refers to the students underperforming in AI literacy when entering their future work. The evaluation expects AI to be integrated in many occupations, highlighting the importance of basic knowledge. According to the evaluation, there is a common positive attitude towards retaining AI knowledge in the education (Umeå University, 2025, p.n/a) The comparison between these two groups was therefore considered useful for exploring whether differences in educational exposure to artificial intelligence may influence whether it is perceived as a threat or opportunity.

Since the respondents participated anonymously, the names are changed and pseudonyms have been used to keep them anonymous and is aligned with 3.9 Research ethics.

Table 4. Respondents and information.

Respondent	**Programme**	**Semester**	**Interview length**
Birch	Business student	8 out of 8	44m 57s
Linden	Business student	8 out of 8	38m 12s
Oak	Business student	8 out of 8	37m 18s
Aspen	Business student	8 out of 8	32m 44s
Maple	Business student	8 out of 8	47m 49s
Pine	Business student	8 out of 8	52m 21s
Alder	Business student	8 out of 8	50m 20s
Rowan	Business student	8 out of 8	50m 9s
Rose	Civil engineering student	10 out of 10	43m 7s

Iris	Civil engineering student	10 out of 10	28m 10s
Sage	Civil engineering student	8 out of 10	37m 27s
Bamboo	Civil engineering student	10 out of 10	33m 26s
Poppy	Civil engineering student	8 out of 10	1h 4m 11s
Moss	Civil engineering student	10 out of 10	31m 8s
Thyme	Civil engineering student	10 out of 10	1h 14m

3.7.3 Recruitment and conduction of interviews

The recruitment of respondents was mainly through personal contact via LinkedIn and personal messaging, which was drawing on the existing network of the researchers, in line with the networking sampling strategy described in section 3.7.1. To mitigate the risk of bias, the researchers deliberately avoided recruiting close friends and instead contacted acquaintances and more distant connections who fulfilled the selection criteria. Collis & Hussey (2021, p.126-127) note that interviewees may give answers they consider "correct" or "acceptable" rather than their genuine views, and that the risk can be heightened when personal relationships exist between the interviewer and the interviewee. Therefore, the intention when recruiting respondents was to maintain a degree of social distance between the researchers and the responders with the aim of reducing this tendency and to encourage more candid responses.

When potential respondents were contacted, they received information about the purpose of the study, what the participation involved and stating the two student groups that where being recruited. We also informed about approximate interview duration, recording and transcription in accordance with their consent and their anonymity throughout the process. A consent form and informational letter produced by the Business Administration department at Umeå School of Business, Economics and Statistics was thereby shared with the respondents that agreed to participate. This ensured an informed consent before the interview took place.

3.8 Data analysis

This section will explain the step-by-step process of our analysis of the collected data, following a data structure that is based on our thematic analysis made on our interviews. The reader will thereby gain relevant information of the thesis handle the collected data material and consequently understand the main findings derived from the conducted semi-structured interviews.

3.8.1 Thematic analysis

To be able to interpret and understand our respondents' viewpoints, we decided to transcript and record our interviews. Consent regarding recording and transcription was established at first contact with the potential respondent and thereby confirmed at the interview. This made our presence at the interview better that helped with asking relevant follow-up questions

and have a conversational interview rather than using fixed interview questions, which is in line with a semi-structured interview style. According to Collis and Hussey, using a semi-structured interview encourages the respondents to discuss the main topics of the interview while keeping an openness to a development to new insights (Collis & Hussey, 2021, p.121). Since we are interested in the respondent's perception and viewpoints, having a semi-structured interview help revealing opinions and valuable insights. To support this further, we the authors took turns holding the interviews, maintaining a one-to-one structure. The benefit of having two interviewers is that it can ensure that the discussion and data fully explored by the respondents, having an extra perspective (Collis & Hussey, 2021, p.194). On the other hand, having more than one interviewee changes the dynamics of the interview which could disturb the angle of opinions (Collis & Hussey, 2021, p.194). This is something we wanted to avoid. In our case, we were conducting 15 interviews which made the decision of doing one-on-one interview the best choice from a time efficiency matter. We felt confidence in this with the support of digital transcription and recording.

The interviews were conducted as face-to-face and online interviews. The benefit of using face-to-face interview is according to Collis and Hussey the value of being able to ask more complex questions that need the complete attention of the respondent (2021, p.120). It is also a question of convenience, as the respondent can choose the most suitable setting for them to be comfortable, potentially resulting in a more honest interview (Collis & Hussey, 2021, p.120). For a further convenience, we offered online interviews on Microsoft teams. The practicality of online interviews allowed the respondents to be able to decide freely where and when the interview shall be conducted. This also contributed to the respondents being more likely to be available, helping is us completing a large number of interviews in a short amount of time. In both cases, face-to-face and online interviews, all interviews were transcribed through Microsoft Teams built-in transcription tool.

Our thematic analysis takes inspiration from Miles and Huberman's general analytical procedure (1994, cited in Collis & Hussey, 2021, p.147), which helped us organize and understand the differences and similarities among our respondents. Due to the high volume of codes, we created the extra step of coding within the studying field as well as for all the students. This was conducted in the following matter:

1. Initial insights and transcription.
2. Coding the interview and compiling with the other interviews in the same studying area.
3. Grouping codes into themes and sub themes.
4. Creating themes for perceptions and experiences.
5. Conclusions and comparisons among the themes.

The transcription of each interview was quickly processed and analysed initially to confirm insights from the interview. We made personal notes and highlighted the transcription record to not lose the relevant and important insights. This also helped finding repetitions and patters, which eases the occurrence of a theme (Bell et al., 2019, p.519). This was particularly important because of the large volume of data we were collecting. Thereby we identified codes which resulted in 179 codes that we thereby categorized. The categories allowed for an overview of patterns, perceptions and other valuable insights to take form. Drawing from the field specific categories, we summarized codes from both groups to compile four themes to further the understanding of the students' perceptions and experiences and how those perceptions affect their knowledge development. The themes and codes were thereby set in a table for a structured overview.

Furthermore, the empirical analysis is derived from our created table were the different themes and insights gave an organised summary of our respondent's perceptions. We compared the two studying fields in their experiences and driving forces and connected it to existing theory, resulting in differences and similarities becoming apparent. The analysis followed an abductive logic, where our empirical material was processed openly, to be able to interpret the codes together with our theoretical framework. This also explains our themes, as they are directly related to our theoretical framework. We discuss the themes and identify the implications of our findings, to which contributes to the existing researched areas. Due to the explorative nature of our study, we decided to create new theme headings that are derived from our interview guide heading. This allowed for the collected data to be fully integrated into the theoretical framework. The analysis of our empirical findings lays the foreground to our discussion where we merge existing theory with our results. Thereby, a cohesive exploration was conducted, resulting in formulation new insights that contributes to research. We analysed the similarities and patterns among the studying fields as well as internally in the groups. We could identify perceptions and the driving forces of AI usage that subsequently explained their behaviours towards the technological change. This led to insights of how technological disruptions affects the perception of the value skills and knowledge for future labour market entrants.

The thematic analysis was conducted in Swedish due to the interviews and transcript records being in Swedish. After the thematic analysis was completed, the citations were translated and added into the empirical finding. The translation was made carefully to match the exact translation in English. This was ensured by both authors translating the quotes separately and thereby comparing the results. This was then followed by a discussion to confirm and assess the translation to be accurate and factual to the respondent's answer. All original citations in Swedish can be found in Appendix 3. This choice was made by the authors to extend the transparency to the reader further, allowing the translations to be reviewed.

3.8.2 Data structure

Figure 2 organises the codes derived from the interviews into subthemes and themes, and provides a visual overview of how the empirical material was interpreted.

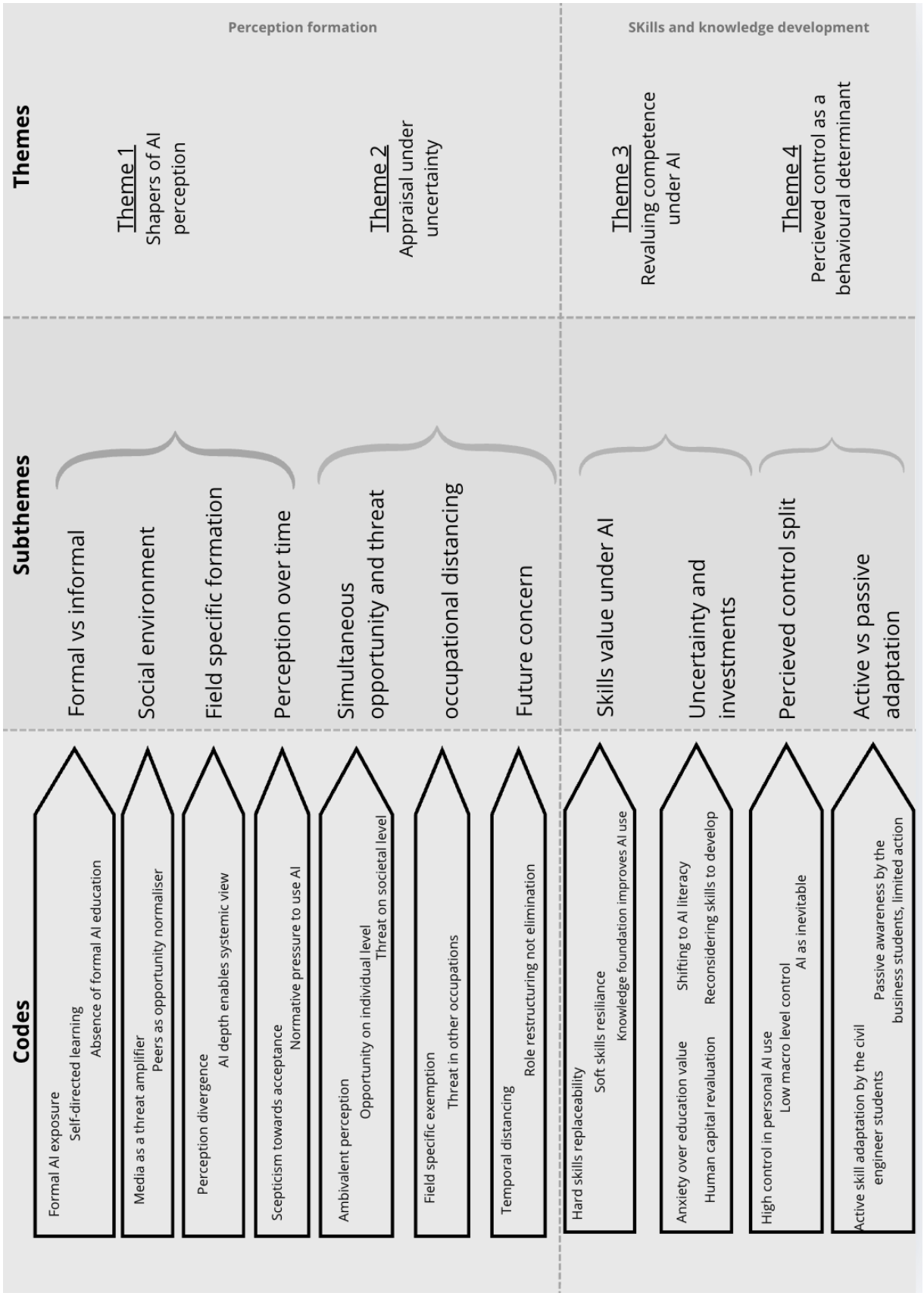


Figure 2. *Data structure.* Source: The authors

3.9 Research ethics

Bell et al (2019, p.110) describes that ethical considerations are the most relevant considerations in a student research project when it involves both researchers and research participants. Therefore, in this section, the stance toward the four ethical considerations till be presented. These ethical principles help the researchers ensure that the ethical risks are minimized (Bell et al., 2019, p.112). The four ethical principles, according to Bell et al., (2019, p.114-124) are avoidance of harm, informed consent, privacy and preventing deception.

The first principle aims to ensure that harm to participants of the study should be avoided. Harm towards participants is not accepted within research, and where issues such as confidentiality and anonymity is considered particularly difficult for qualitative research (Bell et al, 2019, s.116). In this study, this principle is important since the topic of AI use may be perceived as sensitive to some participants, since AI use in academic work can be associated with disciplinary consequences. To reduce the risk of discomfort for the participants, they are informed that their participation is anonymous, participation is voluntary and that they have the right to refrain from answering specific questions.

Secondly, the principle of informed consent is when the potential research participants get as much information as possible in order to make an informed decision if they want to participate in the study or not. This also means that respondents should be informed if recording equipment is being used (Bell et al., 2019, p.118.119). in this study, the researchers have therefore used a consent form and letter of information that is provided by the business administration department at Umeå School of Business, Economics and Statistics, Umeå University, see Appendix 1. This has been done to inform the participants of the thesis, purpose, how data is collected and how the data will be processed and finally where the thesis will be published. The participants are also informed that the interview is being recorded only with their consent.

To protect the respondent's integrity, the third principle processes the protection of privacy (Bell et al., 2019, p.123). This principle is highly linked to the second principle of informed consent since that principle is given based on understanding of what the respondents involvement means. This means that the purpose of the study will be shared with the respondent before they consent to participate. It is also important to know that, even though information has been shared beforehand, respondents may interpret questions as too personal or sensitive which may be impossible for the researcher to know beforehand (Bell et al., 2019, p.123). With this in mind, the questions asked in the interviewed are carefully curated and are formulated appropriately to minimize the risk of a respondent feeling like their integrity or privacy is being violated. The respondents are also informed that they can withdraw at any time without giving any reason.

Finally, the fourth ethical principle is about preventing deception. The principle emphasizes that the research and its findings are presented in a transparent and honest way without deceiving the recipient. It is also important that the respondents are not misled and that the researcher sticks to the original purpose of the study until the end of the research process (Bell et al., 2019, p. 123-124). The researchers have presented the purpose and background of the study in the information letter provided to the respondents, to give as much information as possible so that the respondent has a clear picture of what the aim of the

study is. The researcher has also had the principle in mind when presenting the empirical findings in this thesis, in order to give the most transparent view of the data. No information has been withheld from participants that could influence their decision to participate.

3.9.1 Use of AI in the research process

During the research process of this thesis, generative AI was used in a limited and supportive manner to improve readability and language in parts of the text, as well as giving feedback on the authors' ideas and text written in this thesis. The prompts concerned feedback as well as constructive commentary on the language and structure of the introduction as well as the theoretical background. The response was then carefully reviewed by the authors and compared with the literature if available. The AI tools used were Chat GPT and Google Gemini. AI was not used to generate text or content, analyse empirical material or develop interpretations.

Our usage of AI was indented to improve readability and clarity to our reasonings, improving the overall academic quality of the thesis. AI was used to give feedback on language and chronology of our arguments, rather than formulating arguments and analyses. AI was not used to generate text, content, coding, or for conclusions. After using the AI tools as presented above, the authors edited and reviewed the content of this thesis carefully and as needed.

The usage of AI did therefore not replace the independency by us the authors, instead AI provided limited support to our written material. The reviews given by AI was critically reviewed before making any edits, as a measure to maintain academic integrity. The authors take full responsibility for the content in this thesis.

3.9.2 Truth criteria

The quality of qualitative research is not evaluating trustworthiness using the same criteria as quantitative studies. We will address the trustworthiness of this study through the four criteria's adapted to an interpretivist paradigm (Lincoln & Guba, 1985, cited in Collis & Hussey, 2021, p.161): credibility, transferability, dependability and confirmability.

Credibility addresses whether the subject that is studies was correctly identified and described (Collis & Hussey, 2021, p.161). This was supported by conducting interviews in the respondents' native language (Swedish), using follow-up questions to deepen the understanding, and where the thematic analysis was grounded in a systematic coding process that was reviewed by both authors. *Transferability* refers to whether the findings can be applied to a sufficiently similar situation (Collis & Hussey, 2021, p.161). Statistical generalisability is not claimed as previously mentioned in this chapter, which is consistent with the interpretivist paradigm (Collis & Hussey, 2021, p.118). However, with a detailed description of the sample, setting and analytical process allows the reader of this study to assess the relevance of the findings to their own contexts. *Dependability* refers to the focus of ensuring that the research process is conducted systematically, rigorously and thoroughly documented (Collis &Hussey, 2021, p.161). This criterion was addressed through our transparency of methodological choices which includes sampling strategy, interview design and thematic analysis. *Confirmability* concerns are the findings can be assessed as being derived from data, rather than from the researchers' predispositions (Collis &Hussey, 2021, p.161). How we addressed this criterion is disclosed in section 3.2, the quotes in its original

language Swedish can be found in Appendix 3, this allows for the reader to firmly assess the translations and interpretations.

3.10 Methodological limitations

This section will treat any methodological limitations, which refers to the weaknesses and practical constrains on the research design and therefore may affect how the findings should be interpreted (Collis & Hussey, 2021, p. 100)

A limitation of the sampling is that the study is based on a small, non-random sample. This means that the findings cannot be statistically generalised, however this is consistent with the interpretivist paradigm of this study (Collis & Hussey, 2021, p.118). Under the interpretivist paradigm, research data are not analysed statistically with a view to generalising from the sample to the population, and therefore a random sample is not required (Collis & Hussey, 2021, p.118). Saunders et al. (2023, p.150-151) similarly points out that interpretivism is subjectivist, and focuses on complexity richness, multiple interpretations and meaning-making rather than a universal generalisation.

Another limitation is that the use of networking may have influenced which respondents became accessible to the study. Collis and Hussey (2021, p.118-119) notes that networking makes use of contacts already available to the researcher, which means that the sample may partially reflect who was practically available to the researcher rather them all possible respondents who met the selection criteria. However, finding participants who fit the selection criteria is often difficult for students working under a tight timeline, which further supports why networking became a practical part of the sampling process in this study (Collis & Hussey, 2021, p.118).

The empirical material is only based on two student groups which is considered a delimitation but also a limitation. This made the study manageable within the available timeframe and supported the collection of rich qualitative data, is also means that perspectives from other educational programmes are not represented in the empirical material. This should be considered when interpreting the breadth of the findings.

A further limitation is the study is also the sensitivity of questions related to the use of AI in academic work. Since some forms of AI use in educational settings may be perceived as inappropriate or not allowed, some respondents may have been reluctant to answer such questions honestly with concern that it might reflect negatively on them. This means that a degree of response bias cannot be excluded. Although anonymity was ensured to reduce this risk, it is still possible that respondents adapted their answers due to psychological discomfort or perceived consequences.

Overall, these limitations do not undermine the relevance of the study, but they should be considered when interpreting the findings.

4. Empirical findings

This chapter presents the empirical findings based on 15 semi-structured interviews with business students and civil engineering students. The findings are structured around four themes and their subthemes. The first two themes answer the first part of the research question, how the perception of AI in relation to the future working life is formed. The last two themes aim to examine the second part of the research question, how that perception shape the skills and knowledge development. Table 5 provides an explanation on each theme, while the description under the table provides insight into how the themes connects and builds on each other. The findings follow the data structure, figure 2, explicitly addressing each subtheme and theme.

Table 5: Overview of empirical themes Source: The authors.

Theme	**Summary**
Perception formation	
Shapers of perception	This theme looks at how students' perceptions in relation to their future working life are formed. It presents the difference in formal AI education between the civil engineering students and business students, and the role of self-directed learning as a complement. It further covers how media and peers function as opposing influences on perception, how educational background affect the perception between the groups and how familiarity and norms shift perception over time.
Appraisal under uncertainty	This theme presents the outcomes of the perception formation process. It covers how most of the students hold opportunity and threat perceptions simultaneously, which is driven by uncertainty about how organizations will use AI. It looks at how students distance themselves from the disruption of AI, by either locating it in another field or by placing the threat in the future.
How perception shape skills and knowledge development	
Revaluing competence under AI	The theme addresses how students' perceptions of AI shape their skills and knowledge development. It presents how the groups distinguish between hard skills, which is perceived as vulnerable to AI, and soft skills which is perceived as more resilient. It covers how civil engineering students additionally frame subject expertise as a foundation for using AI. Finally, it examines how anxiety over the value of the education drives revaluation of the human capital investments, such as reconsidering which skills are worth developing and shifting toward AI literacy.

Perceived control as a behavioural determinant	This theme continues to address how students' perception of AI shape their skills and knowledge development. It presents the split between high perceived control over personal use and the low perceived control over the development of AI at the macro-level. It further examines how civil engineering students respond to low macro-level control by skill adaptation and educational investments, while business students more commonly display awareness of the need to adapt without necessarily translating it into action.

Table 5 shows the four different themes and how they are organised into two parts that together aim to answer the research question. The four different themes build on each other and are structured so that each adds a layer to the overall picture. The first two themes address how students' perceptions of AI in relation to their future working life are formed, which is the first part of the research question. Theme 1 is the starting point and identifies the conditions that shape how students form their perceptions of AI in relation to their future working life, where the shaping conditions are educational background, social environment and prior experience. Theme 2 moves from these conditions into the actual perceptual outcomes, showing what kind of perceptions they produce and how students position themselves in relation to the disruption of AI.

The third and fourth theme shift the focus from how perceptions are formed and their outcomes, to what those perceptions lead to. Theme 3 focuses on how the perception, that is identified in theme 2, translates into a revaluation of competencies. Here, students assess which of their skills retain value, and how they adjust their investment priorities according to that. Theme 4 adds a further dimension to this and shows that the revaluation does not give the same behavioural response across all students. The perceived level of control, which differs between the groups but also differs between students in the same group, determines if the revaluation results in active adaptation or passive awareness. Together, these four themes can be seen as building blocks on each other, presenting a progression from the conditions that shape the perception, to the actual perceptions and finally their consequences for adapting to new skills.

4.1 Theme 1 – Shapers of perception

The following theme addresses the first part of the research question, looking how students' perceptions of AI in relation to their future working life are formed. The theme is structured around four subthemes: formal versus informal knowledge foundation, social environment, field-specific formation, and perception over time. It covers the codes formal AI exposure, self-directed AI learning, absence of formal AI education, media as threat-amplifier, peers as opportunity-normaliser, perception divergence between business and civil engineering students, AI depth enables systemic view, scepticism toward acceptance, and normative pressure to use AI.

4.1.1 Formal versus informal

A difference between the two groups is their source of their AI knowledge. Most civil engineering students have received formal AI education through their programme through

specific AI-courses and where AI has been integrated into courses such as programming. Rose describes that in the programming courses the student have been encouraged to learn programming with AI. Iris has taken a course in AI, but also describes that it was not enough be to prepared for the future working life, and that they have not learnt how to think critically or exactly how it works.

> *"Yes, we have. We've taken an introductory course on AI. And we've also been encouraged to use AI in different courses. So even if it might not be an AI course specifically, we've taken, for example, programming. And there, we tried programming with AI. I got help from the teacher with that. It went really well, so yes, I would say so."* – Rose, Civil Engineering

> *"I've taken some course in AI, more about how AI is built and things like that. Basic courses, basically. But otherwise, I haven't learned a lot about exactly how it works. How should you think critically when using AI? We haven't come across anything like that."* – Iris, Civil Engineering

While civil engineering students describe that they have received formal AI exposure, most of the business students expressed the absence of any formal AI education in their programme. When asked whether the university has prepared them for an AI-influenced labour market, all business students describe that they are not prepared and that there is a gap in the education. Linden further describes this by resonating that everyone will use AI in the future and that therefore AI needs to be taught to the students to be prepared for the labour market.

> *"First and foremost, some kind of education that goes through what AI is and how you can best use it to your advantage. Because no matter how you look at it, you're going to need to use AI in the future to get some kind of edge over everyone else, because everyone will be using it. If you don't use AI, I'm sorry, but you're behind the curve."* – Linden, Business

The lack of a formal AI education, the business students strive to gain knowledge in AI through their own self-directed learning. Most business student describe that their learning of AI comes from experimentation and trying to self-learn in different ways. Birch describes this by learning from media and friends, and that the knowledge has gradually come forward over the years since the student first started using it. Linden expresses this in a similar way, that the knowledge about AI comes from self-learning and through YouTube.

> *"A lot of it comes from watching YouTube videos and being on social media, and then hearing from friends: 'Just try this.' 'Yeah, I can try that.' So it's very much learning by doing. Yes. As I touched on a bit earlier as well, when you started with AI, you didn't really have any idea about anything. Instead, it's become... well, you've just learned over time on your own."* – Birch, Business

> *"Mainly from exploring on my own, of course. That's probably where most of it comes from. I kind of understand how is works, and then I've also watched clips on YouTube, mainly to see that you're not behind the curve just yet. And also to get some tips on how to actually get the answer you're looking for, how to communicate with AI specifically, and prompts and things like that."* – Linden, Business

Both groups supplement this with self-directed learning, but the starting point is different. Civil engineering have a more technical background from their exposure in the programme, while business students start from scratch.

4.1.2 Social environment

Looking at how the social environment, there is a clear pattern among both groups in how media as well as peers affect the perception. Media, such as news, LinkedIn and social media, is often described as a source promoting threat-oriented information about AI, giving narratives about job displacements and that there is an uncertain future of AI. Rowan describes this this and notes that media has primarily generated anxiety, but when talking with friends tend to lean more toward an opportunity-oriented framing. Maple similarly points to media and news as a source of concern, describing that news and clickbait headlines frame AI as a threat to jobs.

> *"I think it's mainly the media that has made people feel worried about it and see it as something negative. Family and friends, we haven't talked about it that much, but when talking with friends about it, we try to see it more as an opportunity."* – Rowan, Business

> *"Mainly perhaps media, LinkedIn, discussions between family members. Absolutely. And you understand that it will play a bigger role in the future than you might have thought, especially with news also saying that AI is taking our jobs, blah blah blah. So, a lot of those clickbait headlines, basically."* – Maple, Business

> *"There's a big difference between, well, among, as I said, everything from friends and acquaintances, but a lot of what's in the media is also doomsday writing."* – Alder, Business

The civil engineering students describe this similarly to the business students. Iris describes social media and podcasts as amplifying the negative perspective on AI, while peers tend to be more encouraging and normalizes the view of AI:

> *"Yes, I would say so, but maybe a lot of social media, podcasts and things like that have created more concern about AI in the future. Because they highlight a lot of this negative perspective that it is developing in a way that might not be controllable in the future. And of course that makes you scared, since I don't have all the knowledge in the world about how AI works, what it does, and how it is developing. But then among friends and so on, it's perhaps more encouraging and gives a more positive image of AI."* – Iris, Civil Engineering

Several business students points out that peers see it as an opportunity, which have influenced the respondent to see it the same way. Linden also describe peers as serving as a channel from which one can learn from each other about AI tools and how to use them and stay informed on new developments. Linden describes how the social environment directly influenced the decision about which AI tool is used:

> *"Basically, no matter who you talk to, they use AI. You always talk to them and see how they use it, which tools they use. For example, I've switched to Claude now, from ChatGPT. That's something that came directly from my surroundings. Someone else tried it and said it worked much better. And then I thought, I've gotten*

tired of ChatGPT, so I'll switch to Claude instead. And it has completely influenced me, because I think it's much better. And then you also talk to friends who give you tips, like: this is what I use AI for." – Linden, Business

The civil engineering students describe this similarly to the business students. Sage describes the surroundings as positive to AI, and that the peers are positive to using AI as much as possible but with caution. Bamboo also describes the peer-network as positive, and a source of learning about the tool.

> *"I would say that my surroundings are generally quite pro-AI, which I would say affects my own attitude. If they had been against the internet, then maybe that would have influenced me. But they absolutely are not. The attitude around me is more about using AI as much as we can, but with supervision."* – Sage, Civil Engineering

> *"I would say that I have a group of friends who are quite AI-positive. That's partly why I started using AI myself. And I still feel that I get a lot of knowledge from them."* – Bamboo, Civil Engineering

4.1.3 Field specific formation

The combination of the presence or lack of formal AI education and the social environment creates a difference in perception between the groups. Business students, with no formal exposure and only self-directed learning in combination of media-heavy information, frames AI to a larger extent as more emotionally loaded, such as a source of anxiety or individual opportunity. Civil engineering students, on the other hand, have a more technical background with AI and engages with the phenomenon in a more systemic way by addressing questions of governance, restructuring the workforce and the impact on society alongside the individual views.

The difference is most noticeable when looking at the extent of the students' AI literacy and how it shapes the view of concerns'. Students with a greater technical background, such as Rose, are more likely to extend the threat perception beyond the individual concerns and toward more systemic views:

> *"I think that if you see it as an opportunity, you're probably thinking very individually. And what traditional media does, or media in general, is that they bring up this societal perspective and what the risks of AI are for society. And that is probably what has shaped me, because I see AI from a broader perspective on society, and not just for myself."* – Rose, Civil Engineering

4.1.4 Perception over time

A pattern that emerges across both student groups is that familiarity with AI over time tends to shift perception and the view on how to use it. Several students describe that they were more sceptical or resistant when they first encountered AI, but with increased use and understanding this has led to a gradually changed view and how they relate to it. Aspen describes this shift, where it started with scepticism but has turned to a sense of necessity to use it.

> *"It came two or three years ago. At first, people were very sceptical, but now it has become like: if you don't use it, you're inefficient."* – Aspen, Business

Iris describes an initial resistance to AI and that there was a desire to be independent, but that the view has gradually softened with time. Bamboo also describes a shift in how AI is used, and that it could be seen more as a teaching assistant and that it has become more useful.

> *"I would say that in the beginning, I was even more critical of it. And I don't know, I still had this feeling that I wanted to do things myself I wanted to write my texts myself and didn't want to use AI. And now... that has still shifted a bit, because I trust AI more now."* – Iris, Civil Engineering

> *"In the beginning, it was probably more that you submitted something and hoped you would get a good answer back. But nowadays, it has become more like using it as a kind of teaching assistant. It has moved more in that direction, rather than actually trying to get an answer and then taking that answer and putting it into a text. Instead, it's more about trying to arrive at the right answer yourself. It has become much more useful"* – Bamboo, Civil Engineering

When asked about how the perception has changed since you first started using it, Oak describes it most precisely and concise; that it has changed over time since the trust has been build up whereas in the beginning there was much more scepticism.

> *"I trust it much more to do the job, whereas in the beginning I was much more sceptical"* – Oak, Business

Both student group express an individual shift in perception, but there is also a normative dimension expressed among both groups. AI is commonly framed as an expectation rather than a personal choice to interact with it. If one were to not use AI, it can be considered a competitive disadvantage. Several respondents expressed this norm to require AI literacy. Aspen's quote stated above emphasises this, with Sage formulation this from a civil engineering perspective:

> *"But now I feel that having a basic understanding of AI is so vital in order to be able to make use of it. It has completely changed the way I choose to prioritize things. I choose to prioritize learning AI so that I can learn things faster. It has turned the scale upside down."* – Sage, Civil Engineering

This norm can function as addition to the formation of AI perception mechanism. The student are responding to the standard set by society in having to engage with AI.

The two student groups differ in the source and extent of their AI knowledge. Most business students rely only on self-directed learning through, for instance, social media, peers and experimentation and describe their educational programme as not preparing them for an AI influenced labour market. Most civil engineering students in this sample have encountered AI through their formal education in university, which has been expressed by respondents that it gives them a more technical background. Despite this difference, both groups describe that since starting with AI it has led to greater familiarity which in turn has led to a greater acceptance and more purposeful use over time. Both groups also describe the impact of the social environment in similar ways, where media tends to generate concerns and peers tend to encourage AI use.

4.2 Theme 2: Appraisal under uncertainty

The following theme addresses the first part of the research question, looking at how students perceptions of AI in relation to their future working life are formed. More specifically, this theme presents outcomes of this process, how students interpret AI as both an opportunity and a threat and how they distance themselves from it. The theme is structured around three subthemes: simultaneous opportunity and threat, occupational distancing, and future concern. It covers the codes ambivalent perceptions, opportunity on individual level, threat on societal level, field-specific exemption, threat in other occupations, temporal distancing, and role restructuring not elimination.

The dominant pattern regarding the perception of AI among the respondents is not exclusively seeing it as only an opportunity or only a threat. The majority of the respondents hold opportunity- and threat perceptions simultaneously, however what drives the view of the different perceptions differ among the respondents and the two student groups. What typically separates the views are personal versus societal, short-term versus long-term, their own field versus other fields or if AI is used appropriately vs AI-use in ways that serves organizations in cost-reduction in expense of workers.

4.2.1 Simultaneous opportunity and threat

The ambivalent perception is a coherent response to uncertainty about how AI will be implemented in their future working life and who will benefit from it. Birch explains the mixed orientation most concisely, explaining that the source of the threat is not AI itself but rather how organizations view it and uses it.

> *"Not that it's 100 percent an opportunity. Let's say fifty-fifty then. Because as I said, absolutely, you should use these kinds of tools. They shouldn't be used to replace people. But people don't see it that way. People see that they can save money. And then suddenly there's this threat of replacing workers with AI. So yes, okay. Not 100 percent. Fifty-fifty"* – Birch, Business

The framing that Birch provides, AI as a productivity tool that is being deployed in ways that create job displacement, is a view that is reoccurring among the respondents. Respondents do not in general question the usefulness of AI. What generates the threat perception is the expectations about how organizations will respond to it, not by augmenting the workers but by reducing the workforce. Sage also provides a similar insight while also adding that the implementation choices are beyond the students' control:

> *"It depends a bit on how it's used as well. It could be a huge opportunity, but at the same time it could also be a huge danger. It depends on how people choose to develop it. But that's beyond my control."* – Sage, Civil Engineering

Iris also expresses the mixed perception as a form of degree on how it is used, providing specific examples of when it could be an opportunity. However, when reflecting about the threat narrative, there is an uncertainty in exactly what is the threat and harder to provide examples. Iris illustrates that the future of disruption of AI can be a threat, but that it is hard to know exactly what the threat is.

> *"There's a big opportunity in using AI, for example, getting help when creating presentation material or writing a summary, things like that which just take unnecessary time. Of course, there's also a threat in it. If it becomes, I don't know, too much. Or that… yes, depending on how it's used, basically. And it feels like the market is changing."* – Iris, Civil Engineering

Common for both groups is that the opportunity perception is majorly associated on the individual level, serving as a tool the benefits personal productivity, learning and training. Business students in this study tend to frame AI as more of an opportunity on a personal level, and the threat at occupational or societal level. Birch points out that it is an opportunity, but points out a potential threat if companies start replacing workers with AI. Birch also notes that a secure job is already acquired and that therefore the disruption of AI will not affect the respondent in the near future.

> *"I would say that I personally see it as an opportunity. But then, if the company you're going to work for also sees it as an opportunity, as a way to skip hiring employees, then maybe it suddenly becomes a threat instead. But where I'm going to start working, I know that I have a job. I have a secure job, so to speak. And people there are already using AI. And I think it will become a case of not being replaced, but rather seeing AI as a complement instead"* – Birch, Business

Alder follows with speaking about the opportunities of AI and points out that AI cannot replace relationships between people. When resonating about potential threats, Alder brings up that the future field of work includes working with people, and therefore AI is not a threat.

> *"It's only an opportunity, I have to say. I think that if you're going to work with people, it is between people. Humans will still be needed, and I don't think AI can take a human's role."* – Alder, Business

The civil engineering student also frames AI as an individual opportunity. Rose could be considered one of the most knowledgeable students in the sample regarding AI and describes the framing accordingly:

> *"AI is an incredibly big threat. I think that I… Of course, for me as an individual, it's an opportunity, because I learn faster. But AI for society is a threat, and only a threat."* – Rose, Civil Engineering

The business students primarily concentrate their threat perception on their individual future working life, such as job displacement and increased labour market competition. The civil engineering student on the other hand places the threat more on a societal level. Linden and Maple capture the business student framing by emphasizing that there will be increased competition and that their job might not be needed:

> *"Of course, both opportunities and threats. The threats would probably be that competition becomes much tougher, since I think many jobs will be replaced by AI, but the people applying for jobs will still be the same. So there will be fewer jobs available, but still just as many people applying. That's the threat. That's the absolute biggest threat"* – Linden, Business

> *"The threat would be that in the future I might not get a job, because that job is more or less no longer needed. Because you can do the same type of work with*

> *perhaps fewer employees, since we have a more efficient way of doing it, with AI tools that help us."* – Maple, Business

This perception is extended by the civil engineering student, with Rose in particular expressing direct observation of the disruption of AI rather than speculations:

> *"We lose this human discussion and the human perspective on everything. I think that is very negative for future innovation."* – Rose, Civil Engineering

> *" I have personally witnessed how a task that takes three months and 20 employees, and creates a lot of job opportunities, can now be done with AI. You prompt and iterate for 20 minutes, and then it's finished."* – Rose, Civil Engineering

Several other civil engineer students also raise concerns at a systemic level. Iris brings up the informational perspective, where prior to AI there was a lot more perspectives creating discussion, but now with AI there is only one point of view and that is AI's perspective. Sage further develops this and adds that there is no one keeping track of the fast pace that AI has, and that there is nothing one can do to stop it, which creates an uncertainty and worry.

> "*Yes, partly it could become a threat in the sense that the information that exists and is produced if everyone just uses AI to generate information for material and things like that, and doesn't check it, then it becomes very meaningless, because then we only have one perspective"* – Iris, Civil Engineering

> *"What worries me, I think, is that it's so trendy, and then there's also the fact that no one is really keeping track of how fast it's moving."* – Sage, Civil Engineering

Thyme brings in another dimension of threat, bringing up humanity as a whole. Thyme points about that there should be more awareness, and that if humanity lets it go too far AI could change the world so much that many people may become worthless.

> *I think I should be a bit more nervous for humanity, so to speak, for us as a whole, and what it could do in that sense. I don't think it will become evil and start wiping us out, but rather that it could change the world so much that many people become, in a way, worthless.* – Thyme, Civil Engineering

4.2.2 Occupational distancing

Even though the respondents both express opportunity and threat perceptions at the same time, they do not apply the threat to themselves, instead the students tend to distance themselves from it. The most common distinction that the respondents expressed is the field-specific exemption. Respondents identifies occupations that are highly exposed to be substituted for AI, commonly auditing, financial analysis and data analytics, while arguing that their own career path is not as affected. This pattern was prominent among business students, which is notable as the named occupations falls within the business and economics area, which is the field that the respondents are studying but not the specific career path the have chosen. Birch, Oak and Linden express that AI will affect the labour markets in some way, but that themselves will not be affected due to their chosen work field.

> *"The finance side will take quite a hit, to be completely honest I think there is a lot that could be replaced. Their profession could be wiped out within three years because AI takes it over. But since I'm not going to work in finance, it won't affect me."* – Birch, Business

> *"At least not the kinds of jobs I want to do, I don't think I'll be replaced. I just see it as an opportunity to become even better."* – Oak, Business

> *"I want to work with relationships and sales. It will always be better to do that face-to-face with another person than to talk to a robot."* – Linden, Business

Similar to the field-specific exemption, the pattern extends by location the threat in other field than their own chosen one. Linden, who in the quote above expressed their field to be relationship and sales related captures the logic of distancing themselves from the threat by placing it in other occupational fields:

> *"Economics and finance, a great deal. Construction and things like that, hardly at all. Anything that has to do with the internet, where you sit at a computer and read through things, that will be affected a great deal."* – Linden, Business

Civil engineering students show this pattern less consistently. Several civil engineering students are more willing to acknowledge that the technical tasks within their field are exposed and that their potential future roles will be restructured.

4.2.3 Future concern

A second form of distancing from the potential disruption of AI is temporal, students acknowledge that AI will transform the labour market significantly but locate most of these disruptive changes beyond their near future and immediate planning horizon. This reduces the urgency of AI as a factor in current decisions. Pine and Bamboo illustrate this:

> *"Yes, but in that role, I think that right now it looks like it perhaps couldn't be replaced. But in ten years, absolutely. If AI keeps developing at this pace, then it could. But it will probability be a while"* – Pine, Business

> *"I think customer interaction will always still be handled by humans. At least for the next 50 years, or 25 years, I'd like to think"* – Bamboo, Business

Other students express that the disruptions of AI are not an immediate problem and that it is not a threat right now but rather in the future or that the future of AI disruptions is uncertain. The temporal distancing of the disruption of AI is a common pattern among both groups, Maple and Moss further describes that AI is not something that will affect them right now but rather in the future. However, what that future looks like, is uncertain.

> *"It's not that I see AI as a direct threat to me right now, but in the long term, I can see that it changes what is in demand."* – Maple, Business

> *"It's not that I think it will affect me directly in the near term, but over time, what is valued will change."* – Moss, Civil Engineering

The temporal distancing is commonly paired with a reframing of the threat as a whole among the civil engineering students, changing the anticipation from elimination to restructuring. The role content will transform where the human involvement will shift to more oversight, quality control and judgement. This is express by Sage and Bamboo:

> *"The jobs themselves will change in character, but it won't affect employment opportunities. Consulting work may become less about analysis and more about presentation and sales. It's not that consultants won't be needed rather, they'll be needed in a different form."* – Sage, Civil Engineering

> *"If AI becomes advanced enough, maybe only two quants will be needed: one to prompt it and one to check that everything looks right. So I suppose it's that the amount of work decreases because of AI. Then again, I don't think it will ever disappear completely, but rather that where ten people were needed before, now only two or three are required."* – Bamboo, Civil Engineering

While Bamboo states that the workforce will be smaller as AI can substitute human labour, Poppy builds on the same risk but states that new jobs can emerge from the development of AI. Further, Poppy states that even the future work field would affect themselves, the chosen education is broad offering flexibility in work opportunities.

> *"Sure, there's definitely a risk that AI will take over certain jobs in the world, which could certainly affect me. But it could also be the case that new jobs will emerge, and then of course you have to be a bit flexible about that. Obviously, if you've trained for something specific and AI manages to take over exactly that role, then you might have to consider training for something else or changing career paths. But when it comes to my education, it's quite broad."* – Poppy, Civil Engineer

Taken together, students distance themselves from the disruptions of AI in two ways, by locating the threat in other fields than their own field that they aspire to work in or locating it in the future making the change not as immediate. This allows students to recognize the disruptions of AI while still holding them at arm's length without denying its reality. The students acknowledge the threat broad but real but can be avoided by either occupational choice or time. The comparison between the two student groups reveals that business students more consistently exempt their own future field or occupation from disruptions, and names at-risk occupations that is within their own educational domain but is outside their chosen work path. Civil engineering students, on the other hand, have shown more willingness to acknowledge that there is direct exposure within their own domain. However, they tend to frame this as restructuring the roles and reducing the workforce rather than eliminating the roles completely. Both groups express that the threat is in the future, and employ temporal distancing.

4.3 Theme 3: Revaluing competence under AI

The following theme addresses the latter part of the research question, which examens how students' perceptions of AI shape their skills and knowledge investments and development. The theme is structured under two sub-themes: Skills value under AI and Uncertainty and investment decisions, this covers the codes hard skills replaceability, soft skills resilience, knowledge foundation improves AI use, anxiety over education value, human capital revaluation, shifting to AI literacy, and reconsidering skills to develop.

4.3.1 Skills value under AI

A similarity among all respondents is their concern regarding whether or not their knowledge and skills will still remain valuable in the AI impacted labour market. Their take on the issue is collectively identifying concrete competencies they have developed through their higher education that might be affected by AI. This concern about which skills and competencies will remain valuable, and which might be devalued, is both a cognitive and emotional challenge among the respondents. This theme presents how the respondents evaluate their accumulated competence, what they believe will continue to be of value to the labour market and thereby how that affects the choices they will make further.

Collectively for both civil engineering and business students, is a prominent distinction between two different categories of competence, technical and human skills, where the hard skills are seen as more replaceable. All respondents identified technical and certain task-specific skills to be the most threatened by AI automation. Some examples used were accounting, programming, financial analysis and mathematical modelling. Relational competencies, such as communication, judgement and relationship-building skills were considered to be more spared from being diminished by AI. This was the case for both studying fields, naming multiple interpersonal skills.

Business student frequently expressed financial tasks to be most vulnerable to AI. Linden expressed AI's impact on financial occupations distinctly:

> *"The analysis of a company….one hundred percent. Previously it took days, weeks and almost months to do due diligence for a company. But if you have the right access to the right artificial intelligence, it can do that in just a few minutes."* - Linden, Business

Linden, Rowan and Maple also categorize accounting and revision to the same fate as financial occupations. The possess tasks that are heavily rule constricted and data driven:

> *"I think revision could also be replaced….. you put in company reports, refer to legislations and AI can go through that evaluation at a much faster pace compared if you would do everything manually*" – Linden, Business

> *"I was back in the days interested in learning programming and photoshop and Adobe Illustrator editing. But I am starting to reconsider if it worth learning at this point….. it will not be very sought after in the future after all."* – Rowan, Business

> *"AI-tools outperform many occupations that today need a lot of workforce to complete."* – Maple, Business

Civil engineering students find a similar pattern in their field of work. Sage perspective positions AI to not eliminate these data and rule driven tasks and occupations but reshaping and shifting the human involvement to a controlling position. Thereby, instead of contributing to the production during the process, human involvement is limited to quality control of output:

> *"I believe that the basic analysis assignment.. 'find this information'…will be replaced. It will be more about quality control that is done by humans" – Sage, Civil Engineering*

Both groups also argue for soft skills to be more resilient against AI. Their reasoning is not only because the skills are harder to automate, but also because individuals demand human contact and will not accept AI instead. Aspen articulates this:

> *"I don't think these soft skills will be replaced by AI. But hard skills, absolutely…….I think is about the human relation between two people. For example, when you speak with an AI-chatbot. When you have an issue with anything, like home insurance. You can go insane just because you want to speak with someone who listens and is adaptable. But AI have prompts….a human can draw conclusions that the chatbot cannot do"* - Aspen, Business

Birch and Pine both mention the human factor, capturing the lens of customer relations and customer expectations. They argue for clients to prefer human interaction regardless of how advanced the AI's capabilities may be:

> *"It is the human factor. Clients still appreciate that, it should still be human. You just get 'pissed-off' when you talk to an AI-chatbot."* – Birch, Business

> *"It is the human factor that AI cannot replace. It is things like counselling and leaders…roles like that will be very hard for a chatbot to take over"* - Pine, Business

Civil engineering students find a similar pattern in their field of work. Moss captures this by pointing out leadership and interpersonal skills will be AI resistant:

> "*The fuzzy courses….we have gone through leadership and such…AI will not be able to replace that. The human things are still good"* – Moss, Civil Engineering

Among the civil engineer respondents, there is a related claim that is more common than with the business students. They argue that a deep and advanced knowledge in a certain subject can act as an advantage against being replaced by AI and can also improve the AI use. The logic is that knowledge within a field can generate an effective usage of AI tools. It enables more analytical evaluation, better prompting and greater sense of where AI might be falling short. Rose describes this distinctly:

> *"But what you have to think about are these soft skills you have learned in school. You learn about critical thinking. You learn about that way of thinking. If you don't know anything, then you will not know what to prompt the AI….But if I know something, if I have the basics, then I can make a better prompt"* – Rose, Civil Engineering

This thinking reshapes the role of education as it is affected by AI. Students acknowledge that they cannot compete with AI when it comes to the volume of knowledge. Hence, the value of education shifts into developing contextual understanding, judgement and evaluation to be able to manage AI's output. Iris gives an example of this in practice, speaking on programming:

> *"...right now I have a lot of programming...then I can get help. Like, if I really wanted to learn. That is the difference. If I do not want to learn, then I could really get by easily with AI (glida på ett bananskal). Just sending the assignments and it will spit something out....But if I would want to learn it, then I could use it as a supporting tool instead"* - Iris, Civil Engineering

4.3.2 Uncertainty and investments

There is a clear concern regarding skill devaluation among both studying fields, which poses as anxiety over educational value. This is not only displayed in a reasonable matter among them, but also emotionally. The anxiety regarding their investment in education being valuable and worth the time is a common distinction across many respondents.

> *"But if I think just think about my skills and knowledge....what we have learn according to the curriculum. Sounds awful to say, but the education is then worthless in my opinion. All of these small thing and knowledge things, these are things AI already can do. ...we spend 5 years and take a 500 000 SEK loan to learn something that is already very easily accessible"* – Rose, Civil Engineering

The uncertainty of the value of the education does not remain passive, it drives respondents to reconsider what is worth investing in. This uncertainty has caused respondents to reassess the relative value of different types of knowledge, prioritising competencies that they believe will retain or gain value rather than skills that are perceived as AI-exposed. Rowan describes this shift:

> *"Back in the days I was interested in learning to programme. And also to be able to edit in Photoshop and Adobe Illustrator.... but I am starting to think that it is not worth learning at this point, that it will either way not be very demanded in the future and it is maybe better to educate yourself in AI marketing which I have thought about."* - Rowan, Business

This causes a revaluation of the respondents human capital, leading to reconsiderations of which skills are worth developing. This is expressed by Rowan's quote above, but the pattern is visible across both groups.

Multiple respondents display a change in how they handle their investments in education because of AI's impact, shifting more towards AI literacy. The uncertainty has led Bamboo to an adjustment in priorities, shifting focus from, in Bamboo opinion, AI exposed skills such as programming to seek deeper understanding of AI. Rowan confirms this as well, explaining their shift of focus to new skills they determine to gain value in the future:

> *"I have applied for an AI-course for next semester. I believe I did not feel like that at the start. I thought, when I started studying in 2021, AI was very new. Then I thought I would take a management course at the end to gain the human part. But now the majority of my classmates have chosen to take an AI-course to get a better sense of what it actually does"* – Bamboo, Civil Engineering

Similar to Bamboo, Maple reflects on how AI literacy is becoming more crucial and a fundamental skill to be competitive on the job market:

> *"I connect this a little bit to like Excel, that Excel can be a tool for many employers that you can become extra competitive if you know Excel at a good level. And I think that AI tools in the future can act in the same manner."* – Maple, Business

Comparing the two studying fields, there are apparent similarities and differences. The groups unanimously distinct technical and relational skills to be affected in different

approaches. Both groups express a certain level of anxiety regarding the value of skills in an AI affected environment. The interesting difference between the groups lays in their framing of the defence against AI. Civil engineering students put emphasis on subject expertise to remain competitive and avoid replacement, while Business students argue for soft skills such as relational and interpersonal skills. Commonly for the studying filed is the underlying genuine anxiety towards an uncertainty.

4.4 Theme 4: Perceived control as a behavioural determinant

The following theme continues to address the second part of the research question, looking at how students' perceptions of AI shape their skills and knowledge development. The theme is structured around two subthemes: perceived control split, and active versus passive adaptation. It covers the codes; high control in personal AI use, low macro-level control, AI as inevitable, active skill adaptation by civil engineering students, and passive awareness with limited action by business students.

4.4.1 Perceived control split

Regarding their own use of AI, the students experience a high level of control when it comes to their own personal AI usage, most importantly regarding what tasks they assign to AI and which they do themselves. At the same time they experience low control on the macro level, stating that they have any impact. This pattern is consistent over both civil engineering students and business students, where Linden and Poppy stresses this clearly and applies it to a practical setting with similar reasoning even though they are from different student groups:

> *"I have control over my own AI usage. That I have full control over. But how the development will look like in the world, there I do not have any impact over whatsoever, it is more about adaptation."* – Linden, Business

> *"I absolutely have control over how I use AI and that is something I can decide whether or not I do it and how I do it and to what extent. But I don't have control over how AI will affect the world"* -Poppy, Civil Engineering

However, on a macro-level, the respondent expresses a significantly lower amount of control over AI's development. This includes advancements, decisions regarding regulations as well as how organizations use and expect AI tools usage. This phenomenon is the most agreed finding among near all respondents.

This framing is consistent for both studying groups who have vastly different perspectives and relationships with Ai. This could suggest that their shared view of low control on a macro-level is due to structural factors as opposed to an individual occurrence. It describes the reality of the current situation rather than the individual opinion at a certain occasion. Rowan and Iris express the same uncertainty of control on a macro-level to be at a distance, geographical and a distant future:

> *"Right now, no, it feels like I have little control because you just really can't stop the development.....yes it happens in like Silicon Valley I believe... and big corporations...I have very little control over that."* – Rowan, Business

> *"You get scared because you don't possess all the knowledge in the world about how AI works and what it does and how it is developing."* – Iris, Civil Engineering

Out of all the respondents, Thyme expresses this framing distinctly:

> *"[Regarding perceived control over AI and their future] I think no. Since I'm not involved in the field. Then I don't really have any idea where it is going. Like, it absolutely feels like we have no control over it, what happens will happen no matter what I think or would have wanted to happen."* – Thyme, Civil Engineering

Thyme also emphasizes the lack of control among the developers of AI as well. Thyme believes that nobody can be in control of how AI will develop, not even the developers themselves and mentions a possibility to create a "Superintelligence" that outsmarts the human cognitive capability.

The framing of having low macro-level control is followed by the respondents expressing AI as inevitable. Rose explains this by applying the logic to previous technological shifts to emphasise the bare necessity to adapt:

> *"You don't have any control at all, no. You have no control, instead you must parry according to society. And that applies to everything, as it was with the latest digitalisation as well. You just cannot stop a digitalisation, instead you simply have to ride the wave."* – Rose, Civil Engineering

Aspen acknowledges the positive aspects of how AI development can signify a change. Aspen argues for counteracting and slowing down AI's development will not lead to a diminishing of the technology; it will in one way or another be integrated in society.

> *"Well, control or control. I do not think you have a say in that. Like, Ai will develop and become more and more complicated.....Laws and regulations and environmental impact. Like the amount of data that is used....that is how us humans can affect AI developments. Otherwise, I believe that it will just keep going, becoming better and better. And in that way also take more responsibility."* – Aspen, Business

4.4.2 Active versus passive adaptation

Both groups share the view of the inevitability of AI, but it does not produce the same response across both groups. Civil engineering respondents were more likely to describe an active skill adaptation response to AI, such as investments in relevant courses, AI literacy and actively seeking information about relevant AI developments. Hence, these students strive to compensate their lack of control on societal macro-level with adaptability and knowledge on an individual level:

> *"It is really a shame if you could not do what you would want to because you are scared AI will take over, you might have to adapt your own competencies to make you competitive. As said before, I don't think all jobs will disappear, but maybe there will be less job and if you then have the ability to adapt that job to AI, you can gain competitiveness."* – Poppy, Civil Engineering

Rose, whom has the most extensive practical experience with AI and AI tools, explicitly frames this to be a competitive advantage, having deeper knowledge of AI instead of dismissing the uncertainty:

> *"I believe it is because I am competitive with my knowledge in AI. I think I am the only one I know that have built a company with AI and use as much as I do. And I also this it is a big competitive advantage, that I don't see it as a laziness-tool, rather an optimization-tool."* – Rose, Civil Engineering

The business student present a different pattern, displaying a passive awareness with limited action rather than an active adaptation. They combine the passive awareness with and acceptance that does not translate into action:

> *"At this time, I still feel in control over what I actually know and that I can pick my opportunities, How much I get from this?* [.....] *but going forward it is really difficult to feel like in control over anything [AI development]."* - Pine, Business

> *"I don't think I could say I'm in control really. I don't think so. I think it will become apparent in the upcoming years and see how it affects my future."* - Alder, Business

On the other hand, Alder realises the value of knowledge to be a factor to counteract loss of control. However, Alder does not make claims of actual actions to become more knowledgeable, unlike many of the civil engineers. Maple gives a similar reasoning, acknowledging a low level of control and Maples awareness of this, without curating an adaptation:

> *"It feels like it's not really in my hands, in a way. It will develop 100 percent and yes, I could have felt that it was in my hands if I.... would have felt that it would have benefitted me to learn about the AI-tool.... So yes, you could have felt that it was in your hands, but I'm not particularly interested in, specifically that part."* – Maple, Business

This is commonly occurring among the business students, having awareness but no actual active measure to approach the uncertainty. However, some business student take this opportunity to develop their AI literacy, as they view it as a labour market asset. Oak describes AI knowledge as a concrete asset to provide a future employer:

> *"I see this as an opportunity to be one of those who perhaps learns it. To work with AI already now and be able to offer that to future employers, like: I have knowledge of AI and how to use it for marketing purposes."* – Oak, Business

In a comparison between the two studying fields, there are clear differences that become apparent. Both groups have identified responses against AI and what could minimize the uncertainty, while civil engineering students more commonly display explicit adjustments to their education investments, improving AI literacy to remain competitive. The Civil engineering students express more firmly than the business students about their own capability to adapt to the change and work with AI and thereby accepting the change. Business student on the other hand more commonly expressed an acceptance without any notable actions to improve their control.

5. Discussion

This chapter analyses and discusses the empirical findings in relation of the theoretical framework. The discussion is structured to directly answer both parts of the research question. Section 5.1 addresses the first part of the research question, examining how students' perceptions of AI in relation to their future working life is formed. Section 5.2 addresses the second part of the research question, examining how these perceptions shape students' skills and knowledge development.

5.1 How students' perceptions of AI in relation to their future working life are formed

The first part of the research question asks about how students' perceptions of AI in relation to their future working life are formed. The empirical findings point to two connected answers that interacts with each other. Firstly, that perception is shaped by the conditions students bring into the encounter with AI, such as their educational background, social environment and prior experience. The second concerns how these conditions structure a pattern in the way students appraise AI and position themselves in relation to its consequences. The following two sections examine each dimension.

5.1.1 Shapers of AI perception

The findings reveal that the source and depth of students' knowledge about AI serve as a foundation to the perception formation process, where both selective attention and sensemaking mechanisms matters. There is a structural difference on where the students receive their knowledge from. Business students rely mostly on self-directed learning through social media, peers and "learn by doing". Civil engineering also applies the same methods but has formal AI education in their programme. This does not necessarily create a knowledge gap but rather draws attention to different aspects of AI that students notice, which narratives they find credible and which responses that take into action. In this sense, the absence of formal AI education among the business students is not only a curricular gap between the groups, but also a condition that creates a different environment for how perceptions are formed. This difference in the source of knowledge also reflects a difference in the human capital investment strategy where civil engineering students have access to structured Ai education as part g their programme, and where business students must independently decide whether and how to invest time and effort in developing knowledge about AI, with no institution behind them to guide those decisions (Becker, 1962).

The difference in foundation connects to how selective attention shapes the perception of AI in relation to the future working life. Selective attention, as described by Dearborn and Simon (1958), refers to the tendency for individuals to notice, interpret and prioritise information while also filtering out information that falls outside of the frame of reference. In context to this study, selective attention explains why the two groups of students approaching the same technological phenomenon arrive at different perceptions of what it means and what it threatens. Civil engineering students, having learning experience about AI from the university, tend to direct their attention toward the broader and more systemic aspects of AI, such as how it is developing, what it cannot do yet and what consequences it may have for organizations and society. Business students, that do not have the same

technical background or formal AI education, are more dependent on the framings of AI that the social environment provides. The empirical findings show a clear and consistent pattern across both groups where media tends to present AI as a threat to jobs, where peers tend to present it as a useful tool that should be embraced. The empirical findings suggest that business students are affected by the media driven threat narrative than civil engineering students. This is consistent with Dearborn and Simon's (1958) argument that selective attention is based on prior background, and without technical knowledge to evaluate or give context to claims about AI, business students have fewer cognitive resources to challenge different narratives they encounter. This makes the framing from media and peers more dominant input in their perception formation.

Sensemaking theory helps explain why this matters for how perceptions are formed. Weick (1995) describes sensemaking as the process where individuals construct interpretations of uncertain situations by drawing on the cognitive resources available to them, such as their prior knowledge, experiences and social environment, rather than seeking a single objective truth to form an interpretation that is plausible and actionable. Maitlis and Christianson (2014) further extends this by emphasising that sensemaking is both an individual and social process, stating that people do not make sense alone but rather within their community that provide shared interpretive frames. The empirical findings confirms that students do not form perceptions of AI through independent rational assessment. Instead, they draw on their cognitive resources available to the individual, such as their educational background, peer networks, media narratives and their prior experience with AI tools, to construct interpretations that are plausible even if not fully accurate. Maitlis and Christianson (2014) emphasize that sensemaking is both an individual process and a social process, and this duality of sensemaking can be clearly found in the findings. Within the study programmes, students express that they have shared norms about what AI means and how it should be understood, seem to develop organically among the students. Business students describe that there is a collective sense among the peers that AI is primarily a productivity tool and an opportunity that should be embraced. Civil engineering students, on the other hand, more commonly report programme-level awareness if AI's limitations and the systemic risks. These shared interpretive frames and ways of understanding act as support for sensemaking. They provide a common story and narrative that helps the individual person shape how to interpret that situation. The social environment does not only inform students about AI and its development, but it also provides an interpretive frame in how AI is understood and perceived, and that understanding differs depending on the educational context.

The way students engage with AI knowledge outside of the formal education can be further understood through March's (1991) concepts of exploration and exploitation. Business students who experiment independently with AI tools, such as trying new platforms based on peer recommendation, testing different ways of prompting and gradually expanding how they use AI, are engaging in what March (1991) calls exploration: pursuing new possibilities without a clear way for evaluating what works and why. Civil engineering, on the other hand, can partially apply what they have formally learnt about AI while also having a more technical education, can build further on what they already understand. This can be closer connected to exploitation, where a person develops and refines existing knowledge. However, it is important to note that both groups apply an explorative approach, but civil engineering students are closer to the exploitative approach. One case where this is apparent is with Rose, who is the most knowledgeable person in this sample, also applies the most exploitable approach by integrating it more and more into life, where one example is that AI has control over the main inbox and independently write emails or in the respondents business where AI handles all invoicing. March (1991) argues that exploration and exploitation involve different trade-offs. Exploitation, on one hand, tends to produce more

predictable results in the short term, while exploration have greater potential doe new discoveries but with less certainty of outcome. In the context of how students develop AI knowledge, this tension is visible in the data. Business students' exploratory self-learning tends to produce a quick familiarity, but can have a limited depth of how it works and what it implies. Civil engineering students, who also engages in an exploratory approach, tend to have a more structured foundation to build from, which allows their engagement with AI to develop in a somewhat more grounded direction. These different starting in how the knowledge is acquired and deepened are a primary condition that shapes what students are able to perceive, and thereby also answers the first part of the research question: perception is formed through the interaction between the students educational background, social environment and their prior AI knowledge.

A pattern that appears across both student groups is a gradual shift from scepticism toward acceptance, and more purposeful use as students become more familiar with AI over time. Students who were initially resistant or unsure about AI constantly describe how their view as changes as they have used it more and as the tool has developed, moving from uncertainty about whether to use it at all, toward a clearer sense of how and when it is useful. This is consistent with Weick's (1995) understanding of sensemaking as an ongoing process rather than a one-time judgement, instead the students continue to update their understanding of the situation as they gain new experiences with it. This temporal dimension of perception formation is analytically significant since it demonstrates that perception is not fixed from the first encounter with AI, but rather that is continues to be shaped by experiences and normative pressure. The growing social expectation that AI literacy can be seen as a baseline requirement rather than a personal choice functions as an additional formation mechanism that operates independently of formal education or media influence.

5.1.2 Appraising AI under uncertainty

The conditions described in section 5.1.1, educational background, social environment and prior experiences, are not only prior conditions to perception but it also structures the specific form that perception takes. This section moves from the conditions that shape perception to the actual perceptual outcomes that those conditions produce. More specifically, it looks at how students appraise AI both as an opportunity and a threat, why the framing tends to be concentrated at the individual level while the threat framing extents to the occupational and societal level. Students also distance themselves from these threats, either by placing it in other fields or in the future. These patterns are not incidental but are structured by the educational backgrounds and knowledge conditions that were identified in 5.1.1.

The most common pattern on perception across both student groups is not to have one exclusive perception, but rather to have a threat perception and an opportunity perception at the same time. This finding is consistent with the two-factor appraisal framework by König et al. (2021), which argues that individuals' responses to discontinuous change, like the fast pace of AI developments, should not be understood as a single unified perception, but as the product of two independent factors: gain/loss framing and perceived control. Gain/loss framing refers to whether a change is interpreted as potential benefits or potential harm to the individuals. These two framings can coexist at the same time since different aspects of the same change can carry different implications for the individual. The findings suggest that most respondents can see both sides if the future of AI in relation to their working career. Students in this sample tend to frame the potential benefits toward their own situation suck as increasing efficiency and using it as a learning tool. Potential harm,

or threat perceptions, are directed towards the labour market in a broader sense, towards other occupational fields, or toward societal consequences of AI development. König et al. (2021) further note that the gain/loss framing, and perceived control are independent from each other and can vary. The findings in this study supports this since students typically hold a gain framing for their own personal situation but a loss framing for the collective or systemic level. The mixed response and ambivalence on the gain/loss framing is not a sign of confusion, but rather as response to an uncertain expectation in which different aspects of the future of AI have different implications.

One analytically significant pattern is what the empirical findings describe as occupational distancing. It is the tendency among the students, especially business students, to clearly identify high-risk occupations that are exposed to AI substitution within their own educational domain, while exempting their own chosen future work path from that threat. This pattern is not simply denial, but could be understood through selective attention which, described by Dearborn and Simon (1958), demonstrates that individuals attend to and filter information based on their existing role, background and prior experiences, and also filters out information that is not in their frame of reference. Business students, where their educational background in most cases does not include formal AI education, attend selectively to those aspects of AI's impact that are consistent with they existing understanding of the field. They can identify occupations that are data-driven and rely on routines and see those as a threat to AI within their own educational domain, where auditing and occupations relating to analysis have been mentioned by several students. But when relating to their own chosen career path, which tend to involve relational work, marketing managerial assignments, they are typically protected from the threat expressed in the other occupations and therefore see AI as an opportunity. This distancing mechanism allows for students to acknowledge the reality of the disruption of AI, while seeing their own future as manageable.

Civil engineering students show a different selective attention pattern. The civil engineering students have a more technical education and have engaged with AI in different technical contexts in the education, either du dedicated AI-courses or where AI has been integrated to different courses, for instance programming, they attend to more structural dimensions of AI's impact, which includes governance, perspective on information, environmental consequences and the reorganization of roles. Their threat perception are therefore more likely to extend beyond their individual career concerns and more towards concerns on a societal level, while also simultaneously being more willing to acknowledge that AI has, or is going to have, a direct impact on their own field. Dearborn and Simon's (1958) insight that selective attention is shaped by the prior background and role identity explains why these two groups, when reflecting on the same phenomenon, have systematically different perceptions of what AI means and what it threatens. This finding extends the application of selective attention beyond the organizational setting in which Dearborn and Simon (1958) originally studied it, which demonstrates that the educational field of study can function as a shaping condition for perception of technological discontinuity.

Taken together, the findings discussed in sections 5.1.1 and 5.1.2 answer the first part of the research question. Students' perception of AI in relation to their future working life are formed through the interaction of educational background, social environment and prior experiences with AI, which in turn determines what students cognitively attend to and how they construct interpretations of an uncertain situation. The outcome of this formation process is a pattern of both interpreted gain and loss appraisal shaped by the conditions that each group brings. The perception that emerges is not necessarily individual, it is

differentiated by field of study and it sets the conditions for how students respond through their skills and knowledge development.

5.2 How students' perceptions of AI shape their skills and knowledge development

This section covers the second part of the research question, which asks how perceptions shaped are shaping students' skills and knowledge development. The findings contribute to two mechanisms, students actively assess their competencies they have accumulated due to AI and thereby act according to their perceived level of control. This translates into behavioural responses, influencing if they choose to invest in new skills and knowledge.

5.2.1 Revaluing competence under AI

Commonly for both student groups is the concern about their competencies they have accumulated throughout their higher education will still retain value in an AI-influenced labour market. This finding can be connected to the logic of educational investments, conceptualized by Schultz (1961) and Becker (1962). Human capital theory treats investments in education and training as choices based on the expectation that it will bring future returns. The expectation is that the investment in competencies will generate value in the future labour market. The respondents demonstrate the disruption of this logic due to AI. Because Ai is developing rapidly, the predictability of human capital investments has been compromised, making the outlook shorter. This makes the students assessment of their competencies retaining value more difficult. The student expresses this uncertainty of skills devaluation not only as an emotional and anxiety evoking process, but also as a logical economic process. The students evaluate their currently developing skills and knowledge if they will retain value in the future, where many find this process uneasy and uncertain.

The students identify a distinction between technical and relational competencies, which is closely coherent to the classifications of competencies in the KSAO framework by Ployhart and Moliterno (2011). The four classifications are *knowledge*, which is the understanding of facts and concepts needed for task performing, *skills*, which is the learned capacity to execute a task, *ability,* stable capacities which lays the foundation to be able to perform tasks, and *other characteristics*, which includes personality, attitude and motivations. The respondents identify the hard technical skills to be the most exposed to AI in accordance to this framework. These include tasks such as programming, financial analysis and accounting, which all are aligned with the skills and knowledge components in the framework. Both acquired through training and education, making them more transferable. Relational competencies such as communication, judgement and interpersonal skills are viewed as less susceptible and more resilient towards AI. These are more aligned with abilities and other characteristics in the framework, as they are difficult to formalise and for AI to perform. This perception of soft skills being harder to automate also correlates with Autor et al. (2003) and Acemoglu & Restrepo's (2019) task-based view. The authors positions tasks that are socially and continually embedded are more difficult to automate, in contrast with data-driven and rule-based tasks. This is something the student demonstrate, as they provide examples of data- and rule-based tasks in occupations they view as at risk of being automated, such as accounting and financial analysis.

The students show understanding in the displacement effect, where AI take over tasks that preciously were performed by human labour (Acemoglu & Restrepo, 2019). However, the

reinstatement effect is not commonly established among the students. The reasoning by Acemoglu and Restrepo (2019) is that the technological change does not solely replace the human labour but instead creates new tasks where human involvement can be needed. This results in a shift in labour demand, meaning that the human involvement takes a new form. The dominant perception among the respondents is an understanding that AI reduces and removes the need of human labour in tasks, decreasing the value in certain skills and only a few acknowledges the new need for human labour that can appear. The perceptions of the students in the study are therefore largely shaped by displacement threats that are visible and immediate to them, rather than the reinstatement processes that Acemoglu and Restrepo (2019) identifies as an outcome. This points to a perceived uncertainty regarding anticipating that new task can come out of AI replacing skills in present time. It also reflects an unconscious cognitive bias where students do not identify the reinstatement effect of new tasks emerging, affect their decisions of human capital investments. This can reflect an asymmetry that may result in students underestimating competencies that can in fact remain valuable in the future or a shortcoming of competencies valuable in the future labour market.

Furthermore, the empirical findings suggest an understanding of the augmentation-automation paradox as explained by Raisch and Krakowski (2021). The authors argue that augmentation and automation cannot be opposite to each other in terms of AI usage. When a task becomes AI automated, adjacent tasks can still require human involvement, such as judgement and contextual understanding. This paradoxical view reflects the students reasoning of which competencies they choose to develop. The choice is not only to identify if a skill can be automated but also if the student can adapt their own competencies to adhere to the newly generated task that AI created. Multiple respondents describe the relationship of AI automation of a task can create an adjacent task, where human involvement is required. For example, Sage has observed that basic analytical tasks can be automated by AI, while the quality control will still need human responsibility. Iris has also made distinction of AI tools producing an output and using AI to assist learning in order to develop actual understanding. The student who understands this logic can be considered more strategically understood regarding AI's reshaping of task processes and the demand of human capital in an AI affected workplace. This is in contrast to most students' that only understand the displacement effect, leaving a shallow reasoning regarding this narrative.

Students do a continuous evaluation of their efforts of development as a response to their perception. This directly connects to the second part of our research question. The students reconsider if it is worth developing a certain skill and thereby prioritise and deprioritise accordingly. The skills perceived as AI-exposed are commonly views as less worth to develop. The students position AI literacy as an increasingly valuable asset. This reconsideration is driven by individual perceptions of the students, directly connected to the gain/loss framing as described in section 5.1.2. It suggests that student who have a loss/threat perception on their competencies respond by trying to adjust their investments, highlighting that perception does not only remain cognitive but is translated into choices.

5.2.2 Perceived control as a behavioural determinant

The most consistent finding among the respondent is their view of personal control over their own AI usage and low macro-level control over AI developments. This is directly related to König et al.'s (2021) two-factor appraisal framework regarding perceived control. The level of perceived control and ability to influence outcomes is separate from the gain/loss framing, however equally important factor of how disruptions are interpreted and responded to. According to König et al. (2021), a higher perceived control usually supports

an opportunity perception. When changes feel manageable and controlled, commonly is correlated with optimism. If the control is perceived as low and the situation is diffuse or difficult to affect, a threat perception is more common. However, our findings display a deeper dynamic of how student navigates perceived control. The respondent was unanimous regarding their perception of a high personal control over how they use AI tools, regardless of their education background or AI familiarity. It was also agreed among the students that they have minimal to no control over AI's development on macro-level, including societal or organizational level. The students separate personal control where they remain in agency and accepting the lack of control and power on a macro-level. This allows the students to have an opportunistic perception regarding their personal usage of AI and a threat perception regarding AI's development an impact on a larger scale. Interestingly, this is consistent for both studying fields, despite the difference in AI exposure, formal education and general AI orientation. This could suggest that the segmentation of personal and macro control is structural of how student in present time relate to AI, rather than being an individual phenomenon.

How this perception affects the behaviour of the students, differ vastly between the two student groups. Staw et al.'s (1981) threat rigidity theory describes how a perceived threat can make individuals restrict the information intake in order to retain control and rely on their already attained knowledge. Connelly and Shi (2022) further extends this with three types of responses to a perceived threat, fight, flee, and freeze. The response is determined by the level of perceived vulnerability and severity of the threat. For the business student, many expressed explicitly that if they gained more AI knowledge, they would improve their sense of control. However, they had no explicit plan on how to acquire that knowledge. This reflects the freeze response, where recognition of the threat is generally clear, but the action to minimize the threat remains restrained. This behaviour is also aligned with Fugate et al. (2012) description of how a low perceived control in combination with a threat perception can evoke a passive and avoidant response. This can suggest that the business perception of low level of macro-level control and lack of formal AI education contributes an avoidant behaviour.

The civil engineering student on the other hand, are more commonly demonstrating their active and adaptive response to AI's integration. Poppy suggests that the competitiveness can come from the ability to adapt in response to the low level of control on macro-level, Iris expresses their continuous learning and Bamboo recalibration of educational investments and AI literacy are all examples of Gioia and Thomas (1996) explain as an exploratory response with an opportunity framing. This implies investments in actively trying to retain new knowledge and skills as a response to a perceived change. This is not only due to their AI familiarity being greater than the business students but also reflects a difference in how they interpret their perceived limited macro-level control that formulates a behavioural strategy to adapt.

Furthermore, the findings unveil another behavioural effect as explained by Berntson and Marklund (2007) as the individuals' subjective assessment of their possibilities to obtain new employment on the labour market. This concept captures the objective competence as well as the individual's own perception of their competitive position. The civil engineering students display an active adaptation that goes beyond taking AI courses and self-reliant learning. Rose expresses the competitive advantage of a deep understanding of AI, consciously repositioning themselves to match a future demand by and employer. Iris expresses the value of understanding the demand om the market and adjusting accordingly. These expressions reflects that the students make assessment of their human capital investments in relation to the anticipation of what the future labour market will value.

Business students on the other hand, provide a similar reasoning, although it is less focused on actual subject expertise in AI. The business students overall where motivated more towards AI literacy to be a demonstratable skills, adding to their perceived competitiveness. This reflects an implicit assessment of perceived employability as the student recognise the value of having AI knowledge. It is apparent that the students do not only adjust their developing human capital according to AI but also to other future labour market entrants. Furthering the concept by Berntson and Marklund (2007), this assessment is occurring before the student even enter the labour market. Perceived employability is both a response mechanism to navigate labour market conditions for employed workers as well as future labour market entrants, shaping the human capital investments they make to match future demand.

This repositioning on an individual level can have organizational consequences. Both Ployhart and Moliterno (2011) and Youndt and Snell (2004) emphasise that organizational human capital emerges from individual KSAOs interacting, and how that accumulation can bring forward strategi value on an organizational level, depending on how effectively it is embedded in the organization. In other words, it matters on an individual level for the organization to be able to leverage on the competencies. The students' perception form before entering the labour market, which means that they are not neutral. The perceptions are crucial in what competencies they choose to develop as well as their standpoint on AI related tasks. This can decide on whether or not they are susceptible to engage with AI, depending on the perceived threat or opportunity framing. Therefore, students with limited formal AI education and a passive adaptation orientation may be less prepared to use AI effectively in an organization, making the accumulated human capital less optimized. In contrast, students that have an active adaptation orientation and strong AI literacy are more likely to have created human capital that is better optimized to an AI integration and to work productively. This can have a vital impact on organizations, as the consequences of perception structures developed before entering the organizations affects all levels of the organization. This highlights the relevance of understanding the future labour market entrants' perception is strategically relevant for organizations who are susceptible to AI integration. Campbell et al. (2012) argues that human capital does not automatically give competitive advantages for an organization, but it must be actively worked with and embedded in the organization and the organizational context. The perceptions students develop before even entering the labour market therefore have relevance for organizations, as they impact what competencies the students develop and their view of interacting with AI in their workplace. Altogether, this shapes if the human capital they offer can be effectively embedded and utilized.

To answer the latter part of our research question, our findings articulates that the students' perceptions of AI are not solely cognitive, instead they are translated into patterns of skills and knowledge development. The gain/loss appraisal generates a revaluation of accumulated competencies and changes human capital investments. The perceived control separation between high personal and low macro level unveils the response of the students, active or passive. This is a clear difference between business students and civil engineering students as observed in our findings. Taken together, these two structured processes connect perception to human capital logic, where perceptions form consequences of what human capital the students can offer in the labour market.

5.3 Adjusting the conceptual framework

The conceptual framework presented in section 2.5 was developed deductively from the theoretical literature and guided the initial design of this study. It proposed that the perception of AI is shaped by sensemaking and selective attention as well as produce gain/loss framing and perceived control as appraisal outcomes. The outcome in turn drives human capital revaluation and the behavioural response. While the structure of the conceptual framework was largely accurate in its predictions, the empirical findings have revealed patterns that refines and extends the original model.

Most significantly, the findings show that the educational background does not only function as an input into individual sensemaking but rather as a structured shaping condition. The field of study, which includes the actual curriculum, peer culture and norms, mediates what students notice about AI and how they understand it. This suggests that selective attention does not only operate at the individual level, as Dearborn and Simon (1958) originally proposed, but that the field of study function as a shared conditions that can shape what an entire group of students notice and prioritise in relation to AI.

The findings also revealed patterns that the framework did not fully capture. Occupational distancing emerged as a consistent strategy where students acknowledge the disruption of AI in other occupations while protecting their own future field of work. This is a pattern that the original framework did not specify. The appraisal outcomes further shows that perceived control and gain/loss framing does not operate as a unified dimensions as the framework assumed. Instead, students tend to frame AI as a gain on the individual level, while a loss at the occupational and societal level. The students also expressed high perceived control on the personal level and low perceived control on the macro level.

6. Conclusion and Recommendations

This chapter presents the general conclusion of the study, aimed to answer our research purpose and research question. The conclusions are derived from the collected data, analysis and discussion. Lastly, the chapter will address the limitations to the findings.

6.1 Conclusions

This thesis aimed to contribute to research within business administration by increasing the understanding of how students, that are the future labour market entrants, perceive artificial intelligence in relation to their future working like, and how those perceptions shape their skills and knowledge development. Based on 15 semi-structured interviews with business students and civil engineering students at Umeå University, the purpose of the study sought to answer the following research question:

How are students perceptions of AI in relation to their future working life formed, and how do these perceptions shape their development of skills and knowledge as future labour market entrants?

The findings show that students' perceptions of AI are not formed through independent and rational assessment but rather through the interaction of educational background, social environment and prior experience with AI tools. These conditions determine what students attend to, such as what aspects that they notice, which narratives they find credible and how they construct interpretations of an uncertain situation. Civil engineering students, who have formal AI education as part of their programme, develop a more technical foundation that directs their attention toward systemic implications of AI which extends their threat perception beyond their individual concerns to more societal consequences. Business students, without formal education in AI, may therefore be more dependent on the framing provided by their social environment, where media amplifies the threat narrative and peers act as normalising AI as an opportunity. Both groups arrive at an ambivalent perception, seeing it as both an opportunity and threat simultaneously, but the level at which they locate the threat differs between the groups and their educational background. The opportunity perception is concentrated at the individual level for both groups, while the threat perception extends to societal concerns among the civil engineering students. Both groups choose to distance themselves from the more immediate disruption of AI, by either exempting their own chosen work path from other occupational fields they identify as at risk, or by placing the threat in the distant future. These distancing mechanisms, which can be understood through Dearborn and Simon's (1958) selective attention theory and Weick's (1995) sensemaking framework, allows students to acknowledge the reality of AI's disruption while maintaining a manageable view of their own future employability.

The perceptions formed through this process do not necessarily remain passive, they shape different patterns of skills and knowledge developments. Both student groups actively revalue their competencies given their perception, where technical skills are perceived as more vulnerable to AI automation and relational skills are viewed as more resilient. This revaluation is grounded in the logic of human capital investment, where students assess the expected future returns of their developing competencies and redirect their effort accordingly. Several respondents' states that they consciously shift their efforts toward AI

literacy as the most valuable investment. However, the behavioural consequence of the perception splits the two groups. Civil engineering students respond to the low macro-level control through active adaption by investing time in AI courses, seeking out knowledge and repositioning their competencies to match the anticipated future labour market demand. Business students more commonly acknowledge that they need to adapt and improve their AI literacy without always turning it into concrete action, a pattern which is consistent with Staw et al.'s (1981) threat rigidity theory. This difference is not incidental but is shaped by the same educational background conditions that structure the perception formation, meaning that the presence or absence of formal AI education determines both what students perceive and how they respond. Together, these findings demonstrate that how perceptions of AI are formed has consequences for the human capital that students will bring to the labour market.

6.2 Theoretical contribution

This study contributes to the researching field within business administration by deepening the understanding of how future labour market entrants perceive technological discontinuity and how these perceptions can affect the human capital in an organization. Firstly, the findings in this study contributes to perception theory by extending selective attention and appraisal research beyond their organizational contexts. Dearborn and Simon (1958) established that individuals attend to and interpret information in ways shaped by both their existing role and professional background. König et al. (2021) further developed the understanding of perception by arguing that responses to discontinuous change are structured by two independent factors, gain/loss framing and perceived control, rather than a single unified appraisal. Both frameworks have primarily been applied to organizations and active workers. This study demonstrates that educational field of study functions a shaping condition, producing systematically different perceptual outcomes among students encountering the same technological phenomenon. A further finding extends König et al.'s (2021) framework specifically, where the split between personal control and macro-level control does not vary between students. It is consistent across both groups regardless of educational background or AI familiarity, which suggests it is a structural condition of being a future labour market entrant rather than an individual difference. Together, these findings extend existing perception theories to the pre-entry labour market context, showing that mechanisms through which perception is formed and structured are active before individuals enter the organizational setting.

Secondly, the findings contribute to human capital theory conceptualised by Becker (1962) and Schultz (1961) and extended to the organizational context by Ployhart and Moliterno (2011). Human capital theory treats educational investments as a rational forward-looking calculation of expected future returns. This study demonstrates that these calculations are not made neutrally, rather they are directly conditioned by how students perceive the phenomenon as a gain or loss and by their perceived control over its consequences. This points to an asymmetry in perception that human capital theory has not previously accounted for, investment decisions are not shaped by what is objectively optimal but rather what students are able to perceive, which in the context of AI means the displacement effect dominates while the reinstatement effect (Acemoglu & Restrepo, 2019) is largely unrecognised. The findings therefore contribute to human capital theory by introducing perception formation as a conditioning variable in the investment decisions before entering the labour market. This extends the theory beyond its assumption of relatively informed rational calculation towards more of a understanding to how uncertainty and educational background shape competence development before labour market entry.

6.3 Practical contribution

The primary practical contribution from this study is directed towards organizations. The findings show that students arrive at the labour market with already pre-formed perceptions of AI that are not neutral. These perceptions, that are influenced by educational background, social environment and prior AI experience, may influence how new employees engage with AI-related tasks and organizational change related to AI. Though, the extent to which perceptions actually translates directly into workplace behaviour remains a question for future research.

The behavioural difference identified between the two student groups is practically relevant for organizations. Civil engineering students in this study more commonly show an active orientation toward AI by adjusting educational human capital investments and seeking out AI knowledge. Business students more commonly recognize the need to adapt but does not always turn it into action. Organizations recruiting from both groups should be aware that these orientations are not fixed, but they rather suggest that there should be a structured support for Ai integration in early career stages since. If an organization assumes a self-directed approach to learning AI among new employees, it is likely to produce uneven results.

Organizations that understand the structures of the perception of incoming employees might also be better positioned to embed human capital effectively. Campbell et al. (2012) argues that individual competencies do not automatically generate value for an organization, they must be actively embedded within the organizational context. The perception structures that is presented in this study affect what competencies students choose to develop before entering the labour market. By understanding these patterns can therefore support more informed decisions onboarding and early career development in organizations where AI is becoming a central part of how work is performed.

Although this study is not in the field of education, the findings can carry observations that could be relevant for higher education institutions. Business programmes that do not have any formal education about AI leaves students to form their understanding of AI and the labour market implications from media, peers and their own experimentation. The findings suggest that this gives the person familiarity with the tools with AI, nut not as much of an analytical depth needed to reason about how AI reshapes the skill demand over time. Civil engineering programmes, where AI has been integrated into existing courses to different degrees, appear to support a more grounded perception.

6.4 Societal and ethical implications

Since the integration of formal AI education in different programmes differs, it means that the students enter the labour market with different structures of perceptions depending on their field of study. Business students, without any guidance from the institution, form their understanding of AI's impact on the labour market primarily through peers and media. Civil engineering students, that at least have partial formal exposure, develop more analytically grounded views. This difference is not purely an educational issue, it also has broader implications for how AI-related competence develops across the future workforce and who is better to navigate changes the AI brings to organizations.

This finding can also be connected to an ethical implication in relation to unequal access to AI. UNESCO's framework for AI competency in education emphasise the importance of responsible AI integration, helping student to be critically assessing AI and its implications to use it with responsibility (Miao et al., 2024, p.14). This is not only limited to technical knowledge of AI, but also includes human agency, judgement and ethical responsibility to ensure understanding of societal and environmental impacts of AI (Miao et al., 2024, p.14-15). In our study, we acknowledged the different conditions between business students and civil engineering students of their relation to AI from their education. This displays an uneven AI competence foundation among students, on their course to enter the labour market. This is of ethical relevance, as it suggests that the students that do not have any AI related education are dependent on self-directed learning from informal sources, shaping their perception without necessarily considering critical or ethical implications. Higher education may therefore contribute to student being unevenly prepared for an AI-affected labour market, creating issues of whether or not the education is preparing students to be equally responsible to handle AI in society.

Several students in this study have raised concern about the absence of regulatory frameworks for AI. The findings show that the gap in governance surrounding AI is something that future labour market entrants, students, have concern about and something that they factor into how they perceive their own futures. The historical pattern of regulation not keeping up with the developments is something students are aware of, and it contributes to their sense of low perceived control of the macro-level developments of AI.

The civil engineering students in this sample raised the concern of perspectives becoming too homogenous, expressing that a more widespread use of AI as an information source reduces the amount and diversity of perspectives that drives innovation and analytical reasoning in organizations. While this expressed concern goes beyond what this study can actually evaluate empirically, it reflects a consideration that organizations and researchers in business administration should take seriously as AI becomes more integrated.

Environmental concerns about the energy and water consumption when using AI were also raised by some students, and that AI should therefore not be used as much due to this. This falls outside the scope of this study, but it is worth noting that sustainability considerations are entering how some students perceive AI, which may become relevant for how organizations communicate their AI practices to future employers, and maybe also other stakeholders interested in sustainability.

Ethically, this study engaged with students navigating uncertainty about their futures in a labour market they cannot control. The perceptions captured and researched in this study are not abstract, rather they reflect real anxieties with consequences for the educational and career decisions that the students are currently making. The findings should be read and interpreted with this in mind.

6.5 Limitations

The methodological limitations of this study, which includes the use of a small non-random sample, network-based recruitment and the potential for response bias around AI are addressed in 3.10 Methodological limitations. This section addresses the limitations that are specific to the interpretations of the findings.

Both student groups were recruited from the same university, which means that the observed differences and similarities between the groups may also reflect the culture at the university, local peer norms or labour market expectations that apply to the region, rather than differences and similarities that applies across other institutions and contexts. The findings cannot be generalized beyond this context, and replication in other universities, where there might be other programme structures and other levels of AI integration, would be needed to draw more broad conclusions.

The study captures perceptions at a single point in time. The findings show that perception shifts with time as familiarity with AI increases and as the technology develops, however a cross-sectional design of the study is not able to capture this development. The perception students hold today may differ from what they hold when they enter the labour market, particularly as AI continues to develop rapidly.

The two student groups have differences across several variables, such as programme structure, level of formal AI education and different career orientations, which makes it difficult to understand specifically what drives the differences in perception between the two groups.

Finally, all interviews were conducted in Swedish and then all quotes used to present the findings were translated to English by the authors. Translation was carried out carefully and all original quotes are available in Appendix 3, but there is a risk that nuance may have been lost in some cases, particularly when expressions have been used that do not translate directly between languages.

6.6 Recommendations for future research

The most direct extension of this study would be a longitudinal design that follows the same students of a comparable group as they enter the labour market. The distancing mechanisms that are identified in this study, such as locating threats in other fields or further into the future, are likely to be tested once students are working in organizations where AI is integrated and the impact becomes more immediate and personal. Whether these mechanisms dissolve, persist or transform into different responses Is a question this study is not able to answer.

A comparative study across multiple institutions, fields of study and/or national contexts would help establish whether the patterns that were found in this study are specific to the setting or more general. In particular, it would be valuable to research whether the differences between business students and civil engineering students is that same where business programmes have integrated formal AI education.

Future research could also examine the organizational side of the relationship explored here. This study focuses on how students form perceptions before entering organizations. A suggestion would be to analyse how organizations handle the pre-formed AI perceptions that new employees bring, and whether recruitment, onboarding and practices for developing the early careers are designed in ways that account for differences in perceptions.

Finally, research that examines whether introducing structured AI education into business programmes changes the perception formation process would be of both theoretical and practical value. Specifically, it would be useful to know whether formal AI education shifts

selective attention patters, reduces the dependence on threat narratives from media and supports more analytical reasoning about how AI reshapes tasks structures and skill demand, which is the kind of reasoning that civil engineering students in this study has shown more consistently.

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Appendix 1: Interview guide in English

Introduction

- Present ourselves and describe the purpose of our study.
- Confirm consent to record the interview.
- Emphasize how we collect and handle the data, and ask for confirmation of consent.

Background

- **Establish the respondent's current programme, semester and familiarity with AI.**
- **Q1:** Briefly tell us about yourself and your studies. What are your goals with your education after graduation? Why did you choose your programme? Have you started your job search?

- **Q2:** How would you generally assess your familiarity with AI tools? What was the purpose of your use?
- **Q3:** What did your first contact with AI look like? How often do you use it? Which AI tools do you use?

AI use

- **Understand the respondent's habits regarding AI use in order to identify an augmented or automated usage pattern, without using the actual terminology.**

- **Q4:** Tell us how you use AI in your everyday life and in your studies. What tasks do you usually use it for? Do you use it for feedback or to complete a task? Give a concrete example: what did you let AI take over?
- **Q5:** Has your AI use changed over time? If yes, in what way?
- **Q6:** To what extent do you rely on AI's output? Do you fact-check it, or do you use it directly? Are there any tasks you would not trust AI with? Why?
- **Q7:** Does your programme include any AI-related education? How do you think this has affected how you use AI? Where have you gained your knowledge from? Would you say this has affected how you feel about AI?

Human capital implications

- **Understand the respondent's reasoning about their skills and education, and whether and how they have adapted to an AI-influenced environment.**

- **Q8:** Think about the skills and knowledge you have gained from your education. In what way would you say AI could affect their value in your future career choice and their competitiveness? Can you identify any of your skills that you think AI could replace? Can you identify any skills that you consider less worth developing because of AI?
- **Q9:** Has AI affected how you think and reason about which knowledge and skills you need to have or develop? In what way?
- **Q10:** Do you think your education prepares you for an AI-influenced labour market? What is missing? Have you done anything independently to gain knowledge about AI tools?

- **Q11:** Does AI change how you think about prioritizing your time and effort in your education? Do you feel uncertain about the actual value of certain skills for your future work? How has this been reflected in your choices?

Perceptions: AI as an opportunity or threat

- **Explore perceptions of AI's impact on respondents' future career choices and the mechanisms behind this, including perceived control, sensemaking and selective attention**.

- **Q12:** Think about AI and your future work. Do you see AI as an opportunity or a threat? Something else? Has your view changed? Are there aspects that you see as particularly negative or positive?
- **Q13:** How do you think AI will be integrated into your future workplace within the next 5–10 years? Which tasks will still require humans? What excites you? What worries you?
- **Q14:** Do you feel that you have influence and control over AI and how it affects your future? For example, through developing skills or choosing a certain occupation? Do you feel in control? Can you identify anything that would make you feel more in control over your future career?
- **Q15:** What is your strongest asset in relation to the development of AI? How competitive are your skills and knowledge? Is there anything in your background or education that makes you feel less secure in relation to AI?

Sensemaking

- **Explore how the social environment reflects the respondent's perception, including education, family, friends, media and similar influences.**

- **Q16:** Has your environment, such as your education, family, friends or media, influenced how you think and feel about AI in your future work? Are there differences among you? Is there a certain norm in your study programme?
- **Q17:** Where do you get your information about AI and how it affects your professional field?

Summary / Closing

- Ask whether the respondent has anything to add, either to their contribution or regarding the data collection.
- Thank them for their time and remind them of their anonymity.

Appendix 2: Interview guide in Swedish

Introduktion

- Presentera oss själva och beskriv syftet med vår studie
- Bekräfta samtycke till inspelning av intervjun
- Betona hur vi samlar in och hanterar data, be om bekräftelse av samtycke

Bakgrund

- **Etablera respondentens nuvarande program, termin och kännedom om AI**
  - F1: Berätta kortfattat om dig själv och dina studier – vad är dina mål med utbildningen efter examen? *Varför valde du ditt program? Har du påbörjat din jobbsökning?*
  - F2: Hur skulle du generellt bedöma din kännedom om AI-verktyg? *Vad var syftet med användningen?*
  - F3: Hur såg din första kontakt med AI ut? *Hur ofta använder du det? Vilka olika AI-verktyg använder du?*

AI-användning

- **Förstå respondentens vanor kring AI-användning för att identifiera ett augmenterat eller automatiserat användningsmönster, utan att använda den faktiska terminologin.**
  - F4: Berätta hur du använder AI i ditt vardagsliv och i dina studier – vilka uppgifter använder du det vanligtvis till? *Feedback eller slutförande av en uppgift? Ge mig ett konkret exempel – vad lät du AI ta över?*
  - F5: Har din AI-användning förändrats över tid? *Om ja, på vilket sätt?*
  - F6: I vilken utsträckning förlitar du dig på AI:s resultat? Kontrollerar du fakta eller använder du det direkt? *Finns det uppgifter du inte skulle lita på AI med? Varför?*
  - F7: Har ditt program någon AI-relaterad utbildning? *Hur tror du att det har påverkat hur du använder det? Var har du fått din kunskap ifrån? Skulle du säga att det har påverkat hur du känner inför det?*

Human capital implikationer

- **Förstå respondentens resonemang kring sina färdigheter och utbildning, och om och hur de har anpassat sig till en AI påverkad omvärld.**
  - F8: Tänk på din dina färdigheter och kunskap du fått från din utbildning – på vilket sätt skulle du säga att AI kan påverka deras värde i ditt framtida yrkesval och deras konkurrenskraft? *Kan du identifiera några av dina färdigheter som du tror att AI kan ersätta? Kan du identifiera färdigheter som du anser vara mindre värda att utveckla på grund av AI?*
  - F9: Har AI påverkat hur du tänker och resonerar kring vilka kunskaper och färdigheter du behöver ha eller utveckla? *På vilket sätt?*
  - F10: Anser du att din utbildning förbereder dig för en AI-påverkad arbetsmarknad? *Vad saknas? Har du gjort något självständigt för att skaffa dig kunskap om AI-verktyg?*

- F11: Förändrar AI hur du tänker kring hur du prioriterar din tid och dina ansträngningar i utbildningen? *Kan du känna osäkerhet kring färdigheters faktiska värde för ditt framtida arbete? Hur har detta visat sig i dina val?*

Uppfattningar: AI som möjlighet eller hot

- **Utforskning av uppfattningar om AI:s påverkan på respondenternas framtida yrkesval och mekanismerna bakom detta – upplevd kontroll, meningsskapande, selektiv uppmärksamhet.**
  - F12: Tänk på AI och ditt framtida arbete – ser du AI som en möjlighet eller ett hot? Något annat? *Har din syn förändrats? Finns det aspekter som du ser som särskilt negativa eller positiva?*
  - F13: Hur tror du att AI kommer att integreras på din framtida arbetsplats inom de närmaste 5–10 åren? *Vilka uppgifter kräver fortfarande människor? Vad exalterar dig? Vad oroar dig?*
  - F14: Känner du att du har påverkan och kontroll över AI och hur det påverkar din framtid? Till exempel genom att utveckla färdigheter eller välja ett visst yrke? *Känner du att du har kontroll? Kan du identifiera något som skulle få dig att känna mer kontroll över din framtida karriär?*
  - F15: Vad är din starkaste tillgång i relation till AI:s utveckling? Hur konkurrenskraftiga är dina färdigheter och kunskaper? *Finns det något i din bakgrund eller utbildning som minskar din trygghet gentemot AI?*

Meningsskapande

- **Hur den sociala omgivningen återspeglar respondentens uppfattning (utbildning, familj, vänner, media etc.)**
  - F16: Har din omgivning – såsom utbildning, familj, vänner, media etc. – påverkat hur du tänker och känner inför AI i ditt framtida arbete*? Finns det skillnader bland er? Finns det en viss norm i ditt studieprogram?*
  - F17: Var hämtar du din information om AI och hur det påverkar ditt yrkesområde?

Sammanfattning / Avslutning

- Fråga om uppföljningsfrågor ifall respondenten har tillägg de vill göra till sitt bidrag och/eller angående datainsamlingen
- Tacka för deras tid, informera om deras anonymitet

Appendix 3: Original quotes in Swedish that have been used in Chapter 4

Theme 1 – Shapers of perception

"Ja, det har vi. Vi har gått en introduktionskurs till AI. Och sen så har vi även blivit uppmuntrade till att använda AI i olika kurser. Så även fast det kanske inte är en AI-kurs specifikt - så har vi gått typ programmering. Och där har man testat programmera med AI. Och där har jag fått hjälp av läraren. Det gick jättebra, så ja, det skulle jag säga" - Rose, Civil Engnineer

"Jag har läst någon kurs i AI. Alltså lite mer hur AI är uppbyggt och sånt där. Alltså grundkurser. Men annars har jag inte lärt mig jättemycket om exakt hur det funkar. Hur ska man tänka kritiskt när man använder AI? Inget sånt har vi stött på". – Iris, Civil Engineer

"Först och främst någon form av utbildning som går igenom vad är AI och hur kan du bäst använda det till din fördel? För att oavsett hur man än vrider och vänder på det, du kommer behöva använda dig av AI i framtiden för att få någon form av edge före alla andra, för alla kommer använda det. Om du inte använder AI, jag är ledsen, men du är behind the curve." – Linden, Business

"Det är mycket från att titta på youtubevideos och inne på sociala medier, och så får man höra från vänner: 'Så bara testa det här. Nej, men jag kan testa det.' Så det är mycket learn by doing. Ja. Som jag var inne på lite innan också. Det blir såhär att när man började med AI så hade man inte direkt någon koll på någonting. Utan det har blivit liksom.. Ja, man hade bara lärt sig över tiden av sig själv."– Birch, Business

"Främst av att utforska själv såklart. Det är nog där det kommer ifrån mest. Jag fattar liksom hur de fungerar och sen så har jag även kollat på klipp på Youtube främst för att se såhär, you're not behind the curve just yet. Och så lite tips på hur man faktiskt kan få fram det svaret man är ute efter. Hur man ska kommunicera med just AI och prompts och sånt." – Linden, Business

"Det är väl främst media som har fått den att känna sig oroliga orolig till det och att det är någonting negativt. Familj och vänner så har vi inte pratat så mycket om det, men snarare att vi försöker se det som en möjlighet när man pratar med kompisar om det. Men att det det är ju till största del ganska negativa konversationer om ai." - Rowan, Business

"Framförallt kanske - media, LinkedIn, - diskussioner mellan familjemedlemmar. Absolut. Och att man förstår att - det kommer spela en större roll i framtiden än vad man kanske trodde och framförallt nyheter som också uttalar sig om att AI tar våra jobb, bla bla bla. Alltså mycket de här clickbait rubrikerna typ" – Maple, Business

"Det är stor skillnad från ja, bland bland ja som sagt, allt från kompisar och vänner och men mycket i Media är ju också domedagsskriverier." – Alder, Business

"Ja, jag skulle säga, men kanske mycket sociala medier - och poddar och sånt där, att det har väckt mer oro kring AI i framtiden. För det belyser mycket det här negativa perspektivet - att det utvecklas på ett sätt som kanske inte blir kontrollerbart i framtiden. Och det gör ju såklart att man blir ju rädd eftersom Jag sitter ju inte på världens kunskap

kring hur AI funkar och vad de gör och hur det utvecklas. Men sen bland vänner och så blir det kanske mer en uppmuntran och en mer positiv bild kring AI." (Iris, Civil Engineer)

"I princip vilken person du än pratar med använder AI. Man pratar alltid med dem och ser hur de använder det, vilka verktyg de vänder. Till exempel att jag gått över till Claude nu. Från chatt-GPT. Det är något som kommer direkt från min omgivning. Det var någon annan som testade det och sa att det fungerade mycket bättre. Och då tänkte jag så här, jag har tröttnat på chatt-GPT. Då går jag över till Claude istället. Och det har påverkat mig helt och hållet för jag tycker den är mycket bättre. Och sen så pratar man ju också med kompisar som ger en tips på det här använder jag för att göra AI."(Linden, Business)

"Jag skulle säga att min omgivning är ganska pro-AI generellt. Vilket då påverkar min inställning skulle jag säga. Hade de varit att radera ut internet då hade det kanske påverkat mig. Men det är de absolut inte. Men omgivningen är väl att använda AI så mycket vi kan med en med uppsikt." (Sage, Civil Engineer)

"Jag skulle säga att jag har en kompiskrets som är ganska AI-positiva. Dels också varför jag själv började använda AI. Och känner ändå att jag får mycket kunskap därifrån." (Bamboo, Civil Engineer)

Jag tror att om man ser det som en möjlighet så tänker man nog väldigt Individuellt Och det gammelmedia gör, eller media generellt gör, är att de tar upp - det här samhällsperspektivet och vad riskerna är för samhället med AI. Och det är väl det som har format mig, för jag ser AI som ett helhetsperspektiv - på samhället och inte bara för mig själv. - Rose

Det är två, tre år sedan det kom. Man var väldigt skeptisk till en början men nu har det blivit, Alltså, använder du det inte så är du ineffektiv." (Aspen, Business)

Jag skulle säga att från början var man ännu mer kritisk till det. Och jag vet inte, att man hade kvar att man ville göra saker själv, alltså jag ville skriva mina texter själv och ville inte använda AI. Och nu... Så har det ändå gått över lite, för jag litar mer på AI nu." (Iris, Civil Engineering)

"I början var det nog mer att man skickade in och hoppades på att man skulle få något bra svar tillbaka. Men att det numera blivit som att man använder den som att man har en lärarassistent ungefär. Att gå mer mot det hållet än att faktiskt försöka få svar och sen ta svaren och stoppa in det i en text. Utan mer försöka komma till rätt svar själv, den har blivit mycket mer användbar." (Bamboo, Civil Engineering)

"liter jag mycket mer på den att utföra jobbet medan man i början var mycket mer skeptisk." (Oak, Business)

"Men nu känner jag att det är så pass vitalt att ha en grundförståelse för AI så att man kan nyttja det. Det har helt förändrat sättet jag väljer att prioritera saker. Jag väljer att prioritera att lära mig AI för att kunna lära mig saker snabbare. Det har vänt på skalan upp och ner." (Sage, Civil Engineering)

Theme 2 – Apprasial under uncertainty

Inte är möjlighet till 100 procent. Vi säger fifty-fifty då. För som jag sa, liksom att absolut, du ska använda såna verktyg. Det ska inte användas för att byta ut folk. Men folk ser det inte på det sättet. Folk ser, de kan spara pengar. Och då kommer plötsligt det här hotet om att byta ut arbetare mot AI. Så ja, okej. Inte 100 procent. Fifty/fifty. - Birch

Det beror på lite grann hur det nyttjas också. Det kan ju vara en jättemöjlighet, men samtidigt kan det vara en jättefara också. Det beror på hur man väljer att utveckla det. Men det är liksom bortom min kontroll. – Sage

Det är ju en stor möjlighet i att använda AI. Genom att liksom få hjälp med när man ska skapa, vi säger att man ska skapa ett presentationsmaterial eller skriva en sammanfattning eller sånt som bara tar onödig tid. Absolut så finns det ett hot i det också. Om det blir, jag vet inte, för mycket. Eller att... Ja, beroende på hur man använder det helt enkelt. Och att det känns som att marknaden förändras ju. – Iris

Jag skulle vilja säga att jag ser det personligen som en möjlighet. Sen om man från företaget man ska jobba för ser det som en möjlighet också - att kunna skippa anställda, ja då kanske det helt plötsligt blir ett hot istället. Men där jag ska börja jobba så vet jag att jag har ett jobb. Jag har liksom ett säkert jobb. Och där folk använder AI redan. Och jag tror att det blir, inte att man inte blir utbytt utan man ser AI som ett komplement istället - birch

Är bara möjlighet, måste jag säga. Jag tror att jag ska man jobba med människor, alltså människor emellan. Då det kommer fortfarande behöva vara människorna och jag tror inte att en AI kan ta en människas roll. - Alder

”AI är ett ofantligt stort hot. Jag tror att jag... Det är klart att för mig som individ - så är det en möjlighet, för jag lär mig snabbare. Men AI för samhället är ett hot och bara ett hot”. . Rose

Såklart både möjligheter och hot. Hoten skulle nog vara att konkurrensen blir mycket hårdare i och med att jag tror att många av jobben kommer ersättas med AI, men personerna som söker jobben kommer fortfarande vara densamma. Så det kommer vara färre jobb som ger sig ut, men fortfarande lika många som söker. Så det är hotet. Det är liksom det absolut största hotet. Men möjligheterna är också enorma också. - linden

Det är nog kanske lite både och tror jag. Hotet skulle ju vara att jag i framtiden inte får ett jobb, för att det jobbet inte längre behövs mer eller mindre. För att du kan göra samma typ av arbetsuppgift med kanske färre anställda, för att vi har ett effektivare sätt att göra det på, för att vi har AI-verktyg som hjälper oss. - maple

Vi tappar den här mänskliga diskussionen - och det mänskliga perspektivet på allting. Det tror jag är väldigt negativt för framtida innovation - Rose

”Jag har ju själv bevittnat hur en uppgift som tar 3 månader och 20 anställda, som skapar jättemycket jobbmöjligheter, nu kan det göras med en AI. Man promptar och itererar i 20 minuter och sen är det klart.” - Rose

”Ja men dels kan det ju bli ett hot i att informationen som finns och tas fram, om alla bara använder AI för att ta fram informationen för underlag och sånt där, och att man inte kollar det på något sätt. Då blir det ju väldigt meningslöst, för då har vi bara ett perspektiv” - iris

”Det som oroar mig tror jag är att det är så pass trendigt och sen är det ju den här att det inte finns någon som håller koll på hur fort det går. ”- Sage

Jag tror att jag bör vara lite mer nervös för liksom mänskligheten. Vi som helhet alltså, vad den liksom hade kunnat göra i det. Jag tror att inte den kommer liksom bli ond nu och börjat utrota oss, men bara att den liksom alltså ändrar om världen så pass mycket att många människor blir liksom värdelösa. - Thyme

”Finanssidan kommer ta mycket stryk om jag ska vara helt ärlig, jag tror det finns mycket som kan ersättas. Deras yrke skulle kunna vara utraderat om tre år för att AI tar över det. Men eftersom jag inte ska jobba inom finans kommer det inte drabba mig”. – Birch, Business

”I alla fall inte de arbeten som jag vill göra, tror inte jag kommer ersättas. Jag ser det bara som en möjlighet att bli ännu bättre.” – Oak, business

”Jag vill hålla på med relationer och försäljning. Det kommer alltid vara bättre att det är face-to-face med en annan person än om du får prata med en robot.” – Linden, Business

”Ekonomi och finans jättemycket. Construction och sånt, knappt någonting. Allting som har med internet att göra, att man sitter med datorn och läser igenom saker, det kommer påverka jättemycket.” – Linden, Business

”Ja men i den rollen, tänker jag att just nu ser det ut som att det kanske inte skulle kunna ersättas, men inom tio år absolut alltså. Det är om du utvecklas i den här takten så kan det ju. Det är nog ett tag till i alla fall.” – Pine, Business

”Jag tror kundbemötande kommer alltid ändå göras av människor. I alla fall de kommande 50 åren eller 25 åren skulle jag vilja tro.” – Bamboo, Civil Engineer

”Det är inte att jag tänker på AI som ett direkt hot mot mig just nu, men på lång sikt ser jag att det förändrar vad som efterfrågas” – Maple, Business

”Det är inte att jag tror det kommer påverka mig direkt i närtid, men på sikt förändras ju vad som värderas” – Moss, Civil Engineer

”Jobben i sig kommer att förändras, men inte påverka arbetsmöjligheterna. Konsultarbetet kanske blir mindre analys, mer presentation och sälj. Inte att konsulter inte behövs, utan de behövs i en annan form.” – Sage, Civil Engineering

”Om AI blir tillräckligt avancerat så kanske det behövs bara två quants, en som promptar och en som kontrollerar att allting ser bra ut. Så det är väl att mängden arbete minskar på

grund av AI. Sen tror jag aldrig att det kommer försvinna helt och hållet, men just att det krävdes tio förut och nu Krävs det två eller tre" – Bamboo, Civil Engineering

" Men absolut att det finns risk att AI tar över vissa arbeten här i världen som säkert drabbar mig, men det kan ju också vara så att det tillkommer nya arbeten och då är det klart att då får man vara lite flexibel kring det. Det är klart att har man utbildat sig till någonting speciellt och AI lyckas ta över just precis den så kanske man får fundera på att utbilda sig till någonting annat eller byta bana liksom. Men när det kommer till min utbildning så är den ganska bred"- Poppy, Civil Engineer

Theme 3 – Revaluing competence under AI

"Analysering av bolag. Hundra procent. Förut så tog det, att göra liksom due diligence på ett företag. Det kunde ta flera dagar, flera veckor, nästan månader för ett bolag. Men om du har tillgång till rätt artificiell intelligens, verktyg, då kan du i princip göra det på bara några minuter." - Linden, Business

"Sen egentligen revision också kan jag ju tänka mig. Det är ungefär inne på samma spår. Du lägger in bolagsrapporter, kollar på det, du hänvisar till både lagstiftning och så kan AI egentligen gå igenom den utvärderingen som du bör göra på mycket, mycket kortare tid än om du skulle göra allting manuellt." - Linden, Business

"Ja förr i tiden var jag lite intresserad av att lära mig att programmera. För och typ också rediger så här photoshop Adobe illustrator. Alla dem har jag tänkt på, men jag börjar fundera på om det är en så värt att lära sig vid det här laget att det kommer ändå inte vara så efterfrågat i framtiden." - Rowan, Business

".... själva AI-verktyget, är ju att det faktiskt ändå utkonkurrerar många jobb. Som idag kanske... Behöver ha ganska mycket arbetskraft för att utföra." - Maple, Business

"Och jag tror att de här basala, jag ska göra en analys på "Ta fram den här informationen eller liknande". Jag tror att sådan kunskap kommer att ersättas. Och det kommer vara mer kvalitet, alltså kvalitetskontroll som är det som människor gör." - Sage, Civil engineer

"Såna här soft skills tror jag inte kommer kunna - ersättas av AI. Men hard skills, absolut…….Jag tror det har med människans relationer mellan två personer. Man vet ju själv om exempelvis att när du pratar med en AI-chattbot. När du har ett problem med vad som helst, typ hemförsäkring. Då kan man ju bli tokig för att man vill bara prata med - någon som kan lyssna på en och anpassa sig. Medan AI har ju som, det är promptat. Finns inte lösningen i deras förinställda inställningar, så kommer det inte funka. Medan en människa kan dra slutsatser som inte chatten kan göra." - Aspen, Business

"Ja, det är någonstans den mänskliga faktorn. Och som jag sa lite tidigare, kunder uppskattar fortfarande.. Det ska fortfarande vara mänskligt. Man blir bara "pist-off" när man pratar med en chatbot." - Birch, Business

"Den här mänskliga faktorn som vi liksom har som inte en ai kan ersätta. Det är sådana saker som rådgivning och liksom ledarskaps rollerna alltså sådana saker kommer ju vara väldigt svår att ersätta av en chattbot liksom." - Pine, Business

".... alltså typ de flumiga kurserna. Vi har gått också alltså typ så här ledarskap, lin och sådant det. Det replacerar ju inte riktigt en i alltså typ liksom de mänskliga grejerna är ju fortfarande bra." - Moss, Civil engineer

"Men det som man måste tänka på är de här soft skillsen man lär sig i skolan. Man lär sig kritiskt tänkande. Man lär sig att tankesättet. Om du inte kan någonting så vet du inte heller var du ska promta. Exempelvis om jag ska göra en analys av någonting och jag inte vet någonting. Då kan jag inte prompta upp. Men om jag kan någonting, om jag har grunderna i mig. Så kan jag ju göra en bättre prompt." - Rose, Civil engineer

"... nu är det mycket programmering. Då så. Kan man ju få hjälp. Alltså om jag verkligen vill läramig. Det är det som är skillnaden. Om jag inte vill lära mig, då kan jag verkligen glida på ett bananskal med AI. Att jag bara skickar in uppgifterna, det spottar ut någonting. Och det är ju godkänt liksom. Men om jag skulle säga att jag vill lära mig det, så kan jag ju använda det som ett verktyg istället." - Iris, Civil engineer

"Men om jag tänker bara på mina färdigheter och kunskaper - alltså det vi ska lära oss enligt studieplanen. Så det låter hemskt att säga, men utbildningen är ju då värdelös - känns det som. Alla de här smågrejerna och kunskapsgrejerna - det är ju saker som AI redan kan. Vi lär. Vi spenderar fem år och sedan tar ett lån på 500 000 - typ, för att lära oss någonting - där kunskapen redan finns väldigt lättåtkomligt." - Rose, Civil engineer

"Ja förr i tiden var jag lite intresserad av att lära mig att programmera. För och typ också redigera i så här photoshop Adobe illustrator. Alla dem har jag tänkt på, men jag börjar fundera på om det är en så värt att lära sig vid det här laget att det kommer ändå inte vara så efterfrågat i framtiden tror jag och att det är bättre att kanske utbilda sig inom ai marknadsföring har jag tänkt på." - Rowan, Business

"Jag har sökt en AI-kurs till nästa år, eller till nästa termin. Det tror jag inte riktigt man kände i början. Jag tänker ändå, när jag började studera 2021 så var AI väldigt nytt. Då tror jag man mer tänkte att man skulle läsa en managementkurs - i slutet för att få den humana delen. Men nu har majoriteten av mina... Studiekamrater har valt att läsa en AI-kurs i slutet för att få - bättre koll på vad det faktiskt är man gör." - Bamboo, Civil engineer

"Jag kopplade lite till det här med typ Excel, - att Excel kan ju vara ett verktyg för många arbetsgivare - där du blir extra konkurrenskraftig om du kan Excel på en bra nivå. Och jag tror att AI-verktyg i framtiden - kan spela in lite samma roll - Maple, Business

Theme 4 – Percieved control as a behavioural determinant

"Jag har ju kontroll över mitt användande av AI. Det har jag ju fullständig kontroll över. Men hur utvecklingen kommer se ut i världen, där har jag ingen påverkan överhuvudtaget utan det handlar egentligen bara om anpassning." - Linden, Business

"Jag har absolut kontroll över hur han själv använder ai och det är någonting jag kan bestämma över ifall jag gör det och hur jag gör det och vilken utsträckning men. Jag har ju ingen kontroll över hur ai kommer påverka världen." - Poppy, Civil engineer

"Just nu nej, det känns som att jag har väldigt lite kontroll för att man kan ju inte riktigt stoppa utvecklingen heller…Och att det sker i ja, men kanske mycket i Silicon Valley kan tänka mig och att många stora bolag går ihop och skapar artificiell intelligens nya typer av robotar så att jag har väl. Jag känner att jag har väldigt lite kontroll över det." - Rowan, Business

"Och det gör ju såklart att man blir ju rädd eftersom Jag sitter ju inte på världens kunskap kring hur AI funkar och vad de gör och hur det utvecklas." - Iris, Civil engineer

"jag tror nej. Där eftersom att jag själv inte är inblandad i området. Så har jag liksom ingen.Egentligen ingen aning om vart den är liksom på väg. Alltså. Det känner absolut inte det har någon kontroll över den, utan vad som händer eller händer oberoende av vad jag och så tycker eller hade velat" - Thyme, Civil engineer

"Nej, man har ju ingen kontroll alls. Man har ingen kontroll, utan man måste parera utefter samhället. Och så är det ju med allt, så är det ju med förra digitaliseringen också. Man kan inte stoppa en digitalisering utan man måste rida vågen helt enkelt." - Rose, Civil engineer

"Nja, kontroll och kontroll. Jag tror inte man har så mycket att säga till om. Alltså, AI kommer ju utvecklas och bara bli mer och mer komplicerat. Det är väl typ om, ja, inte vet jag. Lagar och krav och miljöpåverkan. Alltså mängden data som används och sådant där som - skulle kunna påverka, alltså hur vi människor hade kunnat påverka - AI:s utveckling. Men annars tror jag att den kommer bara fortsätta - att bli bättre och bättre om man säger så. Och på så sätt också ta mer ansvar." - Aspen, Business

"Det är väldigt tråkigt att man om man inte skulle göra det man vill för att man är rädd att ai ska ta över det, man kanske mer får anpassa sina egna förmågor för att göra en konkurrenskraftig. Att för som sagt så tror jag kanske inte att alla jobb försvinner, men kanske att det blir färre jobb och om man då har förmågan att anpassa det jobbet till ai så tror jag Att man kan ha stora konkurrensfördelar." - Poppy, Civil engineer

"Det är nog att jag tror att jag är väldigt konkurrenskraftig i mina kunskaper kring AI. Jag är väl den enda som jag känner som skapade ett företag med AI och använder det så pass mycket som jag gör. Och jag tror att det också är en stor konkurrenskraftig fördel, att jag inte ser det som ett lathetsverktyg ett utan effektiviseringsverktyg." – Rose, Civil Engineer

"Än så länge känner jag liksom att jag har kontroll över det och att jag liksom kan. Välja lite tillfällen, så här är det alltså. Hur mycket får jag ut av det här? Är det liksom behöver den här hjälpen, men framåt är det ju väldigt svårt att känna att man alltså tar kontroll över någonting." - Pine, Business

"Det tror jag inte att jag kan säga att jag har kontroll alltså. Det tror jag inte. Så att den jag tror den kommer få visa vad den går för här kommande år och sen hur den påverkar min framtid.." - Alder, Business

"Det känns som att det inte riktigt ligger i mina händer - på något vis. Det kommer att utvecklas - 100 procent och Ja visst, jag hade väl kunnat känna att det ligger i mina händer om jag - hade känt att jag hade känt att jag hade tjänat något på att lära mig AI-verktyget. "Så visst, du hade väl kunnat - känna att det ligger i dina händer, men jag kanske inte heller är så intresserad av - just den delen." - Maple, Business

"Jag ser det här som en möjlighet att vara en av de som kanske lär sig. Att man jobbar med AI redan nu och kunna erbjuda det till framtida arbetsgivare att så här. Jag har kunskap inom AI. I hur man använder det i marknadsföringssyfte." – Oak, Business